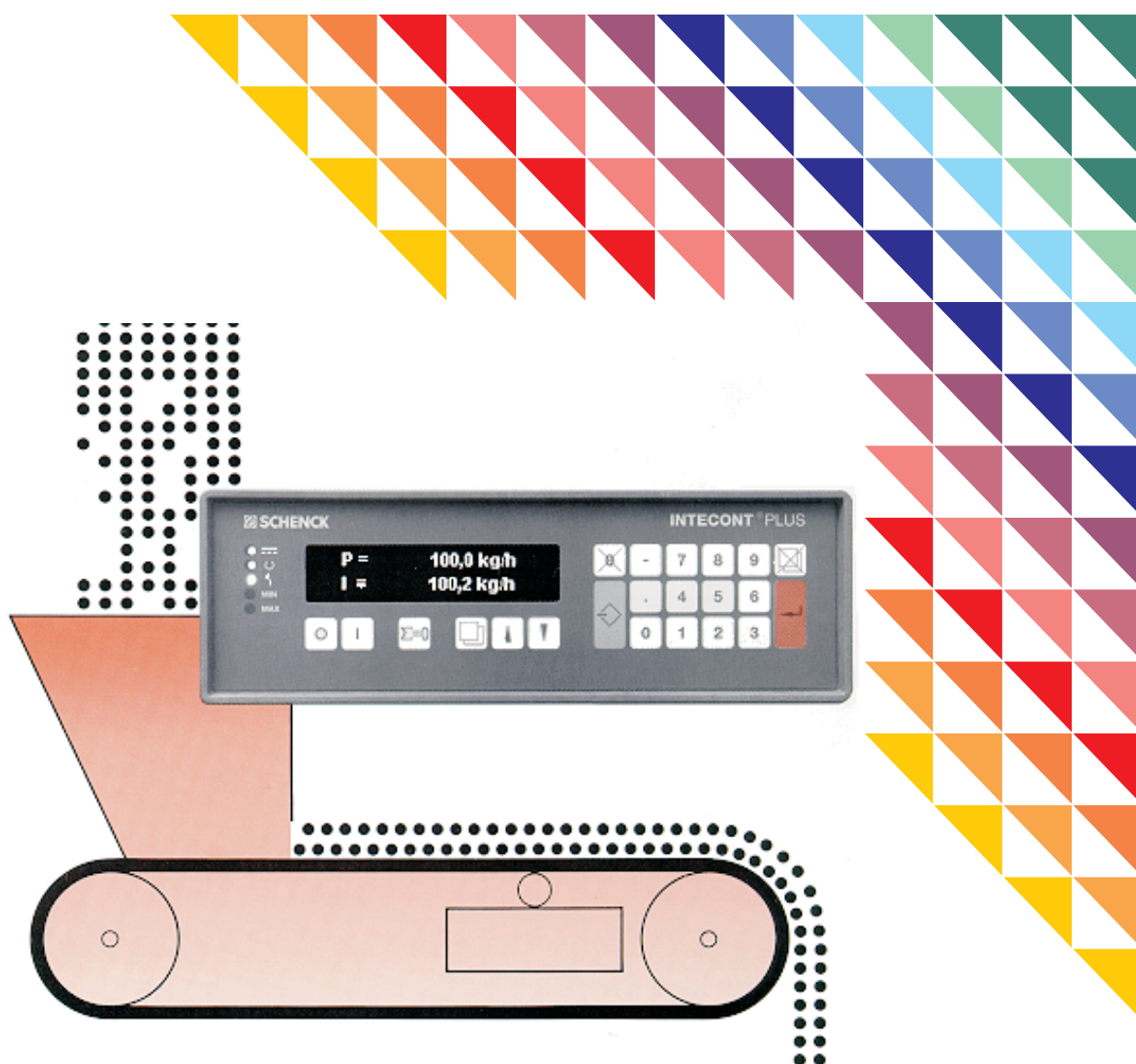




# INTECONT PLUS<sup>®</sup> Weighfeeder Operation Manual

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## 1. Overview

### What Is The INTECONT:

INTECONT PLUS is designed as measuring and evaluation system for

Belt Weighers  
Solids Flow Meters  
Mass Flow Meters  
Mass Flow Feeders  
Weighfeeders  
Loss-in-weight Feeders

The system is supplied in 3 variants:

1. Without option card (standard)
2. With option card VFE 610V  
for additional input/outputs and printer
3. With option card for fieldbus  
VSS021V : Modbus/S5  
VPB 020V: Profibus  
VCB020V: DeviceNet

This manual applies to controlling systems equipped with conveyor belt.

1. Weighfeeder  
Control of feed rate via belt speed
2. Belt weigher with controlled prefeeder  
Control of feed rate via belt load
3. Belt weigher with constant load  
Control of belt load via belt speed

### For Further Reading:

Fieldbus manual      BVH2220  
S5-RK512 (3964R)    FH 458  
Modbus (Comp)      FH 525

### Display:

5 x 7 - dot matrix with 6 mm character height

Upper display    left    : running message  
                         right   : set feed rate in kg/h or t/h

Lower display    left    : event messages  
                         right   : selectable  
                                         feed rate, belt speed, etc.

Display can be changed over to American units.

### LEDs:

2 green and 3 red LEDs

Green LEDs      : Ready

Red    LEDs      : Error or limit value messages

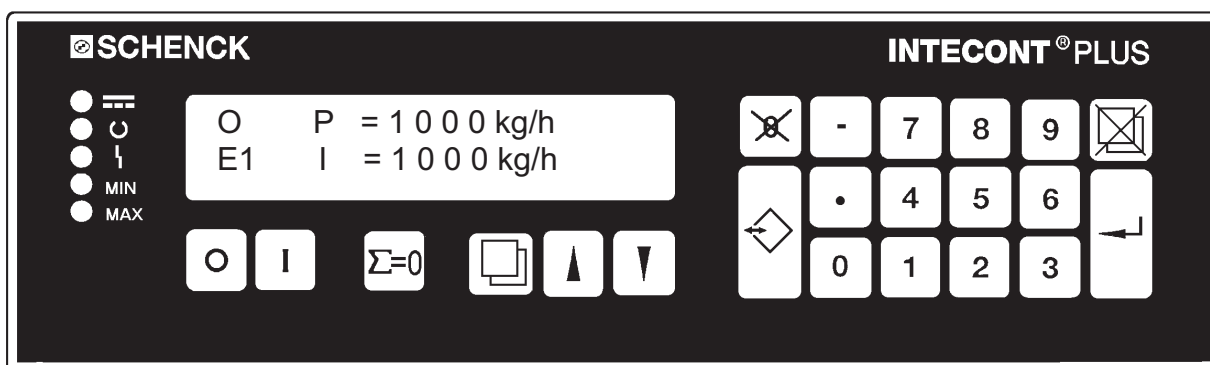
### Keyboard:

Flexible membranes with tactile touch

 Start/stop

 Preselect lower display  
Select functions

 Reset counter



|                                                                                                                                                                    |                              |                                           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|-------------------------------------------|
|                                                                                    | FUNC                         | Call function distributor and event texts |
|                                                                                    | DEL                          | Acknowledge event message, delete input   |
|                                                                                    | ESC                          | Abort function                            |
|                                                                                    | ENT                          | Start function, acknowledge input         |
|                                                                                    | DAT                          | Prepare input, e.g. of setpoint           |
|   | Enter parameters             |                                           |
|   | Enter sign and decimal point |                                           |

Some Definitions:

|                                |   |                                                                                         |
|--------------------------------|---|-----------------------------------------------------------------------------------------|
| I                              | = | Feed rate in kg/h or t/h<br>Material amount discharged from conveyor belt per unit time |
| P                              | = | Feed rate setpoint in kg/h or t/h                                                       |
| Z                              | = | Amount fed in kg or t<br>Amount fed = feed rate x feed time                             |
| V                              | = | Conveyor belt speed in m/s                                                              |
| Q                              | = | Belt load in kg/m<br>Weight of material on one belt meter                               |
| Gravimetric : Controlled mode  |   |                                                                                         |
| Volumetric : Uncontrolled mode |   |                                                                                         |
| Y                              | = | Controller output in mA                                                                 |

Measuring Principle:

Belt load Q and speed V are continuously measured and multiplied with one another. The result is feed rate I used to determine the amount fed out.

I = Q • V • 3600

I

Q

V

in kg/h

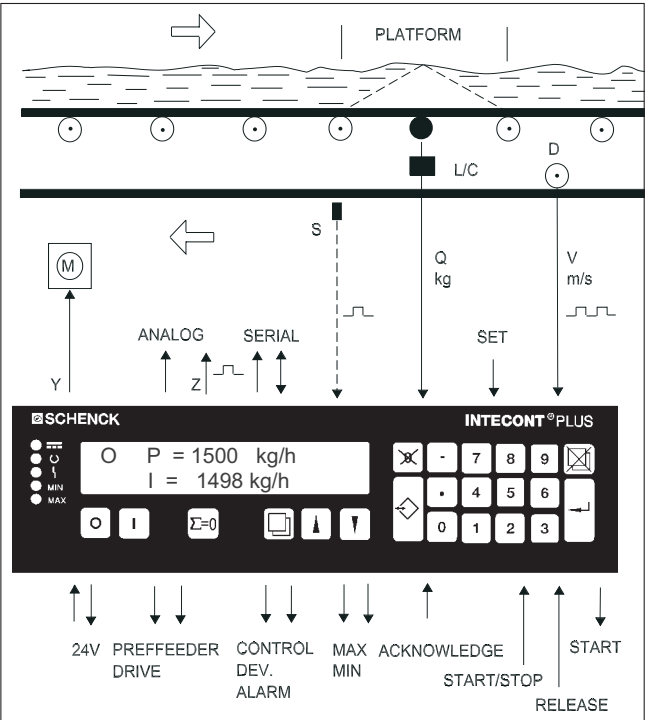
in kg/m

in m/s

Measuring Technique:

- Precision AC voltage amplifier with continuous correction of zero point and range.
- Highly resolving analog-to-digital converter. The integrated method is designed to efficiently suppress mains voltage interferences.
- 16-bit microcontroller
- Unlimited storage of counter readings, configuration and calibration data.

The schematic below shows all inputs and outputs.



|      |   |                                                                                        |
|------|---|----------------------------------------------------------------------------------------|
| L/C  | = | Load cell used to acquired belt load Q                                                 |
| D    | = | Speed transducer                                                                       |
| S    | = | Belt circuit sensor for automatic belt influence compensation BIC (not always present) |
| M    | = | Drive motor                                                                            |
| 24 V | = | Power supply                                                                           |
| MIN  | = | Relay output for lower limit values                                                    |
| MAX  | = | Relay output for upper limit value                                                     |

|            |   |                                                                                                                                                        |
|------------|---|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| ALARM      | = | Relay output for ALARM                                                                                                                                 |
| DEV.       | = | Relay output for DEVIATION fault                                                                                                                       |
| START/STOP | = | Start / stop scale                                                                                                                                     |
| ACK        | = | Acknowledge event message                                                                                                                              |
| SET        | = | Analog setpoint                                                                                                                                        |
| SERIAL     | = | Printer and host computer                                                                                                                              |
| ON         | = | Scale is active.<br>Outputs for control of belt and<br>prefeeder drives, connection of<br>totalizing counter and analog<br>display unit are available. |
| RELEASE    | = | Release                                                                                                                                                |

2. Operation

Power ON

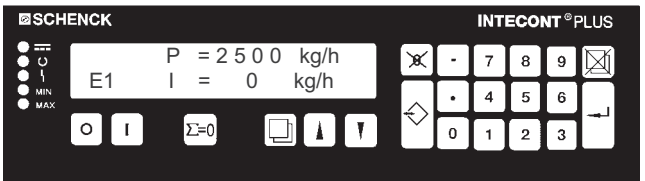
During power failure

all counter readings  
counter pulses still to be output  
preselected normal displays,  
Service values, e.g. ON-time

are stored for an unlimited time.  
The internal clock continues running for approx. 5 days.

After power-up, an automatic display and lamp test is performed, followed by the display version number, then by normal displays.

Message E1



reports power failure .  
Message can be configured as ALARM or WARNING.

ALARM:

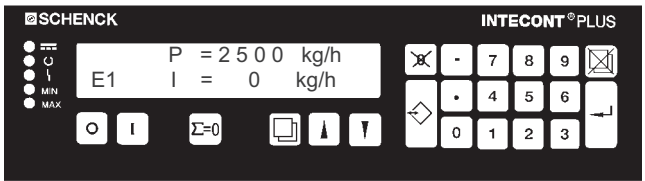
Red lamp  and symbol E1 flash.  
Acknowledge message before  
starting scale.

WARNING:

Signal lamp  remains dark,  
  
symbol E1 does not flash.  
No acknowledgement. Start scale  
manually or automatically.

 Acknowledge event message.

Signal Lamps



-  (green) POWER OK  
lit if all power supplies are OK.
-  (green) CPU OK  
lit if microprocessor is ready to operate.
-  (red) ALARM  
flashes if an Alarm is available.  
In addition, the event code is shown in lower display line.
- MIN (red) MIN LIMIT VALUE  
lit if feed rate limit value is exceeded down.
- MAX (red) MAX LIMIT VALUE  
lit if feed rate limit value is exceeded up.
- READY: The two green signal lamps light and no Alarm is available.

## Normal Display

### Upper Display

**Left** Rotating point as running message.  
Scale is started, amount fed out and feed rate are acquired.  
In STOP state (point stands still) only belt load and belt speed are measured.

Special features in addition to running message:

blank = Gravimetric mode

V = Volumetric mode  
In start-up and clearance modes, "V" flashes.

□ = Keyboard mode

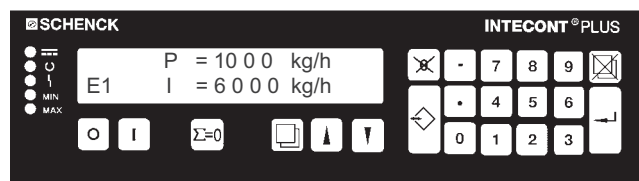
If Keyboard and Volumetric modes are active at the same time, "V" is framed by keyboard symbol.

**Right** Effective setpoint P 1)  
If "=" flashes, Simulation mode is active.

### Lower Display

**Left** Event reported by alphanumeric code, e.g. E1 for power failure.

**Right** Selectable display



Select display

|            |            |      |
|------------|------------|------|
| Counter Z1 | Z1 = 5000  | kg   |
| Counter Z2 | Z2 = 6000  | kg   |
| Counter Z3 | Z3 = 1000  | kg   |
| Feed rate  | I = 1000   | kg/h |
| Feed rate  | Ir = 50.00 | % 2) |
| Belt load  | Q = 10.000 | kg/m |
| Belt load  | Qr = 50.0  | % 2) |
| Belt speed | V = 0.0500 | m/s  |
| Deviation  | Xd = 0.10  | % 2) |

|                    |            |         |
|--------------------|------------|---------|
| External setpoint  | Pe = 1000  | kg/h 3) |
| Percental setpoint | Pr = 50.00 | % 4)    |
| Batch displays     | Zb, ZI, ZD | 5)      |

- 1) With belt weighers controlled for constant belt load, counter 1 is displayed in place of setpoint. All setpoint inputs are omitted.
- 2) Related to nominal value
- 3) If serial or analog setpoint is preselected
- 4) If percental evaluation of serial or analog setpoint is preselected
- 5) See "Batching Mode".

Display formats and units of Z1, Z2, Z3, P, Pe and I can be selected separately.

## Operation In Normal Mode



Start scale.

Condition : External release signal available; if not, message S2 is output. No Alarm available. START source set to OP or Keyboard mode selected.

Feature : Rotating point in left field of upper display.



Stop scale.

Feature : Point stands still after a selectable counter tracking time. Belt load and belt speed continue being measured.



Change setpoint.



Desired digit

Enter desired digits.  
P =----- kg/h  
or  
Pr =----- %



Delete digit.



Acknowledge input.



Abort input.  
Old setpoint remains stored.

Percental (relative) setpoint Pr:

Pr can be used to percentally evaluate an external setpoint Pe (e.g. serial).

$$\text{Effective Setpoint} = P_e \cdot \frac{\text{PR}}{100\%}$$

In Gravimetric mode, P is limited to nominal feed rate; in Volumetric mode, to the triple value of nominal feed rate.



Reset totalizing counters.



Enter number of desired counter (1 or 2).  
FMZ Nr. :

Counter 3 cannot be reset. In case of overflow, counting resumes from 0.



Acknowledge input.



Delete faulty digits.



Abort input.



Select lower display  
(see “Normal Displays”).



Acknowledge event message.



Call further functions, e.g. Display Event, Calibration.



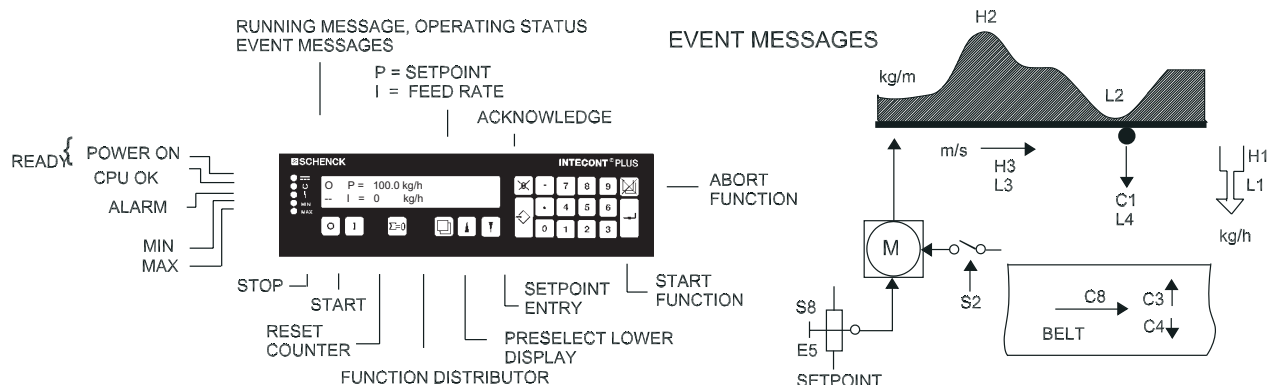
Exit function distributor.

**Note:**

The START, STOP and ENTER SETPOINT commands can also be sent from other operating sources, e.g. serially.

However, the relevant functions in Normal Mode are disabled.

Use “START Keyboard Mode” function to shift all functions to keyboard again.

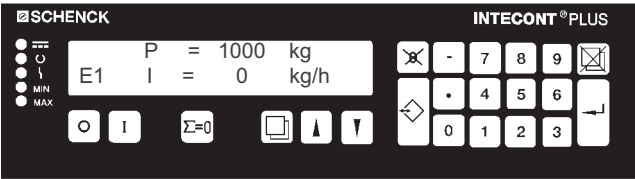


Event Messages

Definitions

All important scale functions are internally monitored. Errors are reported by event message.

For troubleshooting details, see chapter “Event Messages”.



Message E1 Power Failure

Message is output in the left field of the lower display in form of a group letter (e.g. E) and a number (e.g. 1). In addition, an explanatory text can be called up.

The events are organized into 4 classes. The assignment of event to class can be selected using relevant parameter.

- **ALARM** Display flashing and underscored.  
  
In addition, red signal lamp  flashes.  
Totalization cuts off, scale is not operable.  
  
Start:  
First remove cause of fault and acknowledge message.  
  
Acknowledgement:  
If cause has not been removed yet, flashing stops, underscores are deleted, but message is still available.  
If fault is remedied after acknowledgement, message automatically effaces.

- **WARNING 1** Display permanent and underscored.  
Totalization continues.  
  
Acknowledgement:  
If fault has not been remedied yet, underscore is faded out.
- **WARNING 2** Display permanent, not underscored.  
  
Acknowledgement:  
Not required.
- **IGNORE** No monitoring of events, save MIN/MAX messages and deviation via contact outputs and signal lamps.

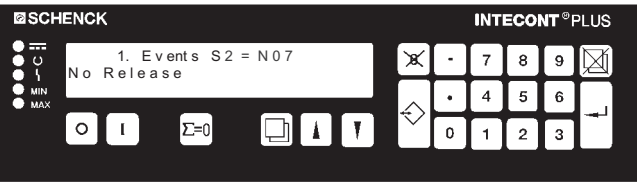
If several events occur at the same time, the most significant event is displayed first.  
Priority: Alarm, Warning 1, Warning 2, sequence of event list.

Operation

-  Acknowledge all available events of classes Alarm and Warning 1.  
If several events are available, acknowledge same one after the other.

Display of event texts:

-  Call function distributor.
-  Scroll “Display Events” into lower display field (normally not required).
-  Select function.



Display shows:

Name                    e.g. No Release  
Code                    e.g. S2  
and hint                e.g. Parameter N07

of respective event.

The hint informs the service staff of the relevant parameter. The consecutive number relates to the priority sequence.



View several events.



Return to normal displays.

List of Events

The list is organized by display priority.

|                     |    |                       |
|---------------------|----|-----------------------|
| System Message S    | S1 | Memory Error *        |
|                     | S2 | No Release            |
|                     | S3 | Maint. STR-meter      |
|                     | S4 | Maintenance Int. Elec |
|                     | S7 | Simulation Active     |
|                     | S8 | Setpoint Limited      |
|                     | S9 | Data Link Host        |
| Material Flow B     | B1 | Out of Tolerance      |
|                     | B2 | Control Deviation     |
|                     | B3 | Controller Limited    |
|                     | B9 | Linearization         |
| Electrical System E | E1 | Power Failure         |
|                     | E2 | Namur Err Tacho *     |
|                     | E3 | Namur Err Belt *      |
|                     | E5 | Stand-By              |
| Calibration C       | C1 | L/C Input *           |
|                     | C2 | Tacho Input *         |
|                     | C3 | Belt Skew             |
|                     | C4 | Belt Drift            |
|                     | C8 | Slip Error            |

|           |    |                 |
|-----------|----|-----------------|
| Maximum H | H1 | I-MAX           |
|           | H2 | Load > MAX      |
|           | H3 | v > MAX         |
|           | H4 | L/C Input > MAX |
| Minimum L | L1 | I < MIN         |
|           | L2 | Load < MIN      |
|           | L3 | v < MIN         |
|           | L4 | L/C Input < MIN |

I        =        Feed rate  
V        =        Belt speed  
L/C     =        Load cell  
Tacho =        Tachometer input for belt speed  
\* Fault in device or cabling.

Contact Outputs

|                   |                                                                                                                                             |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| General Alarm     | available until acknowledgement.                                                                                                            |
| Min/Max outputs   | independent of event class, active also with IGNORE. Limit values are reported as long as event is available provided that scale is active. |
| Control deviation | see Min/Max outputs                                                                                                                         |
| Signal lamps      | See Min/Max outputs                                                                                                                         |

Function Distributor

The function distributor lets you call further functions and/or dialogs.



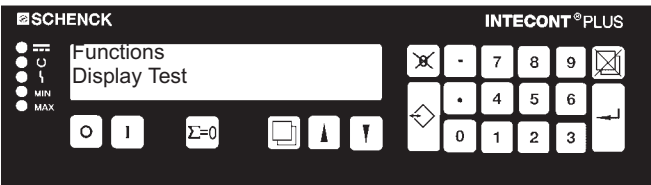
Call function distributor.  
If no event is available, display shows the function selected last. If an event is available, the "Display Events" function is displayed.



Start function.



Return to normal displays and/or abort function.



Move through function distributor.

- Display Events E
- Display Test
- START/STOP Feeder
- Volumetric/Gravimetric
- Volum. Synchron.
- START/STOP Keyb. Mode
- >0 : Zero Set
- Select Batch B
- Activate/Deact. EasyServe
- Programming
- Calibrating Function S
- Print FMZ

- Read Parameters
- Enter Parameters S
- Load Default Param. S
- Print Parameters
- Print Status Report

- START/STOP Simulation
- Tare
- Weight Check
- Imp/Belt Circuit
- Set Time

"Programming" and "Calibrating Functions" lead to further menus you can move through using cursor keys.

B = If batching mode is selected, additional functions can be accessed in the main distributor.

- Abort Batch
- Deselect Batch
- Print Batch

S = Calibrating functions accessible after password input.  
Prompt is output after call of function.

E = Only if an event is available

With opposed functions, e.g. START/STOP, only the one currently available can be reached.

Moving in function distributor does not limit any scale-specific functions.

Display Test and Version Number

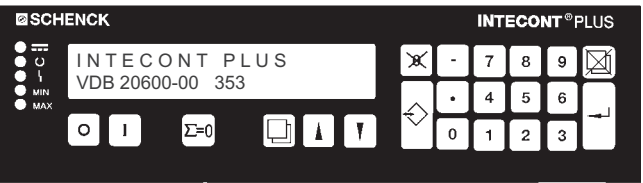
This function lets you check displays and signal lamps.



- Call Display Test function.
- 1. All display elements rhythmically flash.
- 2. Version number is displayed for approx. 10 s.



Test terminates automatically.



Example:

VDB : Weighfeeder  
20600-00 : Version number

Volumetric, Volumetric Synchronous,  
Gravimetric Modes

Gravimetric Mode:

Feed rate actual value is controlled for specified setpoint.  
Maximum possible setpoint corresponds to nominal feed rate. If higher values are entered, Message S8 is output.

Volumetric Mode:

Drive motor for conveyor belt, or material prefeeder, is controlled in proportion to setpoint. Belt load is of no influence.  
At rated conditions, feed rate corresponds to entered setpoint.  
Setpoint is limited to the triple nominal feed rate.  
Setting programs “Zero Setting”, “Taring” and “Belt Circuit” are available only in Volumetric mode.

Volumetric Synchronous Mode:

In contrast to Volumetric mode where the nominal belt load is used for computation of set belt speed, system uses the belt load measured before changeover. This ensures that changeover takes place smoothly.



Call function distributor.  
Select “Volumetric/Volumetric Synchronous/Gravimetric” mode.

Display features:

|                                       |                                 |
|---------------------------------------|---------------------------------|
| Volumetric/<br>Volumetric Synchronous | “V” in left upper display field |
| Gravimetric                           | No feature                      |

Keyboard Mode

This function lets you shift the selected control sources for

START/STOP  
Enter Setpoint

to the INTECONT keyboard.  
After deselection of Keyboard mode, the former sources are active again.  
Upon changeover from external source (e.g. serial) to keyboard, START/STOP status and setpoint are maintained.  
In reverse direction, the external signals are active.



Call function distributor.  
Select “START Keyboard Mode” or “STOP Keyboard Mode” function.

Display feature:

You will know Keyboard mode by the square in the left upper display field.  
External setpoint Pe, if any, can still be viewed.

Activate/Deactivate EasyServe

The “Activate / Deactivate EasyServe” functions let you control your INTECONT also using the EasyServe PC service tool.



Call function distributor.  
Select “Activate EasyServe” function.

To set communication parameters, use Block R.

**Note:** Upon power failure, the current status is stored, i.e. The “Activate EasyServe” function needs not be re-called.

Feeder

The “START/STOP Feeder” function lets you start and stop material prefeeder.  
In started state, the prefeeder is controlled through the INTECONT.  
In Stop state, material flow is interrupted and scale can be tared, etc.



Call function distributor.  
Select “Start Feeder” or “Stop Feeder” function.



Start control sequence.

Printing

All print functions require the option card to be present.

Print FMZ (Quantity Report)



Call function distributor.  
Select “Print FMZ” function.



Start function.



Abort printing.  
If printer has a big data buffer, abort is not effected immediately.

|  |              |                |  |
|--|--------------|----------------|--|
|  | Batch Report |                |  |
|  | 03.03.03     | 11:45:10       |  |
|  | Counter 1    | 12000 kg       |  |
|  | Counter 2    | 90000 kg       |  |
|  | Events       |                |  |
|  | C8           | Slip Error M14 |  |

03.03.02...: Date and time  
FMZ 1: Totalizing counter 1  
FMZ 2: Totalizing counter 2  
Events: Events available during printout

Print Parameters

Lets you print complete parameter list with actual values.  
Parameters deviating from default are identified by \*.  
As appendix, a list of all possible event messages and current hardware equipment is printed.



Call function distributor.  
Select “Print Parameters” function.



Start function.



Abort printing at any time.

|  |                      |                  |  |
|--|----------------------|------------------|--|
|  | PRINT PARAMETERS     |                  |  |
|  | INTECONT PLUS        | VDB 20600-00 353 |  |
|  | 03.03.03             | 7:39:25          |  |
|  | BLOCK A              | Dialog Behaviour |  |
|  | 01 Language          | ENGLISH          |  |
|  | 02 Units             | SI               |  |
|  | BLOCK B              | Rated Data       |  |
|  | 01 Feed Rate Units   | kg/h             |  |
|  | 02 Nominal Feed Rate | * 1,000 t/h      |  |
|  | etc.                 |                  |  |

|   |          |          |    |    |   |            |   |
|---|----------|----------|----|----|---|------------|---|
| ○ | 30.04.03 | 11:45:21 | L2 | W1 | → | Load < MIN | ○ |
| ○ | 30.04.03 | 11:59:01 | L2 | W1 | ← | Load < MIN | ○ |
| ○ |          |          |    |    |   |            | ○ |
| ○ |          |          |    |    |   |            | ○ |
| ○ |          |          |    |    |   |            | ○ |
| ○ |          |          |    |    |   |            | ○ |

|    |   |                                                      |
|----|---|------------------------------------------------------|
| L2 | : | Event code                                           |
| W1 | : | Message class, e.g. WARNING 1                        |
| >  | : | Event occurred                                       |
| <— | : | Cause remedied and event acknowledged, if necessary. |

The event is shown on INTECONT display as long as available.

## Status Report

The status report can be printed at any time and indicates the most significant status values and all events occurred after last printout. If no status report has been printed before, the events after last power-on of INTECONT are counted.



Call function distributor. Select “Status Report” function.



Start function.



Abort printout at any time.

```

STATUS REPORT INTECONT® PLUS

25.11.02   13:37:12
Software                               : VDB 20600-0 913
Hardware version                       : 1
Option card version                   : 1
Station address                       : 1
-----

Mode OP                               : NO
Mode Simulation                       : NO
Scale ON                             : YES
Volumetrisch                         : NO
Alarm                                 : NO
Voltage ON-time                      : 6 h
Scale ON-time                        : 2 h
Counter 1                           : 23 t
Counter 2                           : 23 t
Counter 3                           : 120 t
Belt Slip                           : 0.00 %
Belt Skew                           : 0.20 cm
-----

Events after last status report
    1 * E01 Power Failure
    3 * S02 No Release
  
```

Total voltage ON-time

Total run time of conveyor belt and INTECONT

Number and type of events occurred after last printing of report. If report is printed for the first time, counting starts from power-up.

Totalizing counters 1-3

Variation in belt length in % of conveyor belt total length (Parameter D06 ???).

## Belt off-centre run in cm.

Zero Setting

The zero setting program is designed to acquire the scale's zero point error over one or multiple integer belt circuits.

The acquired value is used to correct the current measuring result in normal mode.

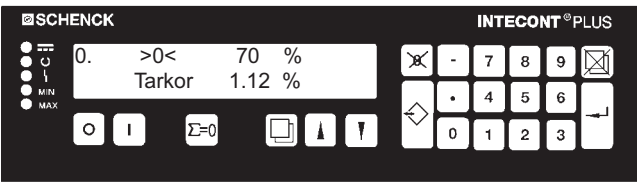
During zero setting, all totalizing counters stop operating.

- Conditions:
- 1. Select Volumetric Mode.
  - 2. Make sure conveyor belt is totally cleared.  
If a prefeeder is present, use "Stop Feeder" function.
  - 3. Make sure conveyor belt is running.  
A respective hint is output automatically.

To call:

-  Call "Zero Setting" program.
-  Start function.
-  If scale is not started, program aborts and prompts for START.

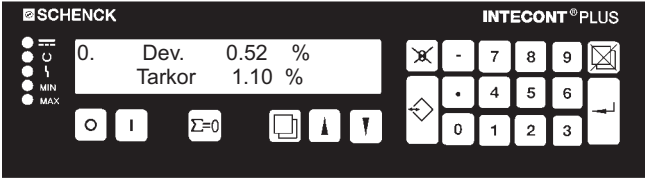
Program running:



- Upper display :      Residual run time in % of total run time
- Lower display :      Zero point error related to nominal belt load

-  Abort program at any time.

Program complete:



- Upper display :      Deviation of zero point from previous zero setting operation in % of nominal belt load
- dev = + : error increased  
dev = - : error decreased
- Lower display :      Deviation of zero point from basic tare in % of nominal belt load.  
Taring program corrects basic tare during calibration.

The result can be accepted or rejected. If rejected, it still serves for information.

-  Acquire result.  
Scale zero point is corrected.
-  Abort program (reject result).  
Restart prefeeder.

Event Messages:

- 1. VALUE > MAX  
output if zero setting result exceeds specified limit.  
Result is rejected.  
Clean scale; if need be, retare.

## Read Parameters

Parameters are defined as variable characteristics or data used to match the INTECONT to the particular application.

Parameter input is protected by password.

The "Read Parameters" function lets you view all parameters without jeopardizing the operating reliability.

The parameters are organized into function blocks A, B... and consecutively numbered within a block.



Scroll "Programming" function into display field and acknowledge.



Select "Read Parameters" sub-function.

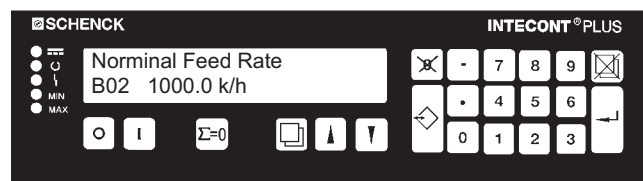
Display shows title of first parameter block.



Scroll through parameter blocks A, B...



Scroll through parameters within a block.



Return to blocks.



Return to normal displays.

## Batching Mode

### Additional Displays



Upper display : If batching mode is selected, totals sign  $\Sigma$

is displayed left of running message.

Lower display:



Use cursor keys to scroll 3 additional values into display field.

1. Zb = Batch setpoint
2. ZI = Batch actual value  
Amount already discharged
3. Zd = Batch residual value  
Amount not yet discharged

$$Zd = Zb - ZI$$

Before start of a batch, ZI and Zd indicate the amounts fed out during previous batch. Batching mode deselected ("Deselect Batch" function), displays are disabled but can be used again upon next selection.

Display format corresponds to the format of Counter 1.

## Select Batching Mode

Selection of batching mode enables batching, however, no batch is started yet.

Condition: Scale OFF



Call function distributor.



Scroll "Select Batch" function into display field.



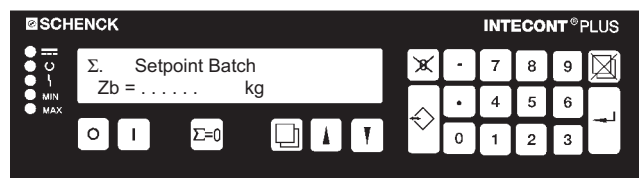
Select function.

System prompts for input of batch setpoint. If batch is selected from external PC, setpoint input is omitted.

## Enter Batch Setpoint

Prompt for setpoint input is output automatically upon selection or can be started in manual when batch is complete (scale OFF).

If batching mode is deselected or a batch is running, no input is possible.



If feed rate setpoint source is set to OP (B07 = OP) or Keyboard mode, program first prompts for input of feed rate setpoint  
P = - - - - - kg/h



Enter feed rate setpoint.



Skip input.  
Old feed rate setpoint is maintained.  
Program automatically prompts for input of batch setpoint.  
Zb = - - - - - kg



Enter batch setpoint.  
Max. 9 digits including decimal point.  
The unit is that of counter 1.



Acknowledge input.



Abort input; old setpoint is maintained.



Delete faulty digits.

## Start and Abort



Start batch, process starts.  
Display ZI : moves from 0 towards setpoint  
Display Zd : moves from 0 towards setpoint



Abort batch.

Batching continues upon next Start command.

## Deselect Batch

Lets you deselect batching mode, i.e. mode is removed from available functions.

Totals sign  $\Sigma$  disappears from upper display, batching values ZI, Zd and Zb are not longer available but remain stored.

Conditions: Batch complete, or aborted using "Abort Batch" function, scale OFF.



Call function distributor.  
Select "Deselect Batch" function.



Acknowledge.

The "Select Batch" function lets you reactivate batching mode.  
The old ZI, Zd and Zb values can be viewed again.

Abort Batch

Lets you abort a running batch, i.e. Batching immediately stops.  
Batching mode remains selected.

New batch starts upon next start command, old batch is not resumed.

- 

Call function distributor and select "Abort Batch" function.
- 

Acknowledge.

You can also abort batch using contact Q2.

Print Batch

This function requires the option card to be present.  
Batch must be completed or aborted.  
Print is possible only in batching mode.

Printed can be started:

1. Manuall via function distributor
2. Automatically, after each complete batch (Parameter I 06).

- 

Call function distributor.  
Select "Print Batch" function.
- 

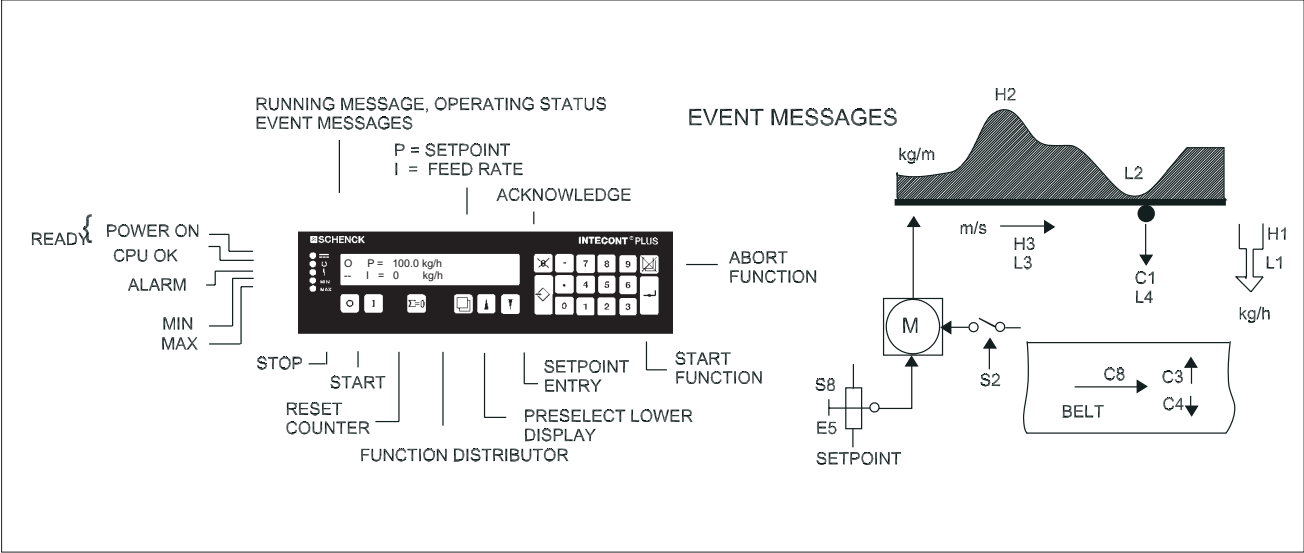
Start function.
- 

Abort printout.

|  |                |                |  |
|--|----------------|----------------|--|
|  | Batch Report   |                |  |
|  | 03.03.03       | 11:45:10       |  |
|  | Batch setpoint | 1000 kg        |  |
|  | Batch setpoint | 998 kg         |  |
|  | Counter 1      | 12000 kg       |  |
|  | Counter 2      | 90000 kg       |  |
|  | Events         |                |  |
|  | C8             | Slip Error M14 |  |
|  |                |                |  |

- 03.03.03 : Date and time
- Batch setpoint : Amount to be batched
- Batch act. value : Amount batched (actual value)
- Counter 1 : Totalizing counter 1
- Counter 2 : Totalizing counter 2
- Events : Events available during printout

3. Control Overview

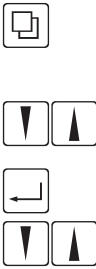


Status displays: upper display field

- ☐ Point rotates
  - ☐ Volumetric mode
  - ☐ Keyboard mode
- Scale is started  
Scale operates in uncontrolled mode  
During start-up and clearance, V flashes.

Modes: Keyboard and Volumetric

- Call function distributor.
- Select "Volumetric/Gravimetric" or "START/STOP Keyboard Mode".
- Activate with ENTER.



Selectable displays: upper display field

- |              |         |                          |                     |
|--------------|---------|--------------------------|---------------------|
| 1. Counter   | Z1...Z3 | 6. Belt speed            | V                   |
| 2. Feed rate | I       | 7. Deviation             | Xd                  |
| 3. Feed rate | Ir in % | 8. External setpoint     | Pe (if preselected) |
| 4. Belt load | Q       | 9. Prozentualer Sollwert | Pr (if preselected) |
| 5. Belt load | Qr in % |                          |                     |

Event messages: lower display field

The most significant event is reported by a letter/number combination, e.g. S 2 (No Release).

To call clear text and further messages, see **FUNCTION** key 

To enter setpoint, see **DATA** key 

To reset counter, see **RESET** key 

To acknowledge events, see **DELETE** key 

I

0

START/STOP

Message S2  
unless scale has been released via digital input.

▼

SCROLL

Select lower display.  
Select function.



Σ=0

RESET

Reset counter.

1. Enter number of desired counter (1 or 2).

2. Acknowledge input with ENTER key, or

3. Abort with ESCAPE key, or

4. Delete faulty input with DELETE key.

FUNCTION

Call function distributor and event messages.

1. Select desired function using SCROLL keys.  
If an event is available, "Display Events" function is already selected.

2. Activate function with ENTER key.

3. Pass through available event messages with SCROLL keys.

4. Return to normal display with ESCAPE key.

DELETE

Acknowledge event message, delete input.

ESCAPE

Abort function, return to normal displays.

ENTER

Acknowledge input, e.g. Setpoint  
Start function, e.g. Display Events

(see DATA)  
(see FUNCTION)

DATA

Prepare input, e.g. setpoint

1. Enter desired setpoint.

2. Acknowledge input with ENTER key, or

3. Delete faulty digits with DELETE key, or

4. Abort with ESCAPE key.

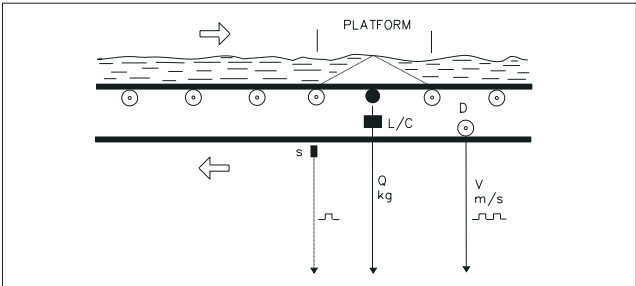
4. Operating Principle

Measuring Principle

A weighfeeder is designed to continuously weigh the material amount transported on a conveyor belt.

Belt Load

The material is guided to a weighing platform arranged under the belt and limited by two carrying idlers. Via one or multiple weighed idlers, the platform load exerts a force on load cell L/C. The deflection for measurement is approx. 0.2 mm. The weighed idlers are connected with the frame construction, e.g. with the use of a leaf spring parallel system.



Proportional to platform load, the load cell output voltage is amplified and transferred to the INTECONT microprocessor with the use of an analog-to-digital converter.

The white triangle indicates the load distribution on a single-idler platform scale. Only half of the material's weight force is applied to the weighed idler.

The translation of load ratio into platform length has become accepted usage in weighing technology.

$$L_{eff} = \frac{L_g}{2}$$

$L_{eff}$  = effective platform length  
 $L_g$  = total platform length

For weighing platforms equipped with multiple weighed idlers, factor 1/2 is different.

Belt load in kg/m:

$$Q = \frac{Q_B}{L_{eff}}$$

$Q_B$  = total load on weighing platform

Belt Speed

A further measure for feed rate is belt speed  $v$ , acquired with the use of speed transducer  $D$  and translated into a corresponding pulse frequency. If belt load is constant and more approximate accuracy will suffice, speed measurement can be omitted.

Feed Rate

INTECONT normalizes physical units kg/m and m/s. Multiplication of the two values results in feed rate  $I$ .

$$I = Q \cdot v = \frac{Q_B}{L_{eff}} \cdot v$$

|           |    |      |
|-----------|----|------|
| $I$       | in | kg/s |
| $v$       | in | m/s  |
| $Q$       | in | kg/m |
| $Q_B$     | in | kg   |
| $L_{eff}$ | in | m    |

$$I[\text{kg/h}] = Q \cdot v \cdot 3600$$

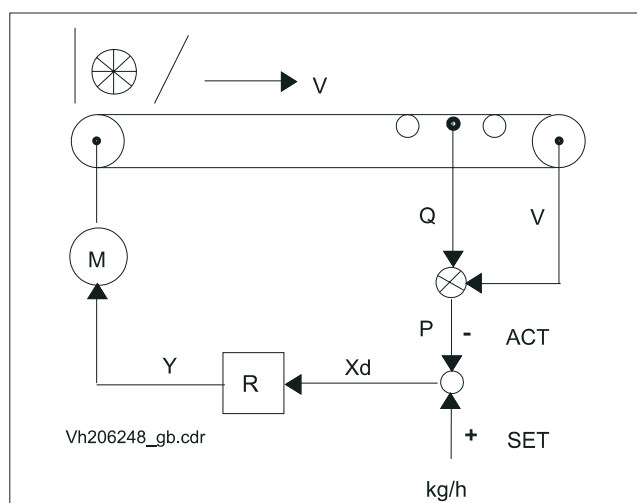
$I$  in kg/h  
else, see above

## Control:

### 1. Weighfeeder

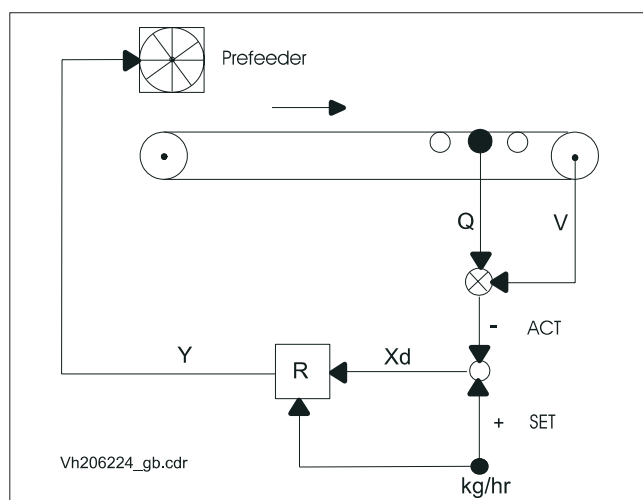
A conveyor belt extracts material from a surge bin.

Actual value is compared to setpoint and the difference is routed to controller R. The latter keeps varying belt speed until actual value corresponds to setpoint.



### 2. Belt weigher with controlled material prefeeder

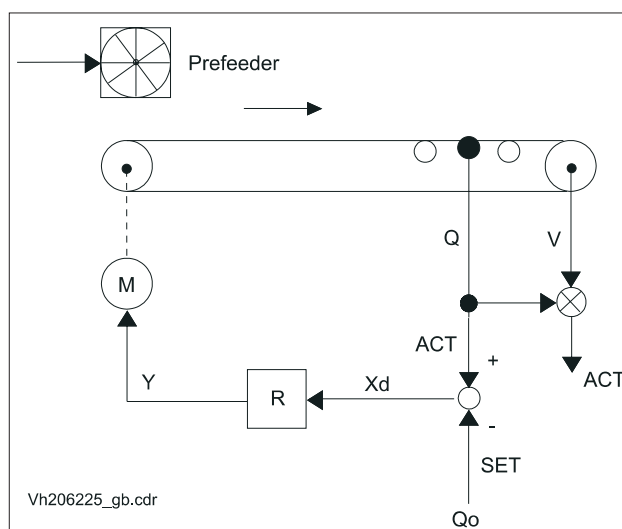
Feed rate is controlled for setpoint by variation of belt load. Belt speed is normally constant.



Position-controlled prefederers require an external three-step controller to be present.

### 3. Belt weigher with constant load

An uncontrolled prefeeder feeds the material to a conveyor belt. Controller adjusts speed so that belt load always corresponds to nominal value  $Q_0$ .



The non-varying working point of the weighing system ensures highest measuring accuracy. Feed rate at point of discharge corresponds to feed rate at prefeeder.

## Power stage:

The drive requires an external power stage with armature voltage or speed control, etc., to be present.

## Operating modes:

1. Gravimetric  
Controlled mode as described above
2. Volumetric  
Uncontrolled mode, i.e. belt speed or prefeeder are controlled in proportion to setpoint. Feed rate value measured or belt load actual value are not considered.

### 3. Volumetric Synchronous

In contrast to Volumetric mode, the belt load measured before changeover is used for computation of current belt speed setpoint in place of nominal belt load. This ensures smooth changeover.

The two modes can be selected in the dialog mode. In addition, special start-up and clearance functions can be activated.

Control sources:

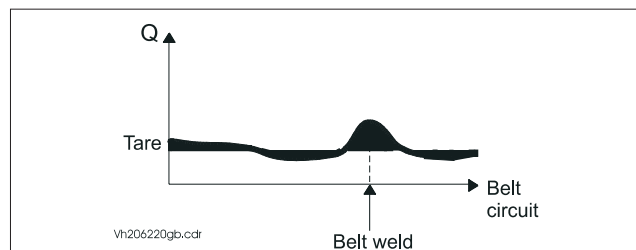
Some operating functions can be started from several sources.

- |    |                            |     |                               |
|----|----------------------------|-----|-------------------------------|
| 1. | Setpoint                   | via | Keyboard<br>Analog<br>Serial  |
| 2. | START/STOP                 | via | Keyboard<br>Serial<br>Contact |
| 3. | Reset counter (and/or)     |     | Keyboard<br>Serial            |
| 4. | Acknowledge event (and/or) |     | Keyboard<br>Contact<br>Serial |

The "Keyboard Mode" control function lets you shift items 1 and 2 from selected sources to keyboard, and vice versa.

### Automatic Belt Influence Compensation (BIC)

Even the best of conveyor belts is not perfectly even; particularly in welding points, the belt will be heavier and stiffer. The figure above shows the situation over one belt circuit in exaggerated fashion.

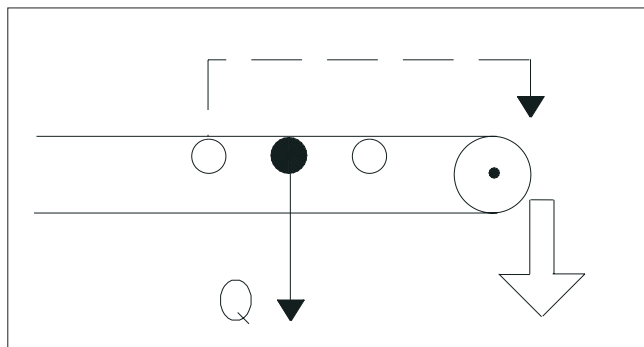


Therefore a high accuracy can normally be ensured only over entire belt circuits, i.e. using the mean value. INTECONT is able to acquire and correct this influence with the use of a belt circuit sensor and a metal belt mark.

This considerably increases the scale's short-term accuracy.

To enable variations to be considered during operation, the belt influence is continuously measured. The compensation adaptively follows up, also when material is fed.

### Control for Point of Discharge (CPD)



For constructional reasons, the weighing platform is located at the point of material discharge direct. Therefore, if belt load varies, the measured feed rate does not exactly correspond to the feed rate at the point of discharge.

A special speed-dependent delay element shifts the measurement to the discharge point.

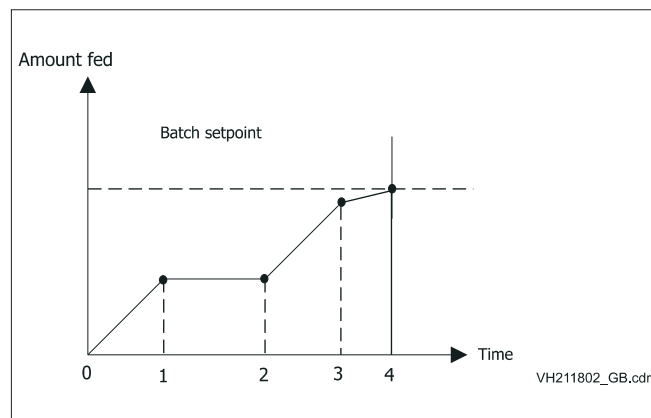
To do so, no special load transducers are required; however, the additional use of the automatic belt influence compensation is reasonable.

Measurement at the point of discharge also controls the material flow at the point of discharge.

## Batching

In batching mode, a preset material amount is fed out. When batch setpoint is reached, conveyor belt and prefeeder stop and batching is complete.

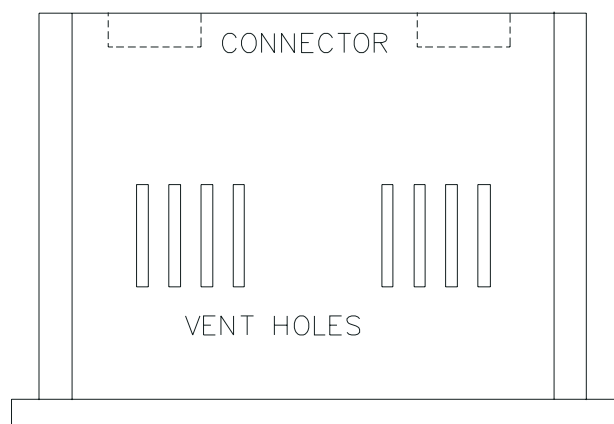
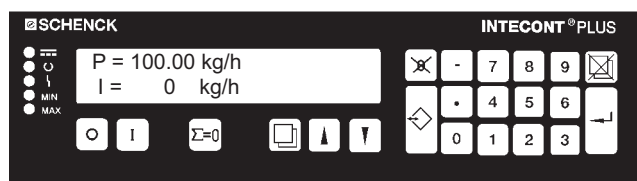
- 0 Start batch
- 1 Abort batch at any time and
- 2 resume.
- 3 Continuously decrease setpoint until preset batch amount is reached.



## 5. Technical Data

### ■ Panel-mounting unit VEG 20600/20610

|                   |                                                                            |
|-------------------|----------------------------------------------------------------------------|
| Front panel       | : 96 x 288 mm                                                              |
| Panel cut-outs    | : 88 x 282, wall thickness max. 10 mm                                      |
| Mounting depth    | : min. 225 mm (including 50 mm for connectors)                             |
| Protected to      | : IP 20, front to IP 65                                                    |
| Material          | : plastic grey to RAL 7022                                                 |
| Display           | : fluorescent, 2 lines of 20 characters, 6 mm high                         |
| Keyboard          | : flexible membranes with tactile touch                                    |
| Signal lamps      | : LEDs, 2 x green, 3 x red                                                 |
| Power supply      | : 24 V(DC) +50 % -25%, max. 1 A, no fuse, pole-reversal protective circuit |
| Ext. power supply | : 90...264V/ 50HZ...60HZ                                                   |



### ■ Surface-mounting unit: with panel mounting unit VEG 20...

### ■ Wall-mounting housing

|                    |                                         |
|--------------------|-----------------------------------------|
| Measuring systems: | width x height x depth<br>410x330x236mm |
| Feeding systems:   | width x height x depth<br>500x600x300mm |
| Protected to       | : IP 65                                 |
| Material           | : steel sheet, 1.5 mm, grey             |
| Voltage supply     | : 90...264V/50Hz...60Hz                 |

Extension : Feed unit controller

### ■ Environment

|                          |                                      |
|--------------------------|--------------------------------------|
| Temperature              | : -25 °C ... +45 °C, tested to -40°C |
| Humidity                 | : Class F:(DIN 40 040) *             |
| Interference immunity    | : to IEC 801                         |
| Interference suppression | : to VDE 871, EN 55 011              |
| Housing safety           | : to IEC 348                         |
| EC recommendations       | : CE label                           |

\* Tropical version (class R) upon request

### ■ Storage temperature

-40 °C ... +80 °C

### ■ Connectors

Plug/screw connectors (Phoenix Combicon), non-confusable

### ■ Control outputs

|             |                                                       |
|-------------|-------------------------------------------------------|
| Number      | : 8 N/O relay contacts                                |
| Base board  | : 3                                                   |
| Option card | : 5                                                   |
| Load        | : max. 230 V(AC)<br>8 A resistive load, 1 A inductive |

If error message "CPU Not OK" is output, all contacts open.

## ■ Control

Number: 5  
Base card : 3  
Option card : 2  
Type : 24 V(DC) +/-30 %, 5 mA, galvanically isolated

## ■ Control inputs (NAMUR)

Number: 2  
Type : NAMUR (DIN 19234), short-circuit and cable breakage monitoring

## ■ Analog output

Number: 2  
Base card: 1  
Option card : 1  
Current : raise ...20 mA impressed  
Raise : 0... <20 mA  
Load : max. 500 ohms  
Reference : 0 V by user, galvanically isolated  
Resolution : 12 bits +/- 1 d

If error message "CPU Not OK" is output, output signal turn 0 mA.

## ■ Analog input

Number : 1  
Current : raise ...20 mA impressed  
Raise : 0... 20 mA  
Load : 500 ohms  
Reference : 0 V internal, galvanically non isolated  
Resolution : 10 bits +/- 1/2 d  
Accuracy: <0.4% over entire temperature range

## ■ External totalizing pulse counter

Frequency : max. 10 Hz  
Pulse length : selectable 50...1000 ms  
Output : 24 V, max. 100 mA  
optionally active low or active high, galvanically isolated, short-circuit-proof, limited to apr. 350 mA, negative amounts are stored internally and must be compensated by positive amounts before new pulse is output.  
This ensures integral accuracy also with minimal belt load.

## ■ Serial printer interface

Number: 1 (option card)  
Type : RS 232 (V24), max. 38400 baud  
Data format : 8 data bits, 1 stop bit, no parity  
Isolation : galvanically non-isolated  
Connectors : Phoenix MSTB, contact spacing 3.5 mm, max. 1.5 mm<sup>2</sup>

## ■ Load cell connection

Voltage supply : +/- 6 V AC (+/-3 %) (apr. 66 Hz), short-circuit-proof  
Measuring cycle : 100ms  
Resolution : 24 Bit  
Load : R (load cell) = min. 80 ohms  
Cable length : max. 30 % voltage drop, max. 500 m  
Measuring range : -20mV...+50mV with VLW: -1.6mV...min. 32mV

## ■ Serial interface for service PC

|               |                                                                                           |
|---------------|-------------------------------------------------------------------------------------------|
| Number :      | 1                                                                                         |
| Type :        | RS232; RS422 or RS 485<br>max. 19200 baud                                                 |
| Data format : | 8 data bits, 1 stop bit, odd<br>parity changeable to even or<br>no                        |
| Isolation :   | galvanically non-isolated                                                                 |
| Connectors :  | Phoenix MSTB, contact<br>spacing 3.5 mm, max. 1.5<br>mm <sup>2</sup> or 9-pole D-Sub HD20 |

## 6. Characteristics

### ■ Power Failure

Scale data, calibration values, counter readings and pulses pending remain stored after power failure for an unlimited period of time. The internal clock continues running for approx. 5 days.

### ■ Dialog Language

for error messages, operation, and service programs.

Eligible : DEUTSCH  
ENGLISH  
FRANCAIS  
ESPANOL  
ITALIANO

### ■ Units (selectable)

SI units : kg, kg/h  
t, t/h  
m, cm  
NON-SI units : lb, lb/h  
t, t/h  
f, inch

(see Parameter Shortlist)

### ■ Start/Stop Totalizing Counter

Select using Parameter B06.

### ■ Enter Setpoint

Select using Parameter B07.

### ■ Release, Emergency Stop

Functions are realized using a digital input.

### ■ Displays

|               |   |                             |    |
|---------------|---|-----------------------------|----|
| Upper display | : | Setpoint                    | P  |
| Lower display | : | Totalizing counters Z1...Z3 |    |
|               |   | Feed rate absolute          | I  |
|               |   | Feed rate relative          | Ir |
|               |   | Belt speed                  | v  |
|               |   | Belt load absolute          | Q  |
|               |   | Belt load relative          | Qr |
|               |   | Batch setpoint              | Zb |
|               |   | Batch actual value          | ZI |
|               |   | Batch residual amount       | Zd |
|               |   | External setpoint           | Pe |
|               |   | Relative setpoint           | Pr |
|               |   | Deviation                   | Xd |

The lower display can be selected using cursor keys.

### ■ Signal Lamps (top down)

|           |       |                     |
|-----------|-------|---------------------|
| VOLTAGE   | OK    | (green)             |
| PROCESSOR | OK    | (green)             |
| ALARM     | (red) |                     |
| MIN       | (red) | Feed rate MIN limit |
| MAX       | (red) | Feed rate MAX limit |

### ■ Event Messages

Events and faults are displayed in form of an alphanumeric code.

Call clear text of events via keyboard.

Events are organized into Alarms, and Warnings with and without acknowledgement. Alarms stop scale, warnings don't.

All alarms are additionally reported via relay outputs.

### ■ Operating Philosophy

Operator prompting through multi-level menus. Important configuration and calibration functions are protected by password.

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## ■ Control Principles

1. Weighfeeder
2. Belt weigher with controlled prefeeder
3. Belt weigher with constant belt load

In addition, belt load and/or belt speed measurement can be deactivated. This enables other types of application.

## ■ Start-up and Clearance Modes

Uncontrolled start-up and smooth changeover to an uncontrolled clearance mode can also be selected.

## ■ Simulation Mode

Special function used to operate your system without mechanical equipment, e.g. for plant testing.

## ■ Zero Setting

Zero setting is performed with empty conveyor belt using an automatic program.

The maximum admissible zero setting elevation is monitored.

## ■ Taring

Designed to zero the basic tare load, the taring program on principle has the same function as the zero setting program. However, tare elevation is not monitored, and the call is protected by password.

## ■ Automatic Belt Influence Compensation (BIC)

Using an additional belt circuit sensor and a metal belt mark, the dynamic zero point influence of the conveyor belt can be compensated during operation (with and without material).

This considerably increases the weighfeeder's short-term accuracy.

## ■ Control for Point of Discharge (CPD)

A special speed-dependent delay elements shifts the measuring point from the weighing platform to the point of material discharge. The material amount discharged is acquired direct.

For display and belt run monitoring, see BIC.

## ■ Zero Drop Out

This function is used to suppress totalization upon measurements around zero, for counter to stop while belt is unloaded.

Deselect function if not desired.

## ■ MIN, MAX Messages and Outputs

Freely selectable limit value monitoring of signal lamps and relay outputs. Any event can be reported by message.

- MIN/MAX for
1. Feed rate
  2. Belt load
  3. Belt speed

Switching threshold, relay outputs and event class can be defined individually for all 6 possibilities.

## ■ Display Filters

can be defined individually for feed rate, belt load, speed, and deviation.

## ■ Maintenance Hints

through display of event

1. Voltage ON-time
2. Scale ON-time

## ■ Batching Mode

with adaptive setpoint ramp

## ■ Adaptions

The electrical system needs not be redesigned nor adapted.

## ■ Calibration

Calibration as such is not required. All you have to do is enter some constructional data of load cells, speed transducer, belt inclination, etc. INTECONT uses these data to compute all requisite display normalisations.

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Check, or recalibration, can be effected in two ways:

## a) Using material

Enter check result into INTECONT, and measurement value will be corrected.

## b) Using check weight

A special check program lets you check normalisation or, if certain technical data are not exactly known (lever ratio, belt inclination, etc.), correct them.

In subsequent operation, this program can be used to detect mechanical changes, e.g. platform distortion after settling of foundation.

## ■ Digital Inputs

| Number | Name            | Default |
|--------|-----------------|---------|
| Q03    | DI Ackn. Events | DI1     |
| Q01    | Release         | DI5     |
| B06    | START Source    | — (OP)  |
| Q02    | DI Stop Batch   | —       |
| M04    | Source Sensor   | DI6     |
| B03    | Tacho Source    | DI7     |
| Q05    | DI Volum. Sync. | —       |

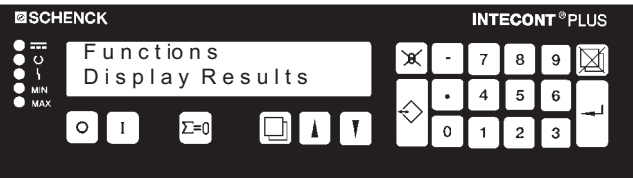
## ■ Digital outputs

| Number | Name            | Default |
|--------|-----------------|---------|
| R04    | DO I-MIN        | DO1     |
| R05    | DO I-MAX        | DO2     |
| R01    | DO ALARM        | DO3     |
| R02    | DO Prefeeder ON | DO6     |
| R03    | DO Feeder ON    | DO7     |
| R11    | DO Full Feed    | —       |
| R12    | DO Dribble Feed | —       |
| M15    | DO Slip Event   | —       |
| M16    | DO Belt Skew    | —       |
| M17    | DO Belt Slip    | —       |
| R06    | DO Q-MIN        | —       |
| R07    | DO Q-MAX        | —       |
| R08    | DO v-MIN        | —       |
| R09    | DO v-MAX        | —       |
| R10    | DO Batch Active | —       |
| R13    | DO Feeder ON    | D04     |
| R14    | DO Deviation    | D05     |
| R15    | DO Volum. Mode  | —       |

7. Setting Programs

The function distributor lets you access all setting programs.

-  Call function distributor.  
If an event is available, "Display Events" is displayed; if no event has occurred, display shows function selected last.
-  Start function.
-  Return to normal displays, or abort function.



-  Move through function distributor.

Display Results E

Display Check

START/STOP Feeder

Service Values

Volumetric/Gravimetric

Volum. Synchronous

START/STOP Keyb. Mode

>0 : Zero Set

Select Batch B

START/STOP EasyServe

Programming

Calibration Functions S

Print FMZ

Read Parameters

Enter Parameters S

Load Default Par. S

Print Parameters

Print Status Report

START/STOP Simulation

Taring

Weight Check

Imp/Belt Circuit

Set Time

Programming and Calibration functions lead to further menus you can move through using cursor keys.

- B = If batching mode is selected, additional functions can be accessed in main distributor.

Abort Batch  
Deselect Batch  
Print Batch

- S = Calibrating functions protected by password. Prompt for password input is output after call of function.

- E = If an event is available

With opposed functions, e.g. START/STOP, only the one currently accessible can be called.

Moving through function distributor has no effect on weighing functions.

Calibration Functions

There are 3 setting programs designed to ease initial calibration and recalibration.

1. Pulses/Belt CircuitLB
2. TaringTW
3. Weight CheckCW

Upon initial calibration, perform functions in the above order of sequence.

All programs can be reached using the CALIBRATION function and are protected from unauthorized use by password.

During program run until acknowledgement

1. no totalization takes place
2. analog outputs are set to elevation values
3. no event messages are suppressed

The operating sequence is the same for all programs.



Call function distributor.

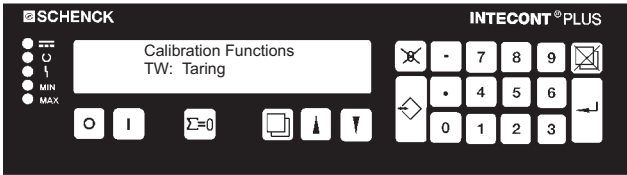


Scroll CALIBRATION FUNCTIONS into lower display field and acknowledge.



Enter password 07734.

Display shows one of the setting programs, e.g. TW: Tare.





Start setting program.



Return to normal displays.  
For apr. 2 min., Calibration menu can be recalled without password input.

Belt Circuit LB

Call setting program LB: IMP/BELT only

1. upon initial calibration
2. after mounting new belt or considerably varying belt tension
3. after having changed Parameter B 04 (Charact. Val.-vs) or B 05 (Nominal Speed) (regardless of whether scale operates with or without speed mesurement).

Program acquires the number of speed transducer pulses for one belt circuit and uses the number of pulses as belt circuit code for the following programs:

- a. Zero Setting>0<
- b. TaringTW
- c. Weight CheckCW

Upon initial calibration, call setting program LB first.

Conditions:

1. Before calling program, measure time of one belt circuit as exactly as possible and enter value into Parameter C 02.
2. Make sure conveyor belt is running (Scale ON).
3. Select Volumetric mode.
4. Deselect Batching mode.

Sequence:



Call function distributor, Calibration functions, password (see "Calibration Functions").



Scroll LB: Imp/Belt into lower display field.



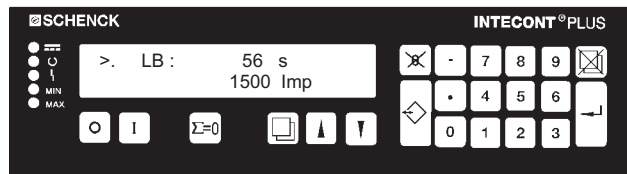
Start program.



Abort at any time.

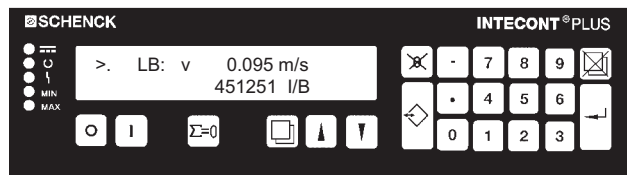
Setting programs do not control belt drive, prefeeder and operating mode.  
If any condition is not met, setting program aborts and a corresponding message is output.

## Program running:



Upper display : Residual run time in seconds  
Lower display : Added speed transducer pulses

## Program complete:



Upper display : Belt speed mean value over total run time  
Lower display : Pulses/belt circuit total



Accept result and enter into Parameter D 06.



Reject result.

The sequence is the same if feeder operates without speed measurement.

## Special Messages:

1. START if belt is not running
2. SELECT VOLUM. if Volumetric mode is not selected

## Attention!

If a belt circuit sensor is present (M03 = YES), program sequence slightly differs.

## Running

message : E-PROG active (lower display)

## Complete

message : ... - I/B  
Values flickers upon every belt circuit.

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## Tare TW

The taring program acquires the weighfeeder's zero point error over one or multiple integer belt circuits and uses value to correct current measuring result in normal mode.

On principle, taring program is identical with zero setting program.

Taring : Acquire basic tare (weight of mechanical system, conveyor belt, etc.) upon commissioning, service and maintenance.

Zero setting : Acquire zero point deviation during operation, e.g. contamination.

Unlike the zero setting program, the correction amount of the taring program is unlimited. After taring, the admissible correction amount of the zero setting program relates to the new reference value.

- Conditions :
1. Conveyor belt totally unloaded. Control prefeeder using the "START/STOP Feeder" function.
  2. Clean mechanical scale system in weighing area. Normal deposits immediately recurring in normal operation need not be removed.
  3. Make sure scale operates in Volumetric mode. A corresponding hint is output automatically.
  4. Deselect Batching mode.

## Sequence:



Function distributor, Calibration functions, password (see "Calibration Functions").



Scroll "TW: Tare" into lower display field.

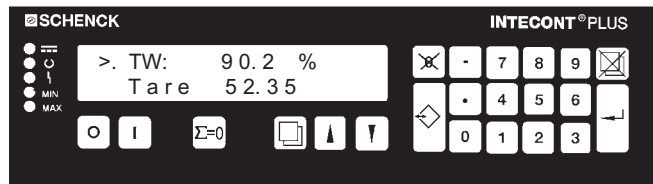


Start program.



Abort at any time.

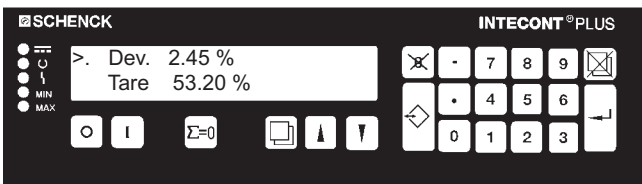
Program running:



Upper display:      Residual run time in % of total run time

Lower display:      Continuously meaned tare in % of nominal belt load

Program complete:



Upper display   :   Deviation of tare value from previous taring operation in % of nominal belt load

                  dev = + : tare increased  
                  dev = - : tare decreased

Lower display   :   Mean value of total tare in % of nominal belt load

-  Accept taring result and enter value into Parameter D 04.  
Tare correction (D05) is set to zero.
-  Reject result, i.e. scale not tared.

Special Messages:

1. START                   if belt is not running
2. SELECT VOLUM.       if Volumetric mode is not selected

Note:

1. Related to nominal belt load, tare value can exceed 100%.
2. If deviation “dev.” exceeds 20 %, check mechanical scale system, e.g. for jammed material particles.

## Weight Check CW

This program lets you check the INTECONT measuring range.

Load platform with a known check weight and acquire platform load mean value over one or multiple entire belt circuits.

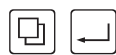
INTECONT compares result with specified setpoint and displays value.

No automatic correction takes place.

Conditions:

1. Tare or set to zero.
2. Enter check weight into Parameter C 08.  
Check weight should amount to 30...100 % of nominal platform load Q0.  
 $Q0 = q0 * Leff$   
 $q0$  = Nominal Belt Load (Parameter D 01)  
 $Leff$  = Eff. Platform Length (Parameter C 05)
3. Apply check weight in provided location.
4. Make sure scale operates in Volumetric mode.  
A corresponding hint is output automatically.
5. Deselect Batching mode.

Sequence:



Function distributor, Calibration function, password (see "Calibration Functions")



Scroll "CW: Weight Check" into lower display field.

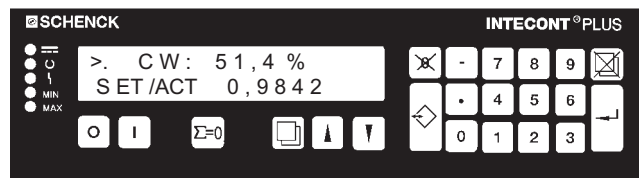


Start program.



Abort at any time.

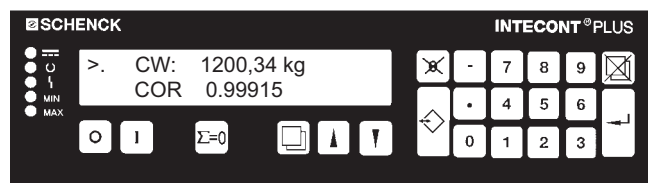
## Program running



Upper display : Residual run time in % of total run time

Lower display : Continuously meaned SET/ACT measuring result

## Program complete



Upper display : Fictitious material amount fed over run time \*.

Lower display : Mean value KOR of SET/ACT over total run time

$$COR = \frac{SET}{ACT} = \frac{\text{Parameter CHECK WEIGHT}}{\text{Measurement Value}}$$



Exit program.  
Since result is not acquired automatically, the two keys are equivalent.

## Special hints:

1. START if belt is not running
2. ABORTED (irrelevant)

\* Lets you check scale zero point without check weight  
Display format: xxxxxxx,yy kg

Evaluating result:

- Error < 1 % : KOR = 0.99...1.01  
Scale OK, no further action required.
- Error < 5 % : KOR = 0.95...1.05  
Enter KOR value into Parameter D 02 .  
This makes good sense only if parameter has not yet considered result of check using material.
- Error > 5 % : KOR < 0.95 or KOR > 1.05  
Multi-percent deviations suggest wrong data input (e.g. belt inclination not exactly known, lever arms) and/or mechanical faults (misalignment, distorsions).

Check does not consider Parameter "Span Correction" D 02, so that check program will indicate same error quotient KOR after check.

Simulation Mode

The Simulation Mode lets you check all scale functions without material at the time of commissioning.  
During simulation, scale is cannot operate in normal mode.



Function distributor, Calibration functions, password (see "Calibration Functions").



Scroll "START Simulation" into lower display field.



Start program.

Display : Event message S7  
In upper line, the "=" symbol flashes.

Function can be cut off in the same way (STOP Simulation).

Features:

1. All operating functions can be performed.
2. Control circuit internally closes; controller controls feed rate actual value for setpoint.
3. Control magniture Y is routed to the outside.
4. Belt load and speed measurement are active.
5. All control inputs and outputs operate normally.
6. Volumetric mode has no influence on simulation.



Set Time

Unlike the other programs, date and time can be changed in Start and Stop states of scale and be read off under SERVICE VALUES.

  Function distributor, calibration functions, password (see "Calibration Functions").

  Scroll "Set Time" into lower display field.

 Call menu.

 Abort at any time.

  Enter year, e.g. 03, month, day, hour, minute, and second.

 Acknowledge every input.  
When seconds have been acknowledged, new date is acquired.

Complete date is displayed for some seconds.

 Delete digit entered last.

After power failure, time remains stored for min. 5 days.

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8. Service Values

The Service table includes detailed system information. Call does not affect weighing functions.



Call SERVICE VALUES function and start.



Move through service values.



Return to normal displays.

Displays:

1. VDB 20600-00 3563

2. HW = 1

3. Opt.K: 0

1
- Version number

Hardware version

(no option card)

(option card available)



4. Date and time

5. Relay outputs switching conditions

I +: DO = 1 1 1 0 0 0 0 0

“1” = Contact closed  
“0” = Contact open  
Inputs 4-8 on option card  
“N” = No option card

6. Inputs switching conditions

I+ : DI = 1 1 1 1 1 + 0  
“1” = Contact closed  
“0” = Contact open  
Inputs 4 + 5 on option card  
“N” = No option card

Inputs 6 + 7 are designed as Namur inputs (belt sensor, tachometer)

“+” short-circuit; “-” cable breakage;  
“1” = sensor covered; “0” = sensor free

7. EL = 22 h

Voltage ON-time  
  
Monitoring: Parameter K 01, Message S4

8. ED = 19 h

ON-time of scale and conveyor belt  
  
Condition : Totalization ON  
Monitoring : Parameter K 03, Message S3

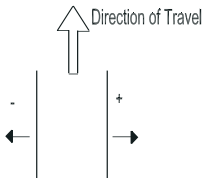
9. Tacho = 96.6 Hz

Speed transducer input frequency  
Frequency should range between 0.5 and 2500 Hz.

10. aw = 30.988 %

Load on load cells related to total of load cell rated capacities.  
If value exceeds 100%, load cells are overloaded.  
From 110 % onward, Message H4:  
“L/C Input > MAX” is output.

11.    **vap = 0.883168 mV/V**  
  
Unnormalised output value of load cell amplifier, delayed by shift to point of discharge (CPD)
12.    **bic = 2.34001 mV/V**  
  
Unnormalised output value of load cell amplifier with belt influence compensation (BIC) active
- 13    **wz = 1.383257 mV/V**  
  
Unnormalised output value of load cell amplifier (Gross)  
  
Value is acquired before belt influence compensation and delay for shift to point of discharge.
14.    **R\_wz = 300 ohms**  
  
Load cell resistance value
15.    **Slip s = 0.25 %**  
  
Change in belt length in % of total belt length (Parameter D 06).  
  
Condition       : Belt sensor active (M03), sensor area on conveyor belt (M04).  
  
Monitoring     : Parameter M13, Message C8
16.    **Belt Drift tr = 0.12 cm**  
The arrows indicate the direction of drift.  
  



  
  
Condition:       See "Slip"  
Monitoring :   Parameters M09-M12, Messages C3 and C4

17.    **Pulses of sensor area Imp.S. = 386**  
  
Measured pulses of sensor area. If belt sensor is properly mounted, the following value should result:  

Imp.S. = 0.5 \* Sensor length \* 0.01 \* vs-Char. Value

  
  
Condition: See "Slip"
18.    **AO1 = 4.15 mA**  
  
Output current of analog output 1
19.    **AO2 = 10.00 mA**  
  
Output current of analog output 2 (on option card)
20.    **ZE = 53**  
  
Pulses output across pulse output for external totalizing counter, counted after command "Reset Counter 1".
21.    **ZO = 1**  
Pulses still to be output.  
If value keeps exceeding zero, pulse frequency is higher than 10 Hz.  
Increase weighting of smallest display digit of Counter 1 (Parameter B 07).
22.    **AI = 12.54 mA**  
  
Input current of analog input
23.    **CS = 48149**  
  
Check sum of scale software
24.    **CS\_b = 48149**  
  
Check sum cyclically recomputed. If value "CS\_b" is unequal to "CS", a serious fault is given. Contact the Schenck Service.

25.

Y = 4.59 mA

Controller output
26.

Qun = 5.133 kg/m

Belt load before linearization

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9. Parameterization

General Information

Parameters are variable characteristics or data used to match INTECONT to the application.

- Display units and format
- Limit values
- Nominal and calibration data, etc.

Some parameters can be set in accordance with requirements (e.g. display formats), others have to be taken from Spec Sheet supplied. During parameter input, all functions are maintained. However, during operation, only parameters with no influence on measurement, e.g. display filters, should be changed.

Default Values

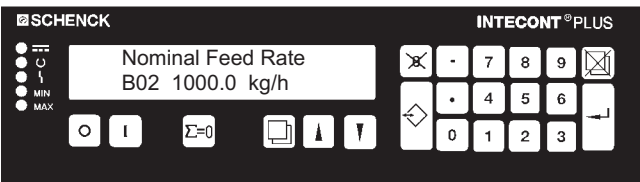
All parameters are preset with default values. These are proven suggestions and can normally used as defined. The default values can be loaded using sub-function “Load Default Parameters”.

Identification

The parameters are organised into function blocks A, B, ... and, within a block, described by

- name
- number 1, 2, 3 ..
- and
- value.

Values differing from default are identified by a \* before the block letter.



There are two types of parameter:

Selection parameters

Let you select from various options, e.g. WARNING, ALARM.

Numeric parameters

Enter numeric value, e.g. nominal feed rate.

Call Parameter Menu



Call function distributor.



Scroll PROGRAMMING into lower display field and acknowledge.

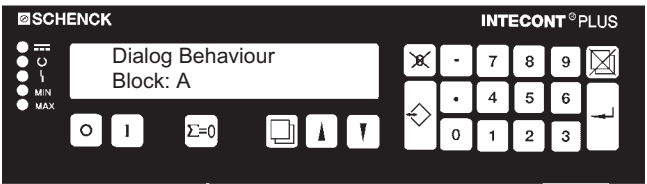


Select ENTER PARAMETER function.



Enter password 07734.

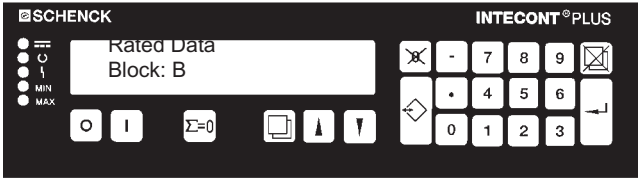
Display shows title of first parameter block.



Return to normal display. For apr. 2 min., Parameter menu can be recalled without password input.

Select Parameters

1. Block Letter

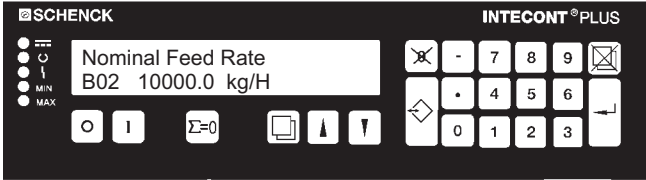


Move through blocks A, B ... up and down.  
Block letter flashes.



Select block.  
Display shows parameter selected last.

2. Parameter Number



Move through parameter numbers within a block.  
Parameter number flashes.



Move through parameter numbers independent of block (UP only).



Return to block titles.  
Return to normal mode.

Enter Parameters



Prepare input.



Selection parameter:  
value flashes.  
Scroll desired value into lower display field.



Go to the left (4) or to the right (6); see "Hardware Configuration".



Numeric parameters:  
Feature -----  
Enter value using number keys.



Acknowledge input.  
Next parameter is displayed.



Delete digits.



Abort input.

OFF key "0" is always active; ON key "I" only if parameter input is complete.

## Hardware Configuration

When parameterising the hardware, the source determines whether unit, channel and level must be indicated or not. The table below will assist you in configuring the hardware.

| Source | Unit | Channel | Level  |
|--------|------|---------|--------|
| FB     |      |         |        |
| OP     |      |         |        |
| v      |      |         |        |
| —      |      |         |        |
| DI     | I+   | DI1-DI7 | HI, LO |
| DO     | I+   | DO1-DO8 | HI, LO |
| AI     | I+   |         |        |

As you can see, with “Fieldbus” (FB), “Operator Panel” (OP), “Speed” (v) and “Not Assigned” (—) no additional input is required. “Digital Inputs” (DI) and “Digital Outputs” (DO) require unit, channel and level to be indicated. With “Analog Input” (AI), the unit is firmly set to I+.

### Sample:

Parameter B06 “START Source” determines from where the scale is started.

| Description      | Source |
|------------------|--------|
| B06 START Source | OP     |
| Alternatives     | FB, DI |

On standard the scale is started from the INTECONT keyboard.

Alternatively, fieldbus or digital input (DI) can be selected. If DI is the desired source, additionally indicate channel and level (see table above). The unit is firmly set to I+.



Keys 4 (go to the left) and 6 (go to the right) let you shift between source, channel and level.



## Load Default Parameters

Lets you load the default setting. Function is protected by password and YES/NO prompt.

**Attention!**  
All changes to parameters and calibration values get lost.



Select PROGRAMMING function, call DEFAULT PARAMETERS.



YES = 1  
NO = 0



Acknowledge selection.

## Power Failure

Parameters remain stored for an unlimited period of time.

## Parameter Overview

**BLOCK A** Dialog Behaviour

|    |          |         |
|----|----------|---------|
| 01 | Language | ENGLISH |
| 02 | Units    | SI      |

**BLOCK B** Rated Data

|    |                   |              |
|----|-------------------|--------------|
| 01 | Feed Rate Unit    | —— kg/h      |
| 02 | Nominal Feed Rate | 10.0000 t/h  |
| 03 | Tacho Source      | DI I+.DI7 HI |
| 04 | vs Charact. Val.  | 10000.00 l/m |
| 05 | Nominal Speed     | 0.1000 m/s   |
| 06 | START Source      | OP           |
| 07 | P-Source          | OP           |
| 08 | Prel EXT. ACTIVE  | NO           |
| 09 | WZ Active         | YES          |
| 10 | FMZ1 Unit         | —— t         |
| 11 | FMZ 1 Pulse Dur.  | 0 ms         |
| 12 | FMZ 2 Unit        | —— t         |
| 13 | FMZ 3 Unit        | —— t         |

**BLOCK C** Calibrating Data

|    |                   |             |
|----|-------------------|-------------|
| 01 | Belt Circuit No.  | 1.00        |
| 02 | Belt Circuit Time | 30.0 s      |
| 03 | L/C Charac. Value | 2.0000 mV/V |
| 04 | L/C Rated Cap.    | 60.000 kg   |
| 05 | Eff. Platf.Lenth  | 0.500 m     |
| 06 | Lever Ratio       | 1.0000      |
| 07 | Angle a           | 0.00 Grad   |
| 08 | Check Weight      | 10.000 kg   |

**BLOCK D** Calibrat. Results

|    |                     |             |
|----|---------------------|-------------|
| 01 | Nominal Belt Load * | 27.78 kg/m  |
| 02 | Span Correction     | 1.0000      |
| 03 | Total Tare          | 0.00 kg/m   |
| 04 | Basic Tare N        | 0.00 kg/m   |
| 05 | Tare Correction T   | 0.00 kg/m   |
| 06 | Belt Circuit Char   | 1000000 l/B |

**BLOCK E** Analog Output

|    |                  |          |
|----|------------------|----------|
| 01 | Source AO 1      | I        |
| 02 | Elevation AO 1   | 4.00 mA  |
| 03 | Limit Value AO 1 | 20.00 mA |
| 04 | Source AO 2      | Y        |

|    |                  |          |
|----|------------------|----------|
| 05 | ElevationAO 2    | 4.00 mA  |
| 06 | Limit Value AO 2 | 20.00 mA |

**BLOCK F** Limit Values

|    |                   |            |
|----|-------------------|------------|
| 01 | Value for I MIN   | 5.0 % lo   |
| 02 | Event Class I MIN | W1 (L 01)  |
| 03 | Value for I MAX   | 120.0 % lo |
| 04 | Event Class I MAX | W1 (H 01)  |
| 05 | Value for Q MIN   | 60.0 % Q   |
| 06 | Event Class Q MIN | W1 (L 02)  |
| 07 | Value for Q MAX   | 120.0 % Q  |
| 08 | Event Class Q MAX | W1 (H 02)  |
| 09 | Value for V-MIN   | 5.0 % V    |
| 10 | Event Class V-MIN | W1 (L 03)  |
| 11 | Value for V MAX   | 120.0 % V  |
| 12 | Event Class V MAX | W1 (H 03)  |
| 13 | Stand-By Limit    | 2 % lo     |
| 14 | Stand-By          | W2 (E 05)  |

**BLOCK G** Filter Setting

|    |                 |       |
|----|-----------------|-------|
| 01 | I Display       | 3.0 s |
| 02 | I Analog Output | 3.0 s |
| 03 | Display         | 3.0 s |
| 04 | V Display       | 3.0 s |
| 05 | L/C FILTER      | 1.0 s |
| 06 | Afterfl.Time    | 3.0 s |

**BLOCK H A** Additional device

|    |            |          |
|----|------------|----------|
| 01 | ZDO Active | NO       |
| 02 | ZDO Limit  | 1.00 % Q |

**BLOCK I** Batch Mode

|    |                   |               |
|----|-------------------|---------------|
| 01 | b.coast-down time | 3 s           |
| 02 | Batch corr.quant. | 0.000 kg      |
| 03 | Adaption fact.    | 0.5           |
| 04 | Batch Tolerance   | 100000.000 kg |
| 05 | Out of Tolerance  | W1 (B 01)     |
| 06 | AUTO Print        | NO            |

**BLOCK J** Printer Setting

|    |                   |      |
|----|-------------------|------|
| 01 | Baud Rate         | 9600 |
| 02 | Print Event       | YES  |
| 03 | Print Page Length | 72   |

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## BLOCK K Maintenance Int.

|    |                   |           |
|----|-------------------|-----------|
| 01 | Maintenance Elec. | 3000 h    |
| 02 | Event Maint. EL.  | W1 (S 04) |
| 03 | Maint. STR-Meter  | 3000 h    |
| 04 | Event Maint. Run. | W1 (S 03) |

## BLOCK L Comm. Fieldbus

|    |                  |             |
|----|------------------|-------------|
| 01 | Protocol Type    | *MODBUS     |
| 02 | Host timeout     | 10 s        |
| 03 | Comm.-Error Host | * W2 (S 09) |
| 04 | Address          | 1           |
| 05 | Resolution       | 4096        |
| 06 | Word sequence    | l:std/L:std |
| 07 | Physics          | * RS232     |
| 08 | Baud Rate        | * 19200     |
| 09 | Format Data      | * 8-O-1     |
| 10 | FLOAT-format     | SIEMENS-KG  |
| 11 | Physics          | RS422       |
| 12 | Baud Rate        | 9600        |
| 13 | Format Data      | 8-E-1       |
| 14 | Address          | 16          |
| 15 | FLOAT-format     | IEEE        |
| 16 | Configuration    | FIXED-1     |
| 17 | Address          | 16          |
| 18 | Baud Rate        | 125K        |
| 19 | Configuration    | FIXED-1     |
| 20 | Byte sequence    | Low - High  |
| 21 | Word sequence    | l:std/L:std |

## BLOCK M Scale Monitoring

|    |                   |               |
|----|-------------------|---------------|
| 01 | CPD Active        | NO            |
| 02 | Platf.Dis.Length  | 0.00 % LB     |
| 03 | Belt Sensor Activ | NO            |
| 04 | Source Sensor     | DI I+ .DI6 HI |
| 05 | BIC Active        | NO            |
| 06 | Sensor Length     | 8.40 cm       |
| 07 | Sensor Width      | 12.00 cm      |
| 08 | Sensor Offset     | 0.00 cm       |
| 09 | BELT DRIFT        | 3.00 cm       |
| 10 | BELT DRIFT EVENT  | W1 (C 04)     |
| 11 | BELT SKEW         | 4.00 cm       |
| 12 | BELT SKEW EVENT   | A (C 03)      |
| 13 | Slip Limit        | 2.00 % LB     |
| 14 | Slip Event        | W1 (C 08)     |
| 15 | DO BELT DRIFT     | —             |
| 16 | DO Belt Skew      | —             |
| 17 | DO Slip           | —             |

## BLOCK N Events

|    |                   |           |
|----|-------------------|-----------|
| 01 | Power Failure     | A (E 01)  |
| 02 | Memory Error      | A (S 01)  |
| 03 | Tacho Input       | A (C 02)  |
| 04 | Namur Err. Tacho  | A (E 02)  |
| 05 | Namur Err Belt    | W1 (E 03) |
| 06 | L/C Input         | A (C 01)  |
| 07 | No Release        | W1 (S 02) |
| 08 | L/C Input MAX     | A (H 04)  |
| 09 | L/C Input MIN     | A (L 04)  |
| 10 | SIMULATION ACTIVE | W2 (S 07) |
| 11 | SETPOINT LIMITED  | W1 (S 08) |

## BLOCK O CONTROLLER

|    |                   |              |
|----|-------------------|--------------|
| 01 | CONTROLLER TYPE   | STANDARD     |
| 02 | P-COMPONENT       | 0.02000 mA/% |
| 03 | I-COMPONENT       | 1.0 s        |
| 04 | CONTR.DEV.FILTER  | 3.0 s        |
| 05 | CONTR. DEV. TIME  | 20.0 s       |
| 06 | MAX. CONTR. DEV.  | 5.0 %        |
| 07 | Deviation factor  | 1.0          |
| 08 | CONTR. DEVIATION  | W1 (B 02)    |
| 09 | CONTROLLER LTD.   | W1 (B 03)    |
| 10 | LOWER LIMIT       | 0.00 mA      |
| 11 | UPPER LIMIT       | 20.00 mA     |
| 12 | CONTR.MAGN.ELEV.  | 0.00 mA      |
| 13 | POSITION AT STOP  | 0            |
| 14 | START-UP          | 0.0 Uml      |
| 15 | CLEARANCE         | 0.0 Uml      |
| 16 | ZERO SETPOINT     | 0.00 mA      |
| 17 | SETPOINT RANGE    | 20.00 mA     |
| 18 | STORE             | NO           |
| 19 | VOLUMETRIC MODE   | Qconst       |
| 20 | BYPASS            | 0.00 mA      |
| 21 | SETPOINT FILT. T1 | 0.0 s        |
| 22 | SETPOINT FILT. T2 | 0.0 s        |
| 23 | SET/ACT COMPAR.   | W-X          |
| 24 | SET/ACT SOURCES   | I            |
| 25 | ADAPTATION 1      | NO           |
| 26 | ADAPTATION 2      | NO           |

## BLOCK P Linearization

|    |                  |           |
|----|------------------|-----------|
| 01 | Linearization ON | NO        |
| 02 | Lin.-S1          | 25.00 % Q |
| 03 | Lin.-I1          | 25.00 % Q |
| 04 | Lin.-S2          | 50.00 % Q |
| 05 | Lin.-I2          | 50.00 % Q |
| 06 | Lin.-S3          | 75.00 % Q |

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|    |                 |            |
|----|-----------------|------------|
| 07 | Lin.-I3         | 75.00 % Q  |
| 08 | Lin.-S4         | 100.00 % Q |
| 09 | Lin.-I4         | 100.00 % Q |
| 10 | Lineariz. error | W2 (B 09)  |

## BLOCK Q Digital Inputs

|    |                  |              |
|----|------------------|--------------|
| 01 | Release          | DI I+.DI5 HI |
| 02 | DI Stop batch    | —            |
| 03 | DI Ackn. Events  | DI I+.DI1 HI |
| 04 | DI Volumetric ON | —            |
| 05 | DI Volum. Sync.  | —            |

## BLOCK R Digital Outputs

|    |                 |              |
|----|-----------------|--------------|
| 01 | DO ALARM        | DO I+.DO3 HI |
| 02 | DO Prefeeder ON | DO I+.DO6 HI |
| 03 | DO Feeder ON    | DO I+.DO7 HI |
| 04 | DO I-MIN        | DO I+.DO1 HI |
| 05 | DO I-MAX        | DO I+.DO2 HI |
| 06 | DO Q-MIN        | —            |
| 07 | DO Q-MAX        | —            |
| 08 | DO v-MIN        | —            |
| 09 | DO v-MAX        | —            |
| 10 | DO Batch active | —            |
| 11 | DO Full feed    | —            |
| 12 | DO Dribble feed | —            |
| 13 | DO Feeder ON    | DO I+.DO4 HI |
| 14 | DO Deviation    | DO I+.DO5 HI |
| 15 | DO Volum. Mode  | —            |

## BLOCK S Comm. EasyServe

|    |                  |       |
|----|------------------|-------|
| 01 | Station address  | 1     |
| 02 | Physics          | RS232 |
| 03 | EasyServe Baudr. | 19200 |
| 04 | Format Data      | 8-O-1 |

## Dialog Behaviour

## A 02 | Units

Range: SI                      Default: SI  
NON-SI

Lets you change over displays and parameter inputs from SI units to American units.

| American | SI unit | Conversion*                |
|----------|---------|----------------------------|
| lb       | kg      | 1 lb = 0.453593 kg         |
| t        | t       | 1 t = 0.907185 t = 2000 lb |
| f        | m       | 1 f = 0.3048 m             |
| inch     | cm      | 1 inch = 2.54 cm           |

also applicable to compound units, e.g. kg/h

## Rated Data

## B 03 | Tacho Source

Range: DI Default: DI (DI7)

“ — ” cuts off belt speed measurement.  
Nominal speed B 05 is used for computation.

Range: 1.0...1000000.0 l/m      Default:  
10 000.0 l/m

Number of speed transducer pulses per belt meter  
Exception: B 03 = —  
Never change characteristic value  
vs; this would change the run  
times of setting programs.

Range: 0.0001..2300000.0 t/h Default: 10.0 t/h

### Reference for limit values and service displays

B 05

Nominal Speed

Range: 0.001...50.0 m/s      Default: 0.1000 m/s

Reference for limit values.  
If BIC and/or VAP is used (Parameter Block M), value should be accurate to some percent.  
If scale operates without speed measurement, Parameter B 05 determines accuracy.

B 06

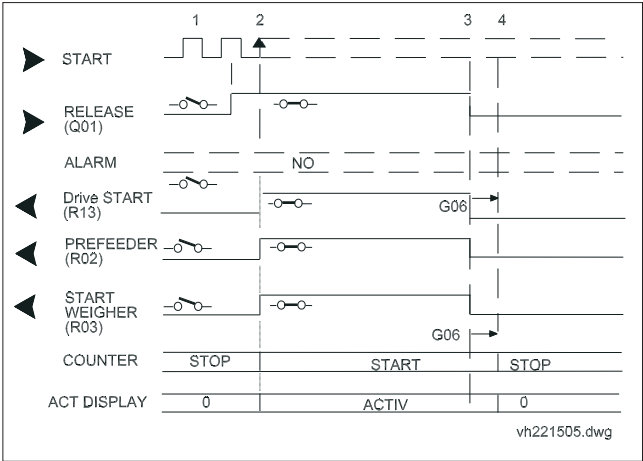
START Source

Range: OP      Default: OP  
FB  
DI

To start/stop totalization, there is no single solution:

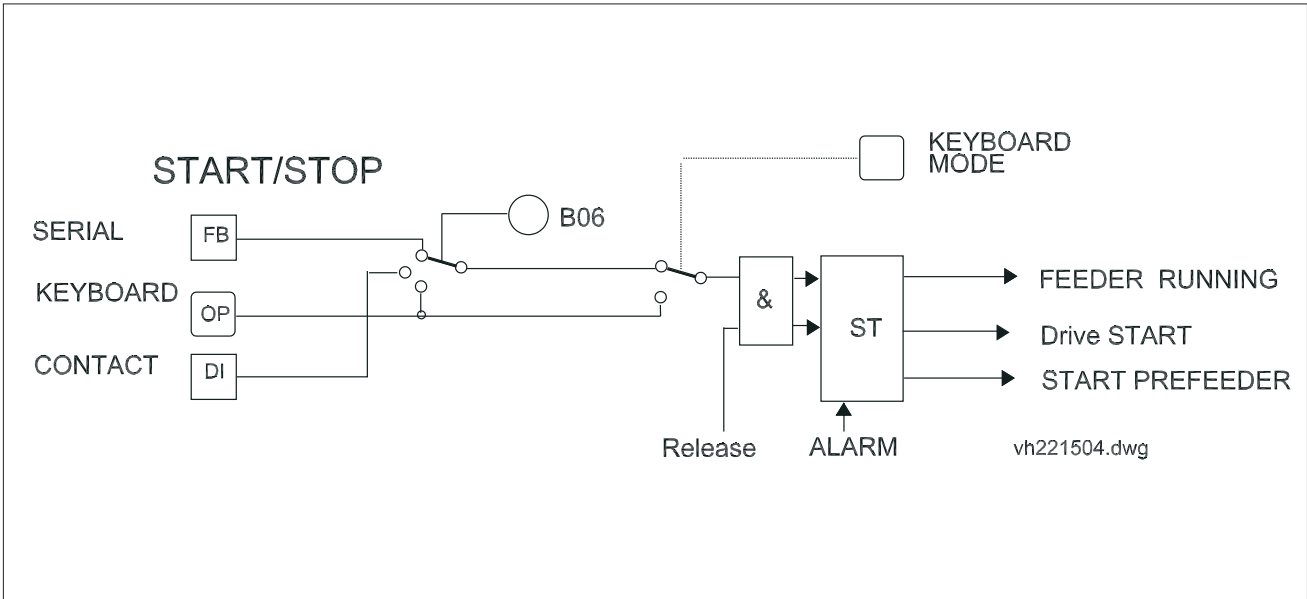
- OP : From INTECONT keyboard
- FB : Via fieldbus interface
- DI : Using digital input. Default value is DI3.  
You can also use any other free digital input. Digital input operates edge-controlled, i.e. if scale fails after Alarm, remove START signal and re-apply.

Dialog function “Keyboard Mode” lets you change over from selected source to keyboard, and vice versa.



**Note**

G06 defines the totalizing counter tracking time.  
If Parameter “Release” Q01 is set to “- -”, no release is required.



B 07P-Source

Range:

OP  
FB  
AI

Default: OP

Feed rate setpoint can be preset from the following sources:

OP : From INTECONT keyboard  
FB : Via fieldbus interface  
AI : Via INTECONT analog input

The “Keyboard Mode” dialog function lets you change over from selected source to keyboard and vice versa.

Independent of B07, batch setpoint can be preset from keyboard and/or fieldbus.

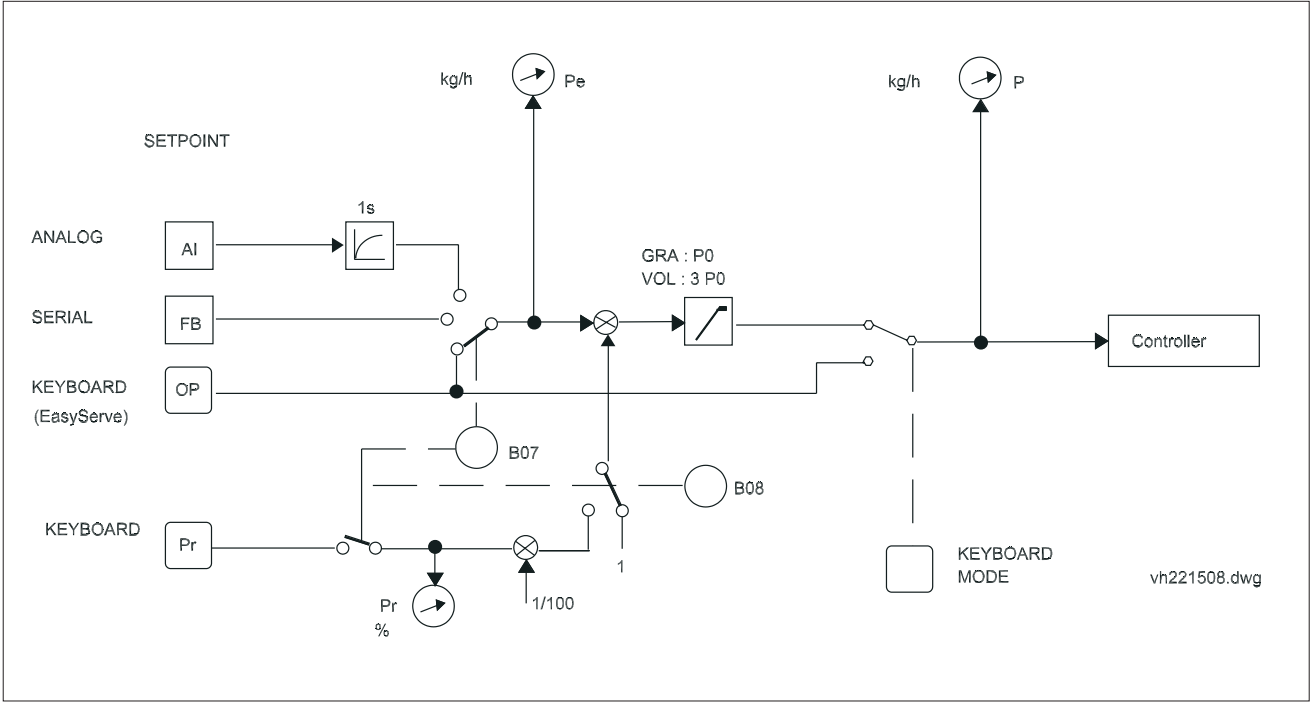
B 08Prel EXT. ACTIVE

Range:

NO  
YES

Default: NO

Percental evaluation of external setpoint via keyboard. Parameter is effective only if B 07 is set to AI or FB .



B 09

L/C Active

Range:    YES  
             NO  
  
Default: YES  
  
"NO" cuts off belt load measurement. Internally, belt load Qo = P0/V0 8 (computed from nominal values) is active.  
D 01 = B 02/B 05  
  
Note     :  
  
Set tare values D 04 and D 05 to 0. Service value wz continues being acquired.

B 10

FMZ 1 Unit

Range:    ----- t                      Default: ----- t  
             ----- .- t  
             ----- .- t  
             ----- .- t  
             ----- .- t  
             ----- t \*10  
             ----- t \*100  
             ----- kg  
             ----- .- kg  
             ----- .- kg  
             ----- .- kg  
             ----- .- kg  
  
B10 determines the display format for counter 1, batching values and pulse weighting for external counter. If no external counter is present, set Parameter B11 to 0.

B 11

FMZ 1 Pulse Dur.

Range:    0....1000 ms                      Default: 0 ms  
  
Length of external counter output pulse. Pulse weighting corresponds to the smallest display digit of counter 1 (Parameter B10).  
  
B11 < 10 ms :            Pulse output is blocked.  
                                 All pulses not yet output are deleted.  
  
Output frequency at nominal feed rate:  

$$f = \frac{PNENN}{3600 \bullet Zmin}$$

  
PNENN= nominal feed rate in kg/h or t/h  
Zmin    = smallest display digit of counter 1 (B 10) in kg or t  
  
Select Zmin (B10) so that f does not exceed 10 Hz. Furthermore, pulse duration B11 should be smaller than 1/f .

B 12

FMZ 2 Unit

As with B10. however, for counter 2

B 13

FMZ 3 Unit

As with B 10. however, for counter 3

## Block C : Calibrating Data

### C 01 Belt Circuit No.

Range: 1...100 Default: 1.00

Determines run time of setting programs (Zero Setting, Taring, Weight Check, etc.).

### C 02 Belt Circuit Time

Range: 1.0...9999,0 s Default: 30.0 s

Determines measuring time of calibration program "Imp/Belt".

Normally, the time of one belt circuit is selected.

### C 03 L/C Charac. Val.

Range: 0.01...9.9999 mV/V Default: 2 mV/V

Load cell characteristic value (transmission factor)

### C 04 L/C Rated Cap.

Range: 0.5000...220000.0 kg Default: 60.000 kg

Total of load cells rated capacities.  
Pivots count as load cells.

### C 05 Eff. Platf. Length

Range: 0.1000...50.000 m Default: 0.500 m

Effective length of weighing platform

### C 06 Lever Ratio

Range: 0.0100...2.0000 Default: 1.000

Lever ratio between force transducing weighed idler and load cell

$$C06 = \frac{F}{Q}$$

Q = Platform load  
F = Load cell load

With weighing modules with leaf spring parallel guidance, lever ratio is always 1.

### C 07 Angle a

Range: 0.0...60 Grad Default: 0.00 degr.

Angle of longitudinal scale axis if load cell is mounted vertically to belt.

### C 08 Check Weight

Range: 0.001...22000.0 kg Default: 10.000 kg

Material load on platform simulated through check weight.

## Block D :                      Calibrat. Results

### D 01    Nominal Belt Load

No input possible                      Default: 27.78  
kg/m

Computed from nominal values B 02 and B 05

Reference for limit values and zero setting program

### D 02    Span Correction

Range:    0.5000...2,0000                      Default: 1.0000

Proportionally influences belt load q measurement

$$q(\text{corrected}) = q(\text{measured}) * D\ 02$$

Function:                      Correction of measuring system  
through check with material.

### D 03    Total Tare

No input possible                      Default: 0.00 kg/m

Total tare = basic tare + tare correction

### D 04    Basic Tare N

No input required                      Default: 0.00 kg/m  
max. 10000 kg/m

Result of taring program

### D 05    Tare Correction T

No input required                      Default: 0.00 kg/m  
max. +- 1000 kg/m

Result of zero setting program

Every taring operation set Parameter D 05 to 0.

### D 06    Belt Circuit Char

No input required                      Default: 1000000  
max. 9E6                      I/U

Result of basic calibration program "Imp/Belt".  
Determines run times of setting programs.

Divided by Parameter B 04 (Characteristic Value vs)  
belt length in m results.

Attention!  
If you enter Parameters D 02 ... D 05 in manual, enter  
D 02 first.

Block E : Analog Output

E 01 Source AO 1

Range: I (feed rate) Default: I  
Q (belt load)  
V (belt speed)  
Xd (deviation)  
Y (control magnitude)  
P (setpoint)

One of these values can be output in analog fashion.  
Filters: Parameters G 02, G 04, G 05, O04

E 02 Elevation AO 1

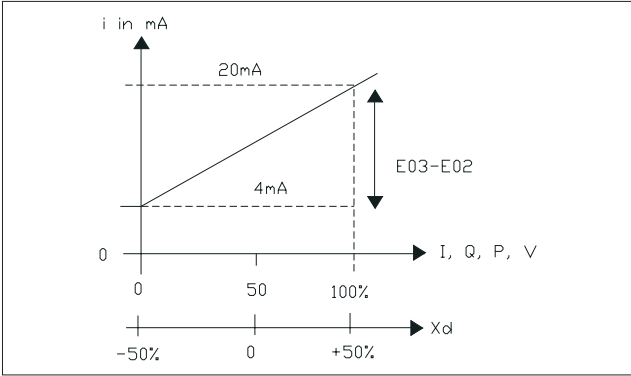
Range: 0.00...20.00 mA Default: 4.00 mA

Output current for values I, Q, P and V of 0%. With deviation Xd, zero point is between elevation and 20 mA (see diagram and formula).  
With control magnitude Y, parameter is irrelevant.

E 03 Limit Value AO 1

Range: 0.00...1000.00 mA Default: 20.00 mA

Output current for values I, Q, P and V of 100 %



$$i = \text{NULL} + (\text{END} - \text{ANH}) \frac{X}{X_{\text{nenn}}}$$

NULL = ANH bei I, Q, P, V

$$\text{NULL} = (20\text{mA} + \text{ANH}) \frac{1}{2} \text{ with Xd}$$

X = I, Q, P, V, Xd

Xnenn = nominal value

- Note:
- 1) Maximum output current is 20 mA.
  - 2) Minimum output current corresponds to elevation E 02.
  - 3) If scale is stopped, outputs for I, P and Xd are set to the elevation value.  
With Q and V, measurement value is output.
  - 4) With control magnitude Y, parameter is irrelevant.

E 04 Source AO 2

Range: I (feed rate) Default: Y  
Q (belt load)  
V (belt speed)  
Xd (deviation)  
Y (control magnitude)  
P (setpoint)

One of these values can be output in analog fashion.

Filter: Parameters G 02, G 04, G 05, O04

E 05

Elevation AO 2

Range: 0.00...20.00 mA      Default: 4,00 mA

Output current for values I, Q, P and V of 0%.

With deviation Xd, zero point is between elevation and 20mA (see diagram and formula).

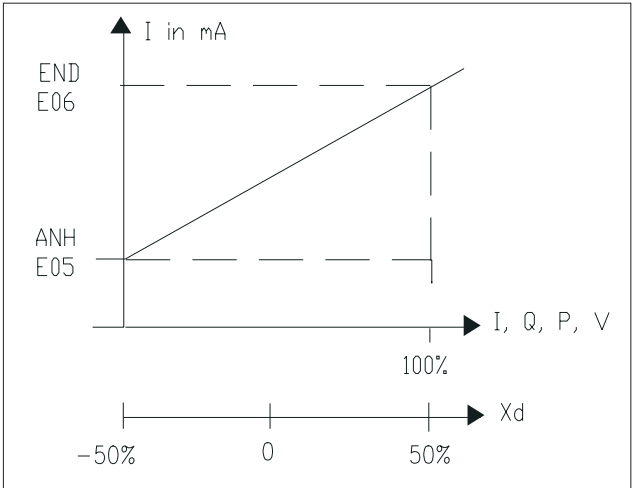
With control magnitude Y, parameter is irrelevant.

E 06

Limit Value AO 2

Range: 0.00...1000.00 mA      Default: 20.00 mA

Output current for values I, Q, P and V of 100%



For explanations and comments, see Parameter E03.

## Block F :

## Limit Values

If measurement values exceed their MIN/MAX limits, a corresponding event message is output (L1... L3, H1... H3).

Monitoring starts 10 s after start-up.

### F 01 Value for I-MIN

Range: -10...200.0 % I Default: 5.0 % I

Reference: Nominal feed rate B 02.

### F 02 Event Class I-MIN L1

Range: WARNING 2 Default:  
WARNING 1 WARNING 1  
IGNORE  
ALARM

### F 03 Value for I-MAX

Range: -10...200.0 % I Default: 120.0 % I

Reference: Nominal feed rate B 02.

### F 04 Event Class I-MAX H1

Range: WARNING 2 Default:  
WARNING 1 WARNING 1  
IGNORE  
ALARM

### F 05 Value for Q-MIN

Range: -10...200.0 % Q Default: 60.0 % Q

Reference: Nominal belt load D 01

### F 06 Event Class Q-MIN L2

Range: WARNING 2 Default:  
WARNING 1 WARNING 1  
IGNORE  
ALARM

### F 07 Value for Q-MAX

Range: -10...200.0 % Q Default:  
120.0 % Q

Reference: Nominal belt load D 01

### F 08 Event Class Q-MAX H2

Range: WARNING 2 Default:  
WARNING 1 WARNING 1  
IGNORE  
ALARM

### F 09 Value for V-MIN

Range: -10...200.0 % V Default: 5,0 % V

Reference: Nominal speed B 05

### F 10 Event Class V-MIN L3

Range: WARNING 2 Default:  
WARNING 1 WARNING 1  
IGNORE  
ALARM

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**F 11 Value for V-MAX**

Range: -10...200.0 % V      Default: 120.0 % V

Reference: Nominal speed B 05

**F 12 Event Class V-MAX H3**

Range: WARNING 2      Default: WARNING 1  
 WARNING 1  
 IGNORE  
 ALARM

**F13 Stand-By Limit**

Range: 0...100 %l      Default: 2 %

Reference: Nominal feed rate B 02

Message : E5 in accordance with event class F 16

Cuts off belt drive and prefeeder (not scale) with setpoints < F13.

Control magnitude is set to the value of Parameter O 12.

Changeover to Stand-By mode is delayed by 3s, switchback takes place immediately.

**F 14 Stand-By E5**

Range: WARNING 2      Default: WARNING 2  
 WARNING1

## Block G : Filters

I = feed rate V = belt speed

### G 01 I Display

Range: 0.0...600.0 s Default: 3.0 s

Feed rate display

### G 02 I Analog Output

Range: 0.0...600.0 s Default: 3.0 s

Feed rate analog output

### G 03 Q Display

Range: 0.0...600.0 s Default: 3.0 s

Belt load display

### G 04 V Display

Range: 0.0...600.0 s Default: 3.0 s

Belt speed display

### G 05 L/C Filter

Range: 0.0...600.0s Default: 1.0 s

Applies to all functions and displays referring to belt load

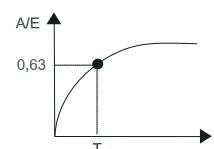
Q = belt load

### G 06 Afterfl. Time

Range: 0.0...2000.0 s Default: 3.0 s

The time totalization goes on after scale stop

All filters are of the first-order type (exponential), i.e. abrupt change of signal, approx. 2/3 of full scale are reached after set time. Long times lead to a stable but slowly refreshing display.



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Block H :

Additional Devices

H 01

ZDO Active

Range:

NO  
YES

Default: NO

Activate Zero Drop Out system (see H 02).

H 02

ZDO Limit

Range:

0.0...10.00 %Q

Default: 1.00 %Q

Reference: Nominal belt load D 01

If ZDO is activated and belt load falls below set limit, feed rate measurement and totalization are suppressed. No symbol appears on display.

## Block I :

## Batch Mode

For detailed batching sequence, see DETAILS chapter.

### I 01 B.coast-down time

Range: 0...3600s Default: 3s

If coast down time is reached, feed rate setpoint moves towards 0 regardless of setpoint source.

### I 02 Batch corr.quant.

Range: 0.000...±1 000 000 kg Urwert: 0.000 kg

Batching error I 02 can be used for correction of next batch (see Parameter I 03).

Note: Set I 02 to 0 at the time of commissioning.

### I 03 Adaption fact.

Range: 0.0...1.0 Default: 0.5

Start of coast down time is adapted automatically so that the system better fits the situation upon next batch.

$$I02(new) = I02(old) - error * I03$$

I 03 = 0 : no adaption

I 03 = 1 : full adaption

Intermediate values cause a filtered adaption which is preferable in plants with unprecise repeatability.

### I 04 Batch Tolerance

Range: 0.0...2200000.0 kg Default: 100000.000 kg

Reference: Batch setpoint

Admissible deviation from setpoint. Check takes place after completion of batch.

### I 05 Out of Tolerance B1

Range: WARNING 1 Default: WARNING 1  
WARNING 2  
ALARM  
IGNORE

If limit value I 04 is exceeded, event message B1 is output .

I 06

AUTO Print

Range:

NO

YES

Default NO

YES :

Autmatic printout after every complete batch

NO:

Batch report can be printed only using control menu.

|                       |                 |                  |     |                       |
|-----------------------|-----------------|------------------|-----|-----------------------|
| <input type="radio"/> | Batch Report    |                  | I   | <input type="radio"/> |
| <input type="radio"/> | 06.03.03        | 16:20:16         |     | <input type="radio"/> |
| <input type="radio"/> | Batch setpoint  | 200 kg           |     | <input type="radio"/> |
| <input type="radio"/> | Batch setpointt | 205 kg           |     | <input type="radio"/> |
| <input type="radio"/> | Counter 1       | 31779 kg         |     | <input type="radio"/> |
| <input type="radio"/> | Counter 2       | 32 t             |     | <input type="radio"/> |
| <input type="radio"/> | Events          |                  |     | <input type="radio"/> |
| <input type="radio"/> | B1              | Out of Tolerance | I05 | <input type="radio"/> |
| <input type="radio"/> | H2              | Load > MAX       | F08 | <input type="radio"/> |
| <input type="radio"/> |                 |                  |     | <input type="radio"/> |
| <input type="radio"/> |                 |                  |     | <input type="radio"/> |
| <input type="radio"/> |                 |                  |     | <input type="radio"/> |

Block J : Printer Setting

J 01 Baud Rate

Range: 2400 4800 9600 19200 38400 Default: 9600

Speed of transmission to printer

J 02 Print Event

Range: YES NO Default: YES

YES : Print event automatically upon occurrence and removal

NO : No printout

|                       |                                        |                       |
|-----------------------|----------------------------------------|-----------------------|
| <input type="radio"/> |                                        | <input type="radio"/> |
| <input type="radio"/> | 30.09.97 11:45:21 L2 W1 --> Load < MIN | <input type="radio"/> |
| <input type="radio"/> | 30.09.97 11:59:01 L2 W1 -->Load < MIN  | <input type="radio"/> |
| <input type="radio"/> |                                        | <input type="radio"/> |
| <input type="radio"/> |                                        | <input type="radio"/> |

J 03 Print Page Length

Range: 9...127 lines Default: 72 lines

Page length for printout of batch report

Line spacing is always 1/6 inch.

Printer is initialized upon change of parameter or power-up.

Block K :

Maintenance Int.

K 01

Maintenance Elec.

Range:

0...10000 hrs

Default: 3000 hrs

Message: S4

(K 02)

Alerts you on maintenance work to be effected as a function of voltage ON-time.

K 02

Event Maint. EL.

S4

Range:

WARNING 1

WARNING 2

IGNORE

Default:

WARNING 1

The total of power ON-times exceeds time K 01.  
If a time interval has elapsed, message S4 is output.

K 03

Maint. STR-Meter

Range:

0...10000 hrs

Default: 3000 hrs

Message: S3

(K 04)

Alerts you on maintenance work to be effected as a function of conveyor belt run time.

K 04

Event Maint. Run

S3

Range:

WARNING 1

WARNING 2

IGNORE

Default: WARNING

1

The total of conveyor belt run times exceeds time K 03.  
If a time interval has elapsed, message S3 is output.

## Block L :

## Comm. Fieldbus

### L 01 Protocol Type

Range: NO Default: NO  
 MODBUS COMP.  
 S5-BCC COMP.  
 MODBUS  
 PROFIBUS DP  
 DEVICENET

Eligible protocol variants.  
 Every protocol requires an optional interface card to be present.  
 "COMP" identifies the compatible protocols.

### L 02 Host Timeout

Range: 0...600s Default: 10s

If timeout value exceeds zero, a message from host system is expected to arrive during set time.

### L 03 Comm. Error Host S9

Range: IGNORE Default: IGNORE  
 W1 (WARNING 1)  
 W2 (WARNING 2)  
 A (Alarm)

If no message is received during the time set by Parameter L02, event message S9 "Data Link Host" is output.

### L 04 Address

Range: 1...254 Default: 1

Slave address for Modbus protocol.

### L 05 Resolution

Range: 1...32767 Default: 4096

Resolution of data for nominal value in Modbus protocol (integer format).

### L 06 Word Sequence

Range: I: std/L: std Default: I: std/L: std  
 I: swp/L: std  
 I: std/L: swp  
 I: swp/L: swp

Determines the word sequence within a data double word in the Modbus protocol.

"I" stands for IEEE-754 values (floating point values)

"L" stands for 4-byte integer values

"std" does not swap the word sequence; "swp" does.

### L 07 Physics

Range: RS232 Default: RS 485  
 RS 422  
 RS 485

Sets interface physics for Modbus

### L 08 Baud Rate

Range: 2400 Default: 9600  
 4800  
 9600  
 19200  
 38400

Modbus baud rate

## L 09 Format Data

Range: 8-E-1 Default: 8-E-1  
8-0-1  
8-N-2  
8-N-1

Modbus data format (data bits - parity bit - stop bits)

## L 10 FLOAT FORMAT

Range: SIEMENS-KG Default: SIEMENS-KG  
IEEE

Determines representation of floating point values for S5 protocol

## L 11 Physics

Range: RS 232 Default: RS 422  
RS 422

Sets interface physics for S5 protocol

## L 12 Baud Rate

Range: 2400 Default: 9600  
4800  
9600  
19200  
38400

Baud rate for S5 protocol

## L 13 Format Data

Range: 8-E-1 Default: 8-E-1  
8-0-1  
8-N-2  
8-N-1

S5 data format (data bits - parity bit - stop bits)

## L 14 Address

Range: 0...126 Default: 16

Profibus DP slave address

## L 15 FLOAT FORMAT

Range: SIEMENS-KG Default: IEEE  
IEEE

Determines representation of floating point values for Profibus DP protocol

## L 16 Configuration

Range: FIXED-1 Default: FIXED-1  
FIXED-2  
NO-PARA-ID  
PARA-ID

Determines structure and size of process image (Profibus DP)

## L 17 Address

Range: 0...63 Default: 16

DeviceNet slave address

## L 18 Baud Rate

Range: 125k Default: 125k  
250k  
500k

DeviceNet baud rates

## L 19 Configuration

Range: FIXED-1                      Default: FIXED-1  
 FIXED-2  
 NO-PARA-ID  
 PARA-ID

Determines structure and size of process image  
 (DeviceNet)

## L 20 Byte Sequence

Range: LOW-High                      Default: LOW-High  
 High-LOW

Determines byte sequence within a data word

## L 21 Word Sequence

Range: I: std/L:std                      Default: I: std/L: std  
 I: swp/L: std  
 I: std/L: swp  
 I: swp/L: swp

Determines the word sequence within a data double  
 word in the DeviceNet protocol.

“I” stands for IEEE-754 values (floating point values)

“L” stands for 4-byte integer values

“std” does not swap the word sequence; “swp” does.

## Block M : Scale Monitoring

### M 01 CPD Active

Range: NO  
YES Default: NO

Start speed-dependent delay for shift of measurement to point of discharge (CPD).

Conditions:

Speed meter available and activated.  
No belt circuit sensor required.

### M 02 Platf. Dis. Length

Range: 0.00...50.00 %LB Default: 0.00 % LB

Input: only required with CPD Active.

Distance between mid weighing platform to material discharge point in % of total belt length LB.  
Resolution is approx. 200.

### M 03 Belt Sensor Active

Range: NO  
YES Default: NO

Belt sensor and sensor area on conveyor belt serve for acquisition of belt slip, belt drift and synchronization of adaptive belt influence compensation BIC.

### M 04 Source Sensor

Range: —  
DI Default: DI (DI6)

Digital input for belt sensor. Function can be controlled using relevant service value.  
Only DI6 can be used. If you select “—”, BIC and belt run monitoring functions cannot be used.

### M 05 BIC Active

Range: NO  
YES Default: NO

Starts adaptive belt influence compensation BIC.

To start BIC:

Dynamic tare component is removed upon start of supply voltage or change to:

- Parameter M 05 “BIC Active”
- Parameter B 04 “vs Charact. Val.”
- Parameter D 06 “Belt Circuit Char”

Brief instructions:

1. Set M03 to YES, enter M 06... M 08.
2. Run setting program “LB: Imp/Belt”.
3. Activate BIC (M05 = YES)
4. Tare.

## M 06 Sensor Length

Range: 0.01...400.00 cm Default: 8.40 cm

Length of sensor area in direction of belt travel.  
See also Service Value "Imp.S.".

## M 07 Sensor Width

Range: 0.01...400.00 cm Default: 12.00 cm

Width of sensor area

## M 08 Sensor Offset

Range: 400.00...400.00 cm Default: 0.00 cm

Offset can be used to correct zero point of belt run monitoring.  
However, this limits the maximum display range.  
Compensate higher differences by shifting of sensor (see "Commissioning" chapter).

New display = old display - offset

## M 09 Belt Drift

Range: 0.01...400.00 cm Default: 3.00 cm

Limit value for admissible belt drift. If exceeded, event message "Belt Drift" C4 is output and digital output M15 "DO Drift" is set.

## M 10 Belt Drift Event C4

Range: WARNING 1 Default: WARNING 1  
WARNING 2  
Ignore  
Alarm

Cause of event:  
Belt leaves specified tolerance zone.

Action:

1. Clean head and tail pulleys.
2. Align belt.

Monitoring as such can also be faulty. Therefore:  
Watch service values of belt circuit sensor; if need be, remedy fault.

## M 11 Belt Skew

Range: 0.01...400.00 cm Default: 4.00 cm

Admissible belt skew limit. If exceeded, event message "Belt Skew" C3 is output and digital output M16 "DO Belt Skew" is set.

## M 12 Belt Skew Event C3

Range: WARNING 1 Default: Alarm  
WARNING 2  
Ignore  
Alarm

Conveyor belt runs too far off centre. Scale is no longer operable.

Action: see "Belt Skew" event

## M 13 Slip Limit

Range: 0.00...10.00 %LB Default: 2.00 %LB

Admissible change in conveyor belt length in % of total belt length.

If limit value is exceeded, event message "Slip Error" C8 is output and digital output M17 "DO Slip" is set.

M 14

Slip Event

C8

Range:

WARNING 1  
WARNING 2  
IGNORE  
ALARM

Default:

WARNING 1

Specified limit value (M13) exceeded

M 15

DO Belt Drift

Range:

DO  
—

Default:

—

Digital output used to report belt drift

M 16

DO Belt Skew

Range:

DO  
—

Default:

—

Digital output used to report belt skew

M17

DO Slip

Range:

DO  
—

Default:

—

Digital output used to report belt slip

| Block N:                                                                                                                                                                           |                       | Events              |  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|---------------------|--|
| N 01                                                                                                                                                                               | Power Failure         | E1                  |  |
| Range: ALARM<br>WARNING 1<br>WARNING 2<br>IGNORE                                                                                                                                   |                       | Default: ALARM      |  |
| Scale does not start automatically regardless of event class.                                                                                                                      |                       |                     |  |
| N 02                                                                                                                                                                               | Memory Error          | S1                  |  |
| ALARM                                                                                                                                                                              |                       | No changes possible |  |
| Scale inoperable                                                                                                                                                                   |                       |                     |  |
| N 03                                                                                                                                                                               | Tacho Input           | C2                  |  |
| Range: ALARM<br>WARNING 1<br>WARNING 2                                                                                                                                             |                       | Default: ALARM      |  |
| Input frequency exceeds 2700 Hz.                                                                                                                                                   |                       |                     |  |
| Condition: B 03 = DI7                                                                                                                                                              |                       |                     |  |
| N 04                                                                                                                                                                               | Namur Err. Tacho      | E2                  |  |
| Range: ALARM<br>WARNING 1<br>WARNING 2<br>IGNORE                                                                                                                                   |                       | Default: ALARM      |  |
| Short-circuit or cable breakage                                                                                                                                                    |                       |                     |  |
| Condition: B 03 = DI7                                                                                                                                                              |                       |                     |  |
| Namur errors set internal speed value to 0.<br>Frequency display (service value) is still active.                                                                                  |                       |                     |  |
| N 05                                                                                                                                                                               | Namur Err Belt Sensor | E3                  |  |
| Range: WARNING 1<br>WARNING 2<br>ALARM<br>IGNORE                                                                                                                                   |                       | Default: WARNING 1  |  |
| Short-circuit or breakage of belt sensor cable (BIC).                                                                                                                              |                       |                     |  |
| Condition: M 03 = YES                                                                                                                                                              |                       |                     |  |
| N 06                                                                                                                                                                               | L/C Input             | C1                  |  |
| Range: ALARM<br>WARNING 1<br>WARNING 2                                                                                                                                             |                       | Default: ALARM      |  |
| 1. Load cell not (or improperly) connected.<br>2. Analog-to-digital converter of measuring amplifier i<br>in saturation (see Service Value wz).<br>3. Supply voltage is below 19V. |                       |                     |  |
| Error must be available for at least 3 s.                                                                                                                                          |                       |                     |  |
| N 07                                                                                                                                                                               | No Release            | S2                  |  |
| Range: WARNING 1<br>WARNING 2<br>IGNORE<br>ALARM                                                                                                                                   |                       | Default: WARNING 1  |  |
| No RELEASE input signal                                                                                                                                                            |                       |                     |  |
| Exception:<br>IGNORE always releases scale                                                                                                                                         |                       |                     |  |

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|             |                           |           |
|-------------|---------------------------|-----------|
| <b>N 08</b> | <b>L/C Input &gt; MAX</b> | <b>H4</b> |
|-------------|---------------------------|-----------|

Range: ALARM                      Default: ALARM  
 WARNING 1  
 WARNING 2

Load on load cell exceeds 110 % of the total of load cell rated capacities (C 04).

Full scale:            approx. 115 % with CSD load cells  
                          160% with HBM load cells

|             |                           |           |
|-------------|---------------------------|-----------|
| <b>N 09</b> | <b>L/C Input &lt; MIN</b> | <b>L4</b> |
|-------------|---------------------------|-----------|

Range: ALARM                      Default: ALARM  
 WARNING 1  
 WARNING 2  
 IGNORE

Load on load cell has fallen below 3% of the total of load cell rated capacities.

|             |                          |           |
|-------------|--------------------------|-----------|
| <b>N 10</b> | <b>Simulation Active</b> | <b>S7</b> |
|-------------|--------------------------|-----------|

Range: WARNING 1                  Default:  
 WARNING 2                      WARNING 2  
 ALARM  
 IGNORE

Simulation mode started

|             |                         |           |
|-------------|-------------------------|-----------|
| <b>N 11</b> | <b>Setpoint Limited</b> | <b>S8</b> |
|-------------|-------------------------|-----------|

Range: WARNING 1                  Default:  
 WARNING 2                      WARNING 1  
 ALARM  
 IGNORE

Gravimetric mode :            Setpoint is limited to nominal feed rate (Parameter B 02)  
 Volumetric mode:            Limitation starts from values exceeding 3 \* nominal value.

Block O: Controller

O 01 Controller Type

Range: STANDARD                      Default: STANDARD  
UNIVERS

STANDARD : Controller is designed for a speed-controlled weighfeeder. Parameters from O17 onward are set to default and faded out; corresponding functions are no longer active.

Attention: Set Parameters B 03 and B 09 to YES.

UNIVERS : All controller functions are available. Various controller systems can be realized.

O 02 P-Component KP

Range: 0...1000.00000                      Default: 0.02 mA/%

Reference: Nominal setpoint

Exception: KP = 0 results in an I controller

O 03 I-Component TN

Range: 0...60 000.0 s  
Default: 1.0 s

Setback time TN is the time during which the I component causes the same control magnitude change as the P component.

Exceptions:

1.KP = 0 : I controller TN is the reciprocal of integration constant KI, i.e. big value = slow controller.  
KI = 1/TN in %/(mA/s)

2. TN = 0 : P controller  
I component is removed

3. KP=TN= 0 : Controller output = elevation

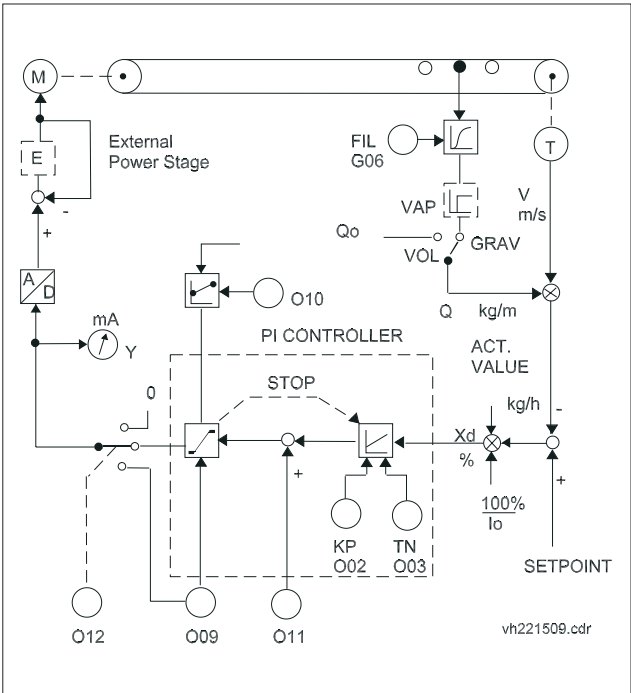


Fig.: Control circuit, FEED type

Never change Parameter O 01 during operation.

O 04 Contr. Dev. Filter

Range: 0.0....600.0 s                      Default: 3.0 s

Filter for display and analog output of deviation, not for monitoring.

## O 05 Contr. Dev. Time

Range: 0.0...600.0 s Default: 20.0 s

See Parameter O 07

## O 06 MAX Contr. Dev.

Range: 0.0...100.0 % Default: 5.0 %

Ref. : Nominal feed rate B 02, or  
Nominal belt load D 01

Details : See Parameter O 07

## O 07 Deviation Factor

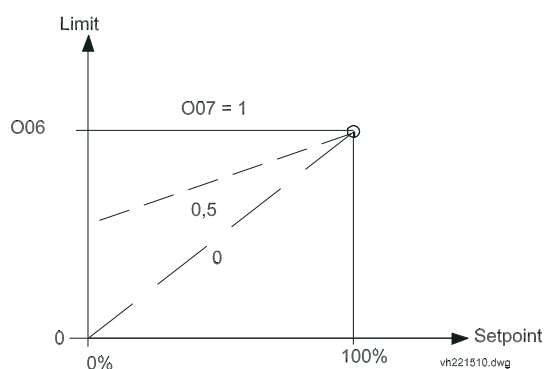
Range: 0.0...1.0 Default: 1.0

If deviation absolute value exceeds limit O06 for time O05, event message B2 (O08) is output.

At 100% setpoint, effective threshold always corresponds to Parameter O06. For smaller setpoints, decrease threshold using Parameter O07.

O07 = 0 : effective threshold = O06 \*  
setpoint/nominal value

O07 = 1 : effective threshold = O06



## O 08 Contr. Deviation B2

Range: WARNING 1 Default:  
WARNING 2 WARNING 1  
ALARM  
IGNORE

Maximum deviation exceeded. Actual feed rate inadmissibly differs from setpoint for a period longer than specified time.

There is no single cause:

1. Material handling error.  
Material flows irregularly or cannot be easily discharged.  
Controller is temporarily limited.
2. Electronics fault.  
Check external controller and motor cables.  
Check feed rate controller for proper setting (current limitation, speed range).
3. Calibration error.  
Upon initial calibration, controller was set so critically that the slightest change to system characteristics can make system unreliable.

Action: Repeat this part of commissioning.

## O 09 Controller Ltd. B3

Range: WARNING 1 Default:  
WARNING 2 WARNING 1  
ALARM  
IGNORE

Event message B3 "Controller Limited" is output if control magnitude reaches upper response threshold O 11.

## O 10 Lower Limit

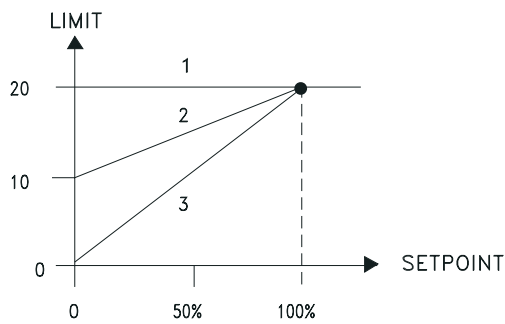
Range: 0...20.00 mA Default: 0 mA

Lower limit of controller control magnitude.

O 11 Upper Limit

Range: 0...20.00 mA Default: 20.00 mA

Upper limit of controller control magnitude. Upon inputs < 20.00 mA, limit depends on setpoint value.



- Examples:
- 1. Upper limit O 11 = 20 mA
  - 2. Upper limit O 11 = 10 mA
  - 3. Upper limit O 11 = 0 mA

O 12 Contr. Magn. Elev.

Range: 0...20.00 mA Default: 0 mA

Elevation of control magnitude by a constant value. Elevation acts before control magnitude limitation. With setting O 10 = O 12, control magnitude cannot fall below elevation.

Control magnitude is output in the form of impressed current of 0...20 mA. At 500 ohmic load, 20 mA correspond to 10 V.

O 13 Position at STOP

Range: 0 Default: 0

Lower Limit

In Stop state of scale, control magnitude can either be set to 0 or to lower limit O 10.

O 14 Start-Up

Range: 0.0...2.0 Uml Default: 0.0

Volumetric start-up cycle after start of scale. After set number of belt circuits, scale goes to Volumetric (uncontrolled) mode.

“v” flashes on display

O 15 Clearance

Range: 0.0...2 Uml Default: 0.0

Volumetric clearance cycle after stop of scale. Material prefeeder immediately cuts off. After set number of belt circuits, belt stops.

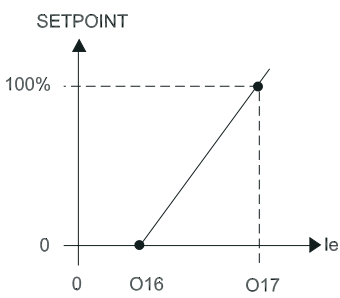
Clearance mode should be used only if prefeeder is controlled by the INTECONT.

“v” flashes on display

O 16 Zero Setpoint

Range: 0...200.0 mA Default: 0 mA

Input value Ie for 0% setpoints. Values smaller than 0 % are limited to 0 %.



You can also enter a voltage setpoint (0...10V). 10V corresponds to 20mA current.

|      |                |
|------|----------------|
| O 17 | Setpoint Range |
|------|----------------|

Range:    0...200.0 mA                      Default: 20.00 mA

Analog input value Ie for 100% setpoint (nominal value B 02).

For details, see Parameter O 16.

The parameters below can be used only if O 01 is set to UNIVERS.

O 01 = STANDARD resets Parameters O18-O26 to default.

## O18 Store

Range: NO YES YES-A  
Default: NO

Stores last controller value after scale stop.

Value is used for next start.

NO : Reject  
YES : Store  
YES-A : Store only if scale has not failed due to Alarm.

Alarms in Stop state of scale do not influence stored value.

## O 19 Volumetric Mode

Range: Qconst Yconst  
Default: Qconst

Selects Volumetric mode

Qconst :  $V \cdot Qconst$  acts as actual value.  
Scale operates speed-controlled.  
 $V$  = current belt speed

Qconst = stored value, nominal belt load, or last measurement value

Yconst : Controller cut off, only setpoint bypass O20 is active.

For more information, see DETAILS chapter.

## O20 Bypass

Range: 0...20.00 mA  
Default: 0 mA

Proportional to preset setpoint, the control magnitude is additively superimposed by a value.  
Parameter indicates bypass value for nominal setpoint.

## O 21 Setpoint Filt. T1

Range: 0...6000.0 s  
Default: 0.0 s

First-order filter in setpoint branch; does not act on bypass.

For more information, see DETAILS chapter.

## O 22 Setpoint Filt. T2

Range: 0...6000.0 s  
Default: 0.0 s

Same as O21, but 2nd-order filter.

## O 23 Set/Act Compar.

Range: W-X X-W  
Default: W-X

W-X : Controller receives SET/ACT deviation.  
X-W : Controller receives ACT/SET deviation.

O24

Set/Act Sources

Range:

I  
Q

Default: I

I

:

Setpoint =

Feed rate in accordance  
with Parameter B 07

Act. val. =

Feed rate measured

Q

:

Setpoint =

Nominal belt load Qo

Act. val. =

Belt load measured

“Q” does not allow any of the sources specified by B 07 to be used for setpoint input.  
Upper display read “Counter Z1”.

O 25

Adaption 1

Range:

NO  
V  
I/Q  
1/W

Default: NO

Adaption on controller input (see DETAILS chapter).

O 26

Adaption 2

Range:

NO  
W

Default: NO

Controller type FEED: O 26 not accessible;  
corresponds to O 26 = OFF

Adaption on controller output (see DETAILS chapter).

For control diagram, see DETAILS chapter.

## Block P: Linearization

### P 01 Linearization ON

Range: NO YES Default: NO

This function lets you linearize belt load measurement.

The 4 linearization points can be acquired through calibration with check weights or check using material. Place points not needed, or measured, outside the measuring range or interpolate. For calibrating details, see Appendix.

### P 02 Lin-S1

Range: 0.01...1000.00%Q Default: 25%Q

Reference: Nominal belt load (Parameter D01)

Linearization point 1: Actual belt load, e.g. check weight or result of material measurement

### P 03 Lin-I1

Range: 0.01...1000.00% Urwert: 25%Q

Reference: Nominal belt load (Parameter D01)

Linearization point 1: Belt load measured by INTECONT

### P 04 Lin-S2

Range: 0.01...1000.00%Q Urwert: 50%Q

Linearization point 2: see P 02

### P 05 Lin-I2

Range: 0.01...1000.00%Q Default: 50%Q

Linearization point 2: see P 03

### P 06 Lin\_S3

Range: 0.01...1000.00%%Q Default: 75%Q

Linearization point 3: see P 02

### P 07 Lin\_I3

Range: 0.01...1000.00%Q Default: 75%Q

Linearization point 3: see P 03

### P 08 Lin-S4

Range: 0.01...1000.00%Q Default: 100%Q

Linearization point 4: see P 02

### P 09 Lin-I4

Range: 0.01...1000.00%Q Default: 100%Q

Linearization point 4: see P 03

### P 10 Lineariz. Error

**B9**

Range: WARNING 1 WARNING 2 ALARM IGNORE Default: WARNING 2

Monitoring for monotonously rising compensation takes place only if linearization is active.

1. Upon start of linearization
2. Upon change to parameter

Block Q:Digital Inputs

Q 01Release

Range: DIDefault: DI (DI5)  
—

Determines digital input used to release scale.

Setting “- - ” means that no release is required.

Default is DI5; however, any other free digital input can be used as well.

Q 02DI Stop Batch

Range: DIDefault: —  
—

Digital input signal used to stop a running batch.

Q 03DI Ackn. Events

Range: DIDefault: DI (DI1)  
—

Determines digital input used to acknowledge events

Q 04DI Volumetric ON

Range: DIDefault: - -  
- -

Digital input used to set scale to Volumetric mode

Q 05DI Volum. Sync.

Range: DIDefault: - -  
—

Digital input used to set scale to Volumetric Synchronous mode

## Block R:

## Digital Outputs

### R 01 DO Alarm

Range: DO — Default: DO (DO3)

Digital output used to report general alarm

### R 02 DO Prefeeder ON

Range: DO — Default: DO (DO6)

Digital output used to START/STOP prefeeder

### R 03 DO Feeder ON

Range: DO — Default: DO (DO7)

Digital output signal used to display START/STOP states of scale

This very output can also be used to START/STOP drive.

### R 04 DO I-MIN

Range: DO — Default: Do (DO1)

Digital output signal used to report that minimum feed rate (F01) is exceeded down. Contact output is independent of event class (F02).

### R 05 DO I-MAX

Range: DO — Default: DO (DO2)

Digital output signal used to report that maximum feed rate (F03) is exceeded.  
Contact output is independent of event class (F04).

### R 06 DO Q-MIN

Range: DO — Default: —

Digital output signal used to report that minimum belt load (F05) is exceeded down.  
Contact output is independent of event class (F06).

### R 07 DO Q-MAX

Range: DIO — Default: —

Digital output signal used to report that maximum belt load is exceeded up (F07).  
Contact output is independent of event class (F08).

### R 08 DO v-MIN

Range: DO — Default: —

Digital output signal used to report that minimum belt speed (F09) is exceeded down.  
Contact output is independent of event class (F10).

**R 09 DO v-MAX**

Range: DO Default: —  
—

Digital output signal used to report that maximum belt speed (F11) is exceeded.  
Contact output is independent of event class (F12).

**R 10 DO Batch Active**

Range: DO Default: —  
—

Digital output signal used to report that a batch is running.  
Signal becomes active upon start of batch, and is reset after batch complete. If batch is stopped, signal continues being available.

**R 11 DO Full Feed**

Range: DO Default: —  
—

Full feed signal is available from start of batch until coast-down time I01 has elapsed. Then batching continues in dribble feed.

**R 12 DO Dribble Feed**

Range: DO Default: —  
—

Dribble feed signal is output when full feed phase is complete (see R11).

**R 13 DO Feeder ON**

Range: DO Default: DO (DO4)  
--

Digital output signal used to report START/STOP states of scale

**R14 DO Deviation**

Range: DO Default: DO (DO5)  
--

Digital output signal used to report that maximum deviation (O06) is reached.  
Contact output is independent of event class.

**R 15 DO Volum. Mode**

Range: DO Default: --  
--

Digital output signal used to report that scale has gone to Volumetric mode.



## Block S:

## Comm. EasyServe

### S 01 Station Address

Range: 1...254 Default: 1

Make sure station address and baud rate of scale and EasyServe are identical.

EasyServe lets you search for proper setting including COM interface (menu: Tools/Options/Communication). The station address is specially designed for plants using a service bus. If the latter is not available, select 1.

### S 02 Physics

Range: RS 232 Default: RS 232  
RS 422  
RS 485

Physics of data transmission protocol

### S 03 EasyServe Baudr.

Range: 9600 Default: 19200  
19200

EasyServe baud rate

### S 04 Format Data

Range: 8-0-1 Default: 8-0-1  
8-E-1  
8-N-2  
8-N-1

EasyServe uses either a 10- or 11-bit character frame.

Example:

8-0-1 means:

1 start bit, 8 data bits, odd parity, 1 stop bit  
(N = No Parity, E = Even Parity)

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## 10. Event Messages

### System Messages S

#### **S1 : Memory Error**

Program and parameter memories are checked in cycles. If an error is detected, scale is normally inoperable.

Action : Contact SCHENCK service.

Parameter : N02

#### **S2 : No Release**

No external RELEASE signal.  
Scale cannot start.

Parameter : N 07

#### **S3 : Maint. STR-Meter**

Total ON-times of conveyor belt and measurement exceed set limit.

Perform requisite service work, if necessary.

Acknowledgement of message does not influence time intervals.

Parameters : K 03, K 04

#### **S4 : Maintenance Int. Elec**

The INTECONT power supply has been switched on for a set time.

Perform requisite service work, if necessary.

Acknowledgement of message does not influence time intervals.

Parameter : K 01, K 02

#### **S7 : Simulation Active**

Operating in Simulation mode, scale is not prepared for normal operation.

The Simulation mode lets you test all scale functions without material at the time of commissioning.

Action : Deselect "Simulation" function.

Parameter : N10

#### **S8 : Setpoint Limited**

Specified setpoint excessive. In Gravimetric mode, limitation starts with values higher than nominal feed rate; in Volumetric mode, with values higher than 3 \* nominal feed rate .

Parameter : N11

#### **S9 : Data Link Host**

Serial communication interrupted for longer than timeout period.

Action : Check cable connections.

Parameter : L 03

Material Flow B (Batching)**B1 : Out of Tolerance**

Batch not completed with preset accuracy.

Tolerance : Parameter I 04

Event : Parameter I 05

Cause : Automatic adaption  
connected  
(Parameter I 03)?  
Material infeed faulty ?

**B2 : Control Deviation**

Actual feed rate heavily differs from  
setpoint for longer than specified time.

There is not single solution:

1. Material handling error  
If belt load is excessive, drive control  
range can be exceeded. Normally,  
further event messages are available  
(H2, H3, L2, L3, H6...).

Action:

Check control magnitude (service value Y)  
At nominal feed rate, value should amount  
to 10...14 mA.

If necessary, correct material infeed.

Heavily varying belt loads cannot be  
precisely corrected.

Action : Improve material infeed; if  
need be, increase  
deviation monitoring  
tolerance.

2. Electrical system fault.  
Check feed rate controller and motor  
cables.

Check feed rate controller for proper  
setting (current limitation, speed  
range).

3. Calibration Fault  
Upon initial calibration, controller was  
set so tight that the slightest change to  
system properties can make system  
operation unreliable.

Action : Repeat this part of  
commissioning.

Parameter : O08

**B3 : Controller Limited**

Controller output excessive. After some  
time, message B2 (Deviation) is  
additionally output.

1. Belt load too small
2. External feed rate controller improperly  
set

Action : Check belt load and/or  
control magnitude (service  
value Y). At nominal feed  
rate, Y should be 10...14  
mA.

Parameter : O09

**B9 : Linearization**

Linearization for belt load measurement  
improperly set.

Message is output only if linearization  
function is active.

Action: Check relevant parameters  
to see if preset curve  
features plateaus or  
turning points.

Parameter : P 10

## Electrical System E

### **E1 : Power Failure**

Power failed or cut off.  
It is possible that totalization has stopped during this time.  
Action : Acknowledge message  
Parameter : N 01

### **E2 : Namur Err Tacho**

Short circuit or breakage in speed transducer cable. Scale is inoperable.  
Action : Check speed transducer cable.  
Deselect speed measurement for a short time using Parameter B 03.  
Parameter : N 04

### **E3 : Namur Err Belt Sensor**

Short circuit or breakage in belt sensor cable. (Belt sensor is available only on scales using automatic belt influence compensation BIC.)  
Action : Check sensor cable.  
Parameter : N 05

### **E5 : Stand-By**

Scale is in Stand-by mode, i.e. setpoint is smaller than set limit value. Belt drive and material prefeeder are cut off.  
Parameter : F14

## Calibration C

### **C1 : L/C Input**

1. Load cell cable broken, not (or improperly) connected  
2. Supply voltage too low  
Action : Check cabling.  
If OK, check load cell amplifier.  
Parameter : N 06

### **C2 : Tacho Input**

Speed transducer output frequency exceeds 2700 Hz, most probably due to subsequent change of belt speed (e.g. gearbox).

Action : Check speed transducer for proper dimensioning. If need be, measure pulse frequency using oscilloscope.  
Parameter : N 03

### **C3 : Belt Skew**

Conveyor belt runs too far off-centre. Scale is no longer operable.  
Action: See C4  
Parameters: M11, M12

### **C4 : Belt Drift**

Belt has left specified tolerance zone.  
Action: 1. Clean head and tail pulleys.  
2. Realign belt.  
3. Watch service values of belt circuit sensor.  
Parameters: M09, M10

**C8 : Slip Error**

Conveyor belt length has excessively changed, or slip between speed transducer and conveyor belt.

Error is acquired only if an additional belt circuit sensor is present.

Action : Check belt tension; if need be, correct.  
Check sensor for proper setting.

Parameters : M13, M14

If events C 8, C3 and C4 occur at the same time, belt circuit sensor is located outside sensor area.

All 3 causes are possible.

Maximum H**H1 : I > MAX**

Current feed rate exceeds set limit value.

Action : Normally none, unless message H4 is additionally available or system-specific limits have to be observed.

Parameters : F 03, F 04

**H2 : Load > MAX**

Current belt load exceeds set limit value.

Normally, weighfeeder is designed for +33% maximum belt load variation.

This is critical only with small setpoints (minimum admissible drive speed possibly exceeded down).

Action : Normally none, unless message H4 is additionally available or system-specific limits have to be observed.

Parameters : F 07, F 08

**H3 : v > MAX**

Current belt speed exceeds set limit value.

Action : System-specific unless C2 is additionally available.

Parameters : F 11, F 12

**H4 : L/C Input > MAX**

Scale overloaded. Errors in measurement can occur.

Action : Check material infeed (belt load excessive).

Parameter : N 08

Minimum L**L1 : I < MIN**

Current feed rate has fallen below set limit value. Error in measurement possibly out of tolerance.

Action : System-specific unless L4 is additionally available.

Parameters : F 01, F 02

**L2 : Load < MIN**

Current belt load has fallen below set limit value.

Normally, weighfeeder is designed for +33% minimum belt load variation.

This is critical only with small setpoints (maximum admissible drive speed possibly exceeded down).

Action : see L1

Parameters : F 05, F 06

## L3 : V < MIN

Current belt speed has fallen below set limit value.

Action : Check to see if belt is running.

Parameters : F 09, F 10

## L4 : L/C Input < MIN

Scale underloaded; errors in measurement possible.

Action : Check mechanical weighing system, weighed idler alignment, and load cell cable.

Parameter : O 09

## Signal Lamps

•=== (green) POWER OK  
LED not lit:  
1. No power supply  
2. Device defective  
3. All other functions OK, LED probably defective. Call display test.

•○ (green) CPU OK  
If display is pending or flashing, processor system is faulty, Scale is inoperable.

⌋ (red) ALARM  
flashes if an Alarm is available. Display additionally shows event code.

Action:  
see relevant event code.

MIN (red) MIN LIMIT VALUE  
lit if feed rate limit value is exceeded down.

Action:  
see L1: I<MIN

MAX (red) MAX LIMIT VALUE  
lit if feed rate limit value is exceeded up.

Action:  
see H1: I>MAX.

## No power supply:

Upon short circuit on the secondary side of the power supply, internal overload protection responds. Disconnect power for approx. 1 min., and it is ready to operate again.

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## 11. Commissioning

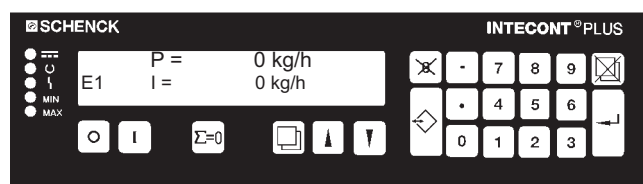
### How To Operate Your INTECONT

There is no way out.

Who wants to put INTECONT into service has to work himself through the Operating manual.

To make learning easier, INTECONT can be operated without mechanical equipment, load cells and speed transducer. Although no feed rate is displayed, all operating functions can be performed.

1. Connect power supply and turn on (see DETAILS chapter).
2. The upper two green LED report "Ready to Operate", the red LED in the middle flashes and reports Alarm E1.



Upper display: Totalizing counter Z1

Lower display: Event message E1 "Power Failure"  
Feed rate I



Acknowledge message E1.  
Display shows event message E2  
reporting that no speed transducer is  
connected.

3. Save "Feeder ON" and zero setting, all functions described in Operating Manual can be performed, e.g. "Display Events".



Call function distributor.



Select "Display Events" function.



Use cursor keys to view all available events.



1. Namur Err. Tacho E2  
Speed transducer not connected, or  
cabling faulty.
2. L/C Input C1  
Load cell not connected, or cabling  
faulty.
3. L/C Input < MIN L4
4. No Release S2  
No external Release signal

4. If you wish to turn off the flashing signal lamp or try the setting programs, change some parameters as described in PARAMETERIZATION chapter.

- (a) Set Parameter N05 to IGNORE E2
- (b) Set Parameter N06 to WARNING 1 C1
- (c) Set Parameter N09 to WARNING 1 L4

The residual parameter let you change the class of the 4 possible events.

5. Set "Release" contact (Parameter Q01).  
Scale can now be started. However, no setting program is available and no actual value can be acquired.
6. Set Parameter B03 to NO.  
B03 lets you deactivate belt speed measurement.
7. Select Simulation Mode.

Control circuit internally closes, and actual value is controlled for setpoint. Belt speed display "v" indicates nominal value; belt load display "Q" any value. To be able to check belt load display, simply use the test plug described at the end of manual.

8. Now you can run all setting programs.  
As described in Chapter 6, call LB program first and  
acquire result using



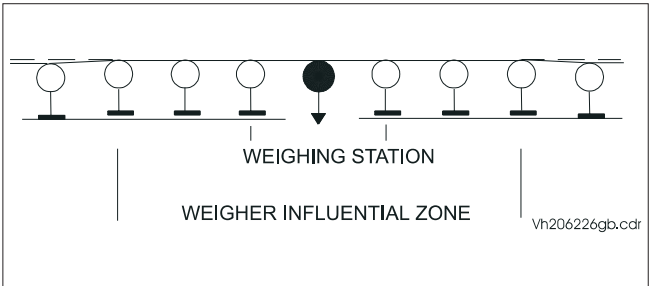
key.

**Remember:**

- 1. Set Parameters N05, N06, N09, B03 to default values.
- 2. Deselect Simulation mode.

**Mechanical Prerequisites**

Mount mechanical equipment in accordance with the instructions given in separate manual. Check weighed and carrying idlers in the weigher influential zone for correct alignment and out-of-round.

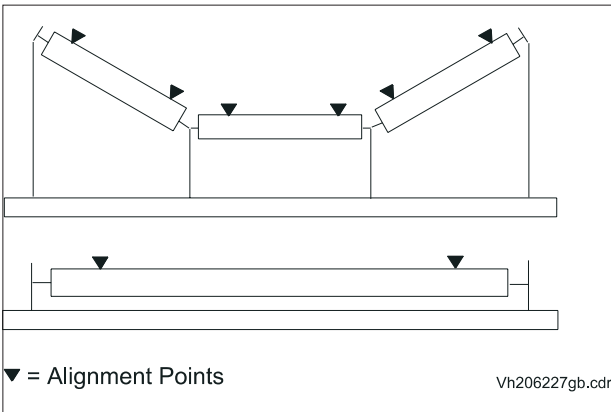


**Weigher influential zone**

All weighed idlers and three carrying idlers upstream and downstream of weighing station.

**Alignment**

Raise idlers in the influential zone to approx. 3...5 mm above remaining idlers. Align then against one another to a height difference below 0.4 mm.



Depending on idler set equipment, idlers have to be aligned in various points.

## Out-of-round

Ensure that the out-of-round error of idlers in the weigher influential zone do not exceed 0.4 mm.

## Securement

After alignment, secure idler sets against displacement.  
Mark idlers and sets to avoid mistakes during maintenance.

## Influences

Incorrect alignment causes a zero point error normally compensated during calibration.  
However, changes in belt tension increase the error size.  
For this reason, ensure correct alignment and constant belt tension for accurate weighing.

## Maintenance

If idlers are dismantled for maintenance, make sure they are remounted in their former locations.

## Electrical Prerequisites

Run and connect cables as described in Chapter DETAILS.

## Enter Parameters

All important rated and calibration data are shown in the supplied Spec Sheet.  
In addition, see hints given below. We recommend to read through all parameters and to select the best possible setting.

### Parameter Block A:

Dialog language and display unit.  
Parameter values are automatically converted if the unit is changed.

### Parameter Block B:

Characteristic value vs B 04 is important for calibration. Adjust units to nominal feed rate. To ensure stable display, resolution of feed rate display (B 01) should not exceed 4...5 decimals.  
Display format of totalizing counters should not exceed 6...7 digits. The decimal point is one digit.

Example:

|                   |           |
|-------------------|-----------|
| Nominal feed rate | 10 t/h    |
| Unit B 10         | —.— t     |
| Max. value        | 999.999 t |

$$\text{Max. counter run time} = \frac{999,999 \text{ t}}{10 \text{ t/h}} = 99,9 \text{ h}$$

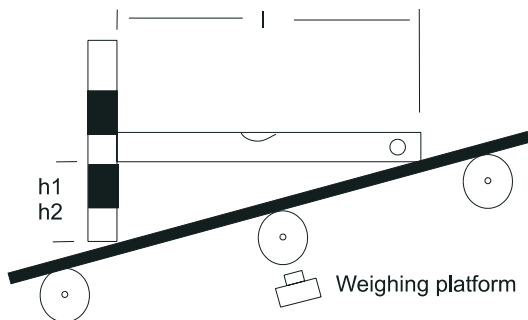
Then counter is reset.

If an external totalizing counter is connected, output frequency must not exceed 10 Hz (see Parameters B 07, 08 and Service Displays Z0 and ZE).

### Parameter Block C:

Ensure correct input of calibration and evaluation data as shown in relevant Spec Sheet. Measure effective belt length and inclination at site.  
For definition of belt length, see Chapter DETAILS.

Enter belt inclination only if load cells are arranged right-angled to belt. If load cells are mounted vertically, the effective angle is 0. For definition of check weight, see Data Flow Diagrams in Chapter DETAILS.



Determine belt inclination by using spirit level and measuring element. Repeat measurement with spirit level turned and calculate mean value of height h.

Angle C 07 is then  $a = \tan (h/l)$

For angles  $< 20^\circ$ :  $a = 55 h/l$

## Parameter Block D:

Calibration results need not be entered. During initial calibration, range correction D 02 should be set to 1.

Check to see that, with rated data entered, load cell rated capacity is not exceeded.

$$F = q * L * H * \cos(a)$$

|   |   |                      |      |            |
|---|---|----------------------|------|------------|
| q | = | Nominal Belt Load    | D 01 | in kg/m    |
| L | = | Eff. Platform Length | C 05 | in m       |
| H | = | Lever Ratio          | C 06 |            |
| a | = | Angle                | C 07 | in degrees |
| F | = | Force on load cell   |      | in kg      |

F should be the total of load cell rated capacities C 04. If not, load cells are too small for the requested nominal feed rate.

## Parameter Block F:

Select limit values and event classes in accordance with application requirements.

## Parameter Block G:

Adjust display filters, if required. Normally, the default values can be used.

## Parameter Block H:

See "Automatic Zero Setting". Maintain default values for the time being. Do not start automatic zero setting before calibration is effected.

## Parameter Block K:

For calibration, the maintenance parameters are irrelevant. The default values are useful suggestions.

## Parameter Block L:

Description of fieldbus interfaces.

## Parameter Block M:

Normally, the control for point of discharge (CDP) should be activated always (M 01, M 02). For automatic belt influence compensation BIC and slip indication, a special belt circuit sensor and a metal sensor area on belt are required. If these are not available, set Parameter M 03 to NO. Perform initial commissioning without BIC.

## Parameter Block N:

The classes of event messages are useful suggestions for a standard scale. If you have changed messages N05, N06 and N09 for learning, remember reset to default after commissioning.

## Parameter Block O:

Parameter O 01 lets you select between standard and universal weighfeeder controller.

- O 01 = STANDARD: Weighfeeder with belt speed control. Parameters O 18...O 25 are no longer available.
- O 01 = UNIVERS: Belt weighers with belt load control, or control for constant belt load. See DETAILS chapter, Applications item.

## Control

1. Set external feed rate controller.
  - a) Set setpoint ramp, if any, to smallest possible value.
  - b) I-R compensation for 0, if any.
  - c) Set current limitation to approx. 2 times the motor rated current.
  - d) Set controller parameters O02 and O03 (KP, TN) to 0.  
Set Parameter O12 (Elevation) to 10...15 mA.  
If need be, control magnitude can also be entered from a local control unit.
  - e) Set external controller amplification so that weighfeeder nominal speed results. At a 20mA control magnitude, speed should not exceed motor rated speed.
  - f) Enter setpoint steps into Parameter O12 and optimize external controller behaviour.
  - g) Select elevation so that motor just stops turning (normally 0 mA).
  
2. Set internal controller
  - a) Reset controller parameters O02 and O03 to default and select Volumetric Mode.
  - b) Enter setpoint steps; if need be, optimize controller parameters.  
Default values normally slow down the transient response of the speed.
  - d) Check settings in Gravimetric mode using check weights.

The setting instructions are tailored to weighfeeders but analogously apply to other systems as well.

## Functional Check

1. Start scale.  
Starting depends on selected start source B 06.
2. Interpret event messages, if any, in accordance with Chapter 10 and remedy fault.
3. Call service messages.

### a) Tacho

If measurement is ON (B 03 = DI7), speed transducer output frequency must be below 2500 Hz. Compare value with Spec Sheet. If frequency heavily varies at constant belt speed, there is too much space between speed transducer NAMUR sensor and gearing. See also "Check of Belt Speed".

### b) wz

The unnormalized output value of load cell amplifier must not exceed 2.85mV/V, not even with nominal platform load. Value changes with varying load on platform (e.g. check weight).

Value excessive:

- Load cell overloaded
- Tare compensation weight (if any) improperly set

Value negative:

- Load cell improperly connected
- Tare compensation weight (if any) improperly set

### c) aw

Load on load cell in percent related to the total of load cell rated capacities. Interpret as under Item b). At nominal belt load, aw should not exceed 100%.

4. If all else fails, see "Test Plug" item.

## Calibration

Calibration as such is not required. After input of rated and calibration data, scale is calibrated.

Simply call:

1. Setting program "Belt Circuit LB" determining the basis of one belt circuit for zeroing and taring programs
2. Taring program TW
3. Zero setting program for training. Zero point error should be small after taring. .

Always observe the above order of sequence.

For first program, measure exact time of one belt circuit and enter value into Parameter C 02. For operation and message interpretation, see Chapter 3.

Two further checks should be effected:

1. Check using check weight
2. Check of belt speed



If something goes wrong:

our service department is always prepared to help. Maybe only some small error or maloperation is the cause of trouble. Check individual operations one by one. Normally, the issue can be eliminated.

## Check Using Check Weight

A check using check weight is useful because

1. mechanical or electrical faults can be given
2. calibration data may be uncertain, e.g. lever ratio not exactly known, inclination hard to measure
3. incorrect data can have been entered.

The check weight should range between 30 %...100 % of nominal platform load and be exactly known.

1. Enter effective check weight into Parameter C 08. If weighing modules with leaf spring parallel guidance are used, the check weight corresponds to the actual value.  
In other cases, see Spec Sheet or calculate check weight in accordance with Chapter DETAILS / Data Flow Diagrams. For effective check weight, belt inclination is irrelevant.
2. Call program "Weight Check CW", as described in Chapter "Setting Programs".

For evaluation of results, see operating sequence description.

### Possible causes of faults:

Distorsion of platform, insufficient alignment, interferences on load cell cable.

In case of suspicion, repeat check with another weight.

Nonlinearity suggests the faults above; linear errors are most probably due to incorrect data.

## Check Belt Speed

The speed transducer frequency generator can be driven by

1. friction wheel in the lower belt run
2. drive motor shaft

In the first case, the entered characteristic value vs (Parameter B 04) depends on the exact friction wheel diameter; in the second, on the belt underslung factor. Therefore check speed measurement for accuracy.

1. Acquire belt speed with stop watch and tape measure.  
To be on the safe side, conduct various measurements and calculate mean value.
2. Read INTECONT speed display over the same period of time.
3. Compare both values with one another and, if necessary, correct characteristic value vs B 04.

$$B04(\text{new}) = B04(\text{old}) \frac{V_g}{V_a}$$

old = old parameter value  
new = new parameter value  
V<sub>g</sub> = speed measured  
V<sub>a</sub> = speed read off

This method is also applicable with unknown characteristic values.

If characteristic value vs is known and deviations are excessive, the space between sensor (Namur transducer) and gearing is too big. Check by switching ammeter into sensor circuit.

|                   |     |   |               |
|-------------------|-----|---|---------------|
| Tooth over sensor | : i | = | 0.1...1.65 mA |
| else              | : i | = | 1.65...8 mA   |

Neither level should exceed limit value 1.65 mA.

## Check Using Material

Checks or calibrations with check weights cannot reproduce the actual circumstances to perfection. Highly accurate weighing results can be achieved only by various measurements with material and subsequent correction.

For correction, use Parameter D 02.

Always observe the following items:

1. Make sure the route from platform to material collecting point is clean.
2. Ensure that material diverters, if any, do not divert any material.
3. If feed screws or air slides are used between platform and collecting point, feeding starts approx. 30 min. before check measurement so that normal build-ups are given.
4. Set dedustings to a minimum.
5. Ensure that hoppers or vehicles used for transport of material to a legal-for-trade weigher are cleaned and weighed before every filling operation (tare).

Example:

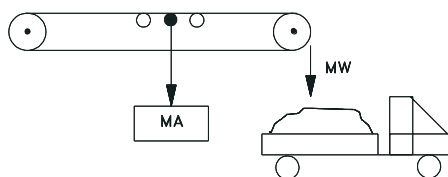
Old value of Parameter D 02 = 1.

Within 15 min. a material amount of MW = 4.9 t is fed. Difference in counter reading MA read on INTECONT is 5.0 t.

Enter computed value into Parameter D 02 .

$$D02(\text{new}) = D02(\text{old}) \cdot \frac{MW}{MA}$$

$$= 1 \cdot \frac{4.9}{5.0} = 0.98$$



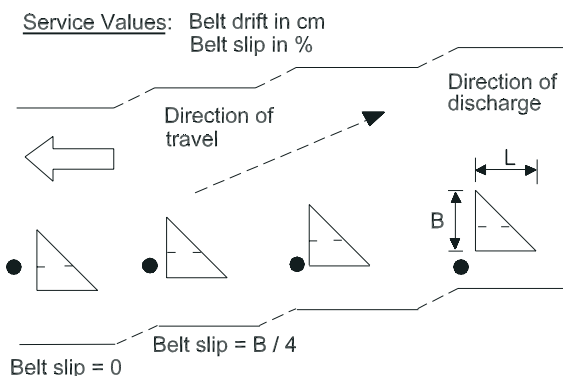
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## Belt Run Monitoring

INTECONT comes with belt a belt run monitoring function designed to detect belt drift and belt slip. Activate monitoring using Parameter M03 "Belt Sensor Active".



The belt run monitoring functions can be used only if a belt circuit sensor (Namur transducer) and a triangle metal sensor area on belt area are available (see Fig. below). If not, set Parameters M03 "Belt Sensor Active" and M05 "BIC Active" to "NO".



To activate belt run monitoring:

1. Set Parameter M03 "Belt Sensor Active" to "YES".
2. Enter sensor length L into Parameter M06 "Sensor Length" (s. Belt Run Monitoring figure).
3. Enter sensor width B into Parameter M07 "Sensor Width" (s. Belt Run Monitoring figure).
4. Set Parameter M08 "Sensor Offset" to 0.
5. Call calibrating function "LB: IMP/Belt".
6. Acquire result after min. 2 belt circuits.
7. Adjust sensor.

**Note:** Service value "Imp.S" indicates the number of pulses measured for the covered sensor. If the belt sensor is located right in mid sensor area, the following value results:

0329

$\text{Imp.S.} = 0.5 * L (M06) * 0.01 * \text{vs\_Charact. Val (B04)}$

**Example:** M06 = 8.40 cm and  
B04 = 1000 l/m result in Imp.S. = 42

**Note:**

Monitor service value "Belt Drift Tr" indicating the deviation of mid sensor (and belt) from sensor position in cm.

**Example:**

Belt drift = 0.20 cm: Parameter M08 "Sensor Offset" = - 0.20 cm indicates belt slip 0.00 cm.

8. If necessary, determine event classes of Belt Slip and Belt Drift messages (Parameters M09...M17).

**Note:**

Normally, the default values can be used in actual practice.

9. Check messages.

**Note:**

To check, adapt Parameter M08 "Sensor Offset" accordingly and remember to reset value to default after check.

## Belt Influence Compensation BIC

The automatic Belt Influence Compensation BIC is designed to monitor and correct the belt influence. As is with belt run monitoring, a sensor and a sensor area are required. To commission BIC, first start belt run monitoring.

**Note:**

Perform initial commissioning without active BIC.

**Features:**

After start of the belt influence compensation, the influence of the conveyor belt on measurement will be compensated after approx. 10 belt circuits.

**Conditions:**

- Belt run monitoring commissioned
- Belt run monitoring started

**To start BIC:**

Set Parameter M05 "BIC Active" to "YES".

## Control For Point of Discharge (CDP)

Upon measurement at the point of discharge, the measuring point of the weighing platform is shifted to the point of material discharge using a delay element. The amount discharged is measured direct.

**Note:**

If CPD is active, Automatic Belt Influence Compensation BIC should be started as well.

**Conditions:**

- Length ratio from mid platform to material discharge point and total conveyor belt length is known.
- Length ratio in percent is entered into Parameter M02 "Platf.Dis.Length".

**To start CPD:**

Set Parameter M01 "CPD Active" to "YES".

## Linearization

Normally belt load needs not be linearized. Linearization makes good sense only with strong belt load variations and simple mechanical weighing systems.

### Calibration Using Check Weights:

1. Apply check weight Q1, start scale and call zero setting program.
2. Note result of zero setting program (lower display) and enter into Parameter Lin-I1 later. Abort program; do not overwrite result.

3. Enter into Parameter Lin-S1.

$$q1 = \frac{Q1}{L} \cdot \frac{100\%}{q0}$$

|    |   |                                          |         |
|----|---|------------------------------------------|---------|
| Q1 | = | Check weight                             | in kg   |
| L  | = | Eff. platform length<br>(Parameter C 06) | in m    |
| q0 | = | Nominal belt load<br>(Parameter D01)     | in kg/m |

4. Repeat steps 1...3 for residual linearization points gradually increasing check weight.

### Calibration Using Check With Material:

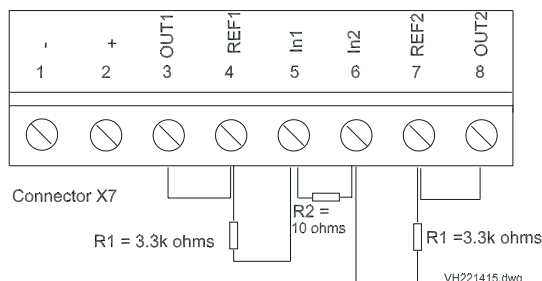
1. Perform check with material using belt load q1.
2. Read off mean value of belt load q1 (a) from INTECONT. Enter value into Lin-I1.
3. Enter into Parameter Lin-S1.

$$q1(s) = q1(a) \frac{Ms}{Ma}$$

|    |   |                                           |       |
|----|---|-------------------------------------------|-------|
| Ms | = | Material amount fed                       | in kg |
| Ma | = | Material amount read off<br>from INTECONT | in kg |

4. Repeat steps 1...3 for residual linearization points gradually increasing check weight.

## Test Plug



Almost any possible error in operation is reported by event message. For interpretation, see Chapter 10.

Check INTECONT for proper operation by using a simple test plug. This helps to see whether the fault is with load cell, cabling or electrical equipment.

1. Switch plug as shown in drawing.
2. Withdraw load cell connector X7 and replace by test plug.

At the indicated resistance values, the following service values result:

1. wz apr. 1.5mV/V
2. Load cell apr. 75 % (C03 =2mV/V)  
utilization aw apr. 52% (C03 =2.85mV/V)

If wz is in specified range, measuring amplifier operates properly.

Note:

With different R1 and R2 resistance values, the following raw measured value results:

Raw measured value ~ 500 \* R2 / R1 [mV/V].

12. Details

Setpoint and Start Sources

Sources

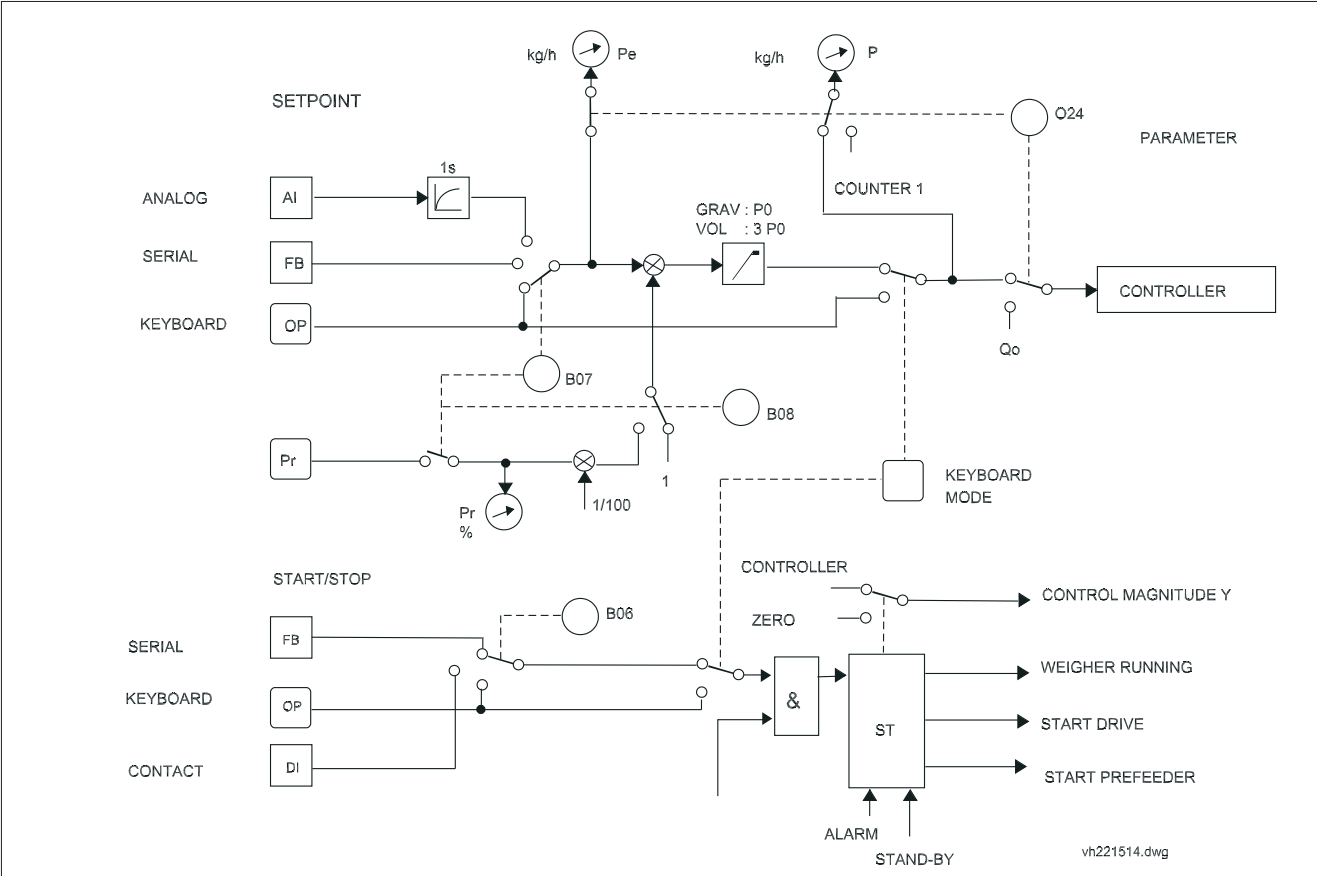


Fig.: Setpoint and Start Sources

The “Enter Setpoint” and “Start/Stop” functions can be started from different control sources preselected via parameters.  
The “Keyboard Mode” dialog function lets you shift from preselected sources to keyboard and vice versa.

Parameters:

|      |             |                                                     |
|------|-------------|-----------------------------------------------------|
| B 06 | START Souce | FB (fieldbus)<br>OP (keyboard)<br>DI (contact)      |
| B 07 | P-Source    | OP (keyboard)<br>FB (fieldbus)<br>AI (analog input) |

|      |                   |                                                                               |
|------|-------------------|-------------------------------------------------------------------------------|
| B 08 | Prel. Ext. Active | NO<br>YES                                                                     |
| O24  | Set/Act Sources   | I = feed rate<br>Q = belt load<br>(normally in the position shown in drawing) |

Setpoint displays:

|    |                                                    |
|----|----------------------------------------------------|
| P  | Effective setpoint                                 |
| Pe | External setpoint<br>(only if preset with B 07)    |
| Pr | Percental evaluation<br>(only if preset with B 08) |

In Keyboard Mode, activated Pe and PR displays remain stored.

## Setpoints Keyed In:

- P Absolute setpoint  
(only if no external setpoint is preselected)
- Pr Percental setpoint  
(only if B 08 = YES and analog or serial input is preselected with B 07).

## Setpoint Synchronization:

Upon changeover from external setpoint source or keyboard, external setpoint is acquired. In reverse direction, the external source is active.

## Start/Stop Synchronization:

Upon changeover from external source to keyboard, Start/Stop status remains stored. In reverse direction, the external source is active.

## Power OFF:

Keyed in setpoints are stored.

## Control Inputs and Outputs

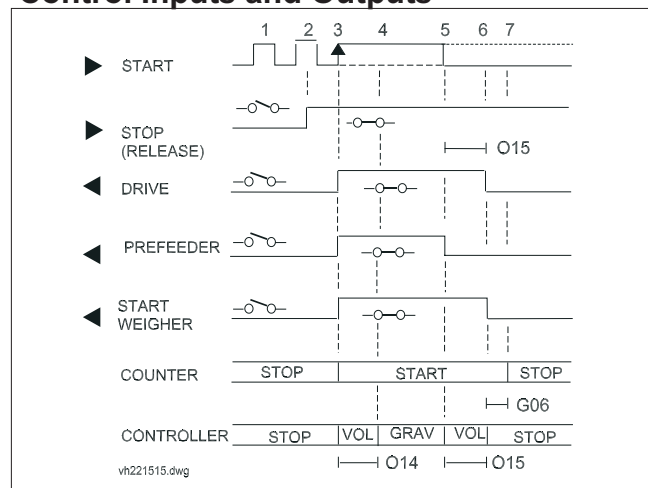


Fig.: Control Inputs and Outputs

The Start/Stop and Release signals are designed as inputs; the residual signals, as outputs or status features.

## Parameters:

- O14 Start-up in belt circuits
- O15 Clearance in belt circuits
- G 06 Afterfl. Time in s

In many cases, O14 and O15 are set to 0.

Operating sequence:

- 1 As long as no Release signal is available (contact open), no start command is effective.
- 2 Release contact is available, scale is released.
- 3 L-H edge of Start input or Start command from another source starts scale.  
During start-up period 014, control operates in Volumetric mode.
- 4 Automatic changeover to Gravimetric mode takes place.
- 5 Cut-off command  
Prefeeder immediately stops, control for clearance period 015 goes to Volumetric mode.
- 6 Clearance is complete, scale stops.
- 7 Feed rate acquisition and totalization continue for time G06.

## Stand-By:

If setpoint falls below limit F15 for more than 3s, "Drive" and "Prefeeder" outputs go to OFF state. Controller control magnitude (Parameter 012) is set to lower value.

Controller does not operate; integral component can be deleted or remain stored (Parameter O18).

Measurement is still active, status display and "Start Feeder" output continue reporting ON.

If setpoint exceeds limit F15, scale immediately goes to normal mode.

## Alarm:

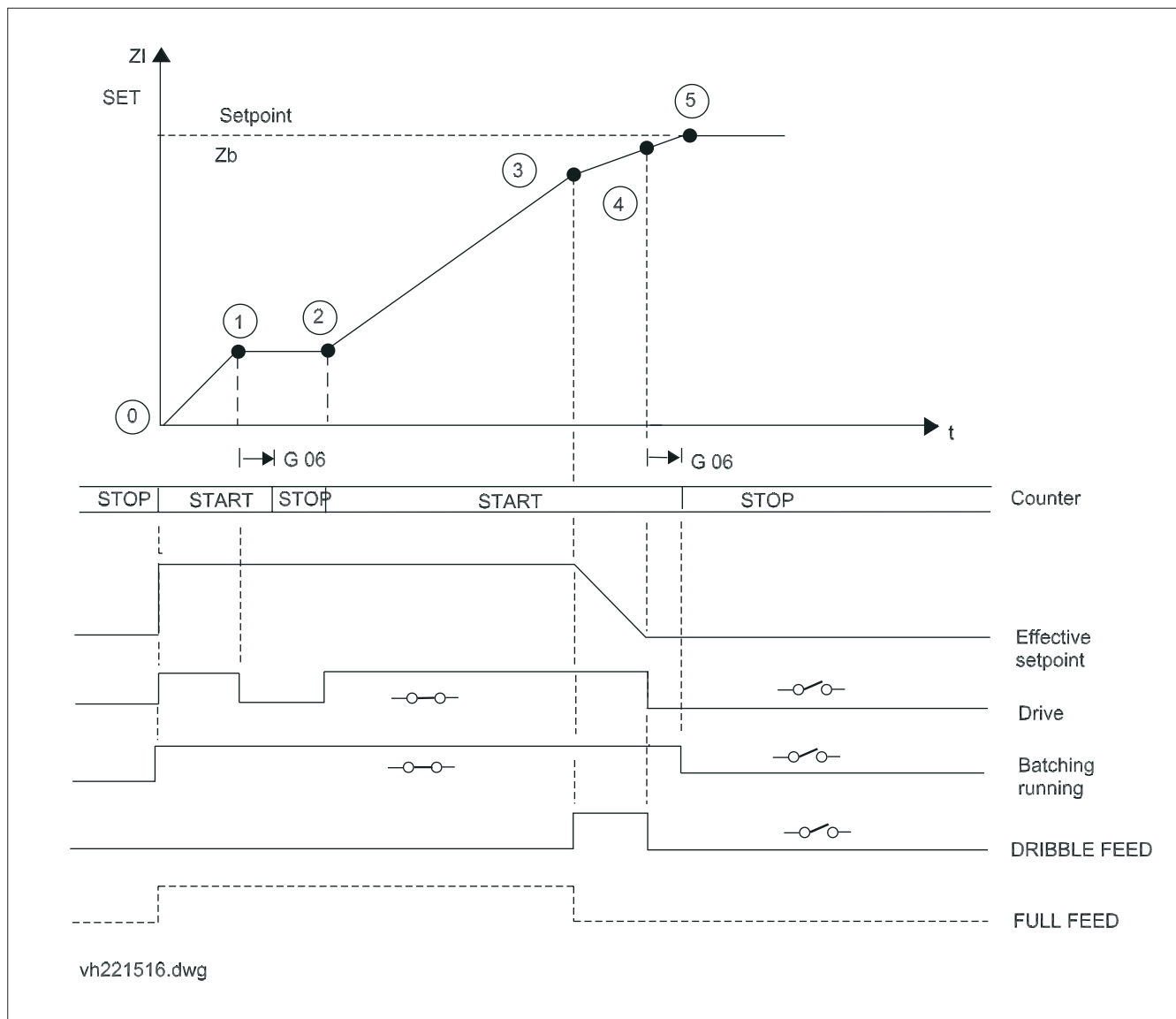
If scale cuts off after Alarm, Clearance mode is omitted. Counter tracking time G 06 is effective.

## Measurement and Displays:

Belt load and belt speed are measured both in Start and Stop state of scale; feed rate and totalized amount are acquired only in Start state and during afterflow time G 07.

The running display (rotating point) reports all phases during which feed rate acquisition and totalization are active.

## Batching



If required, the outputs signals (batching in full feed and dribble feed) can be activated in place of the MIN/MAX values.

ZI = Batch amount  
Zb = Batch setpoint  
I = Feed rate

Sequence:

- 0 Start batch using Start command.
- 1 Stop batch using Stop command.  
Totalizing counter remains active for time G06.
- 2 Continue batch using Start command.  
This is also possible during time G06.
- 3 If coast down time has elapsed (I01), feed rate setpoint moves towards zero. Full feed and dribble feed signals change level.

4 Cut-off point  
Zb correction amount (I02) is reached, batch is complete. Feed unit and prefeeder cut off.

5 Material afterflow.

New batch can be started only if afterflow time G06 has elapsed.

Adaption:

After every complete batch, INTECONT uses the missing material amount to compute a correction amount (I02). Switching points 3 and 4 automatically shift upon next batch.

$$K(\text{new}) = K(\text{old}) + \text{adap} * \text{error}$$

K(new) = New correction amount I02  
adap. = Adaption factor I03

Errors smaller than smallest display digit are not corrected.

DI Complete Batch:

The Complete Batch contact acts like a Stop command with additional abort of batch. Next Start command starts new batch.

Alarm:

Batch is stopped and can be resumed after removal of cause and acknowledgement of event message.

Power OFF:

Batch aborts, batching mode remains selected. All counter readings and amounts batched are stored for an unlimited period of time. Next Start command starts new batch.

Stand-By Mode:

During coast down time, scale can go to Stand-by mode. Relevant message is suppressed.

Clearance Mode:

Batching with clearance (Parameter O15 > 0) is useful only with certain restrictions. Control sequence slightly differs.

1. Prefeeder cuts off (Item 4) as shown in drawing.
2. Drive and dribble feed stop after set clearance period.
3. "Batch Active" message remains available for time G06 after stop of drive.

Data Flow Diagrams: Measurement

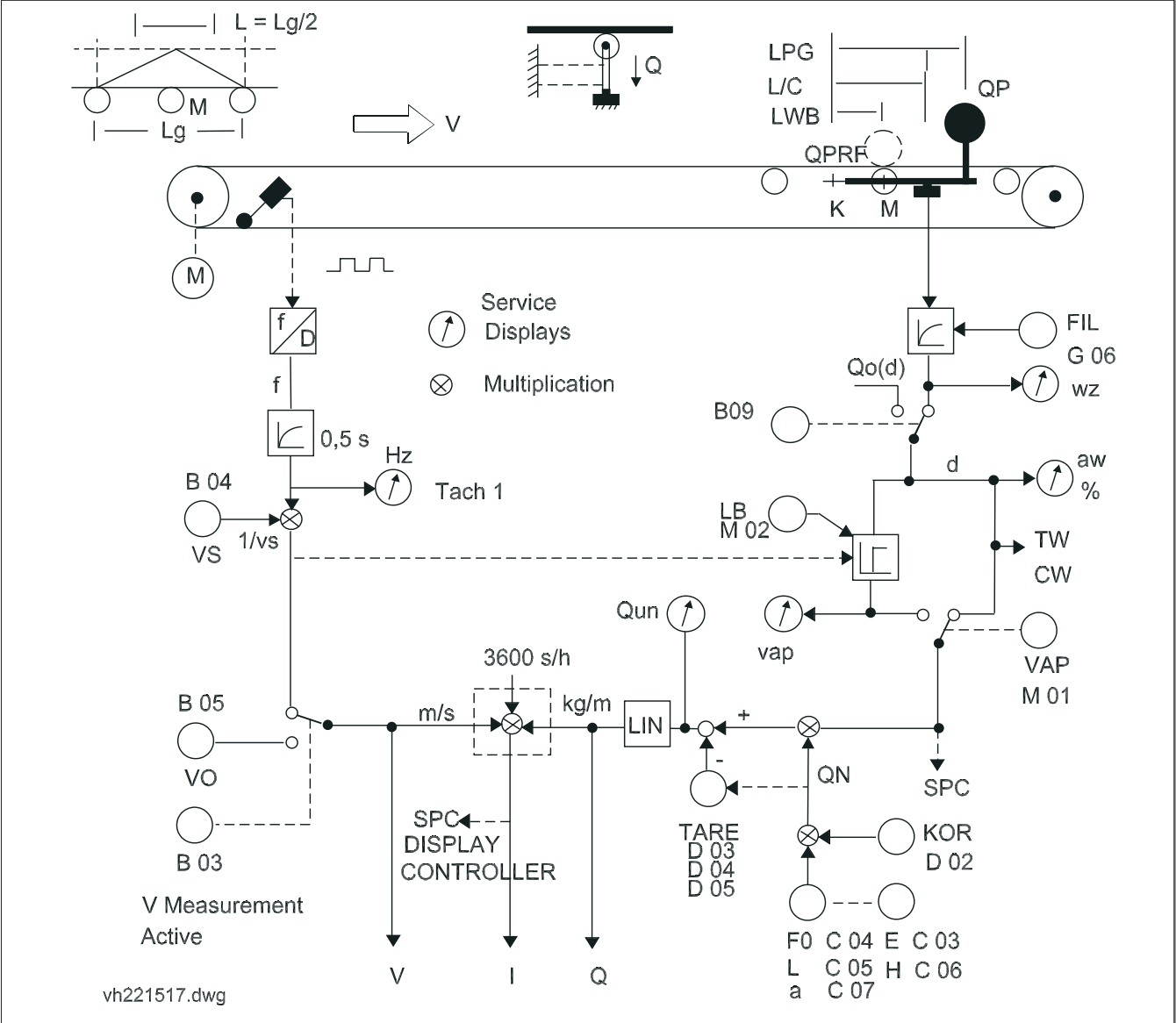


Fig.: Measurement

Note:

In normal INTECONT use, the data flow diagrams need not be known.

In special cases, they can be used to trace internal normalizations important for precalibration (theoretical calibration).

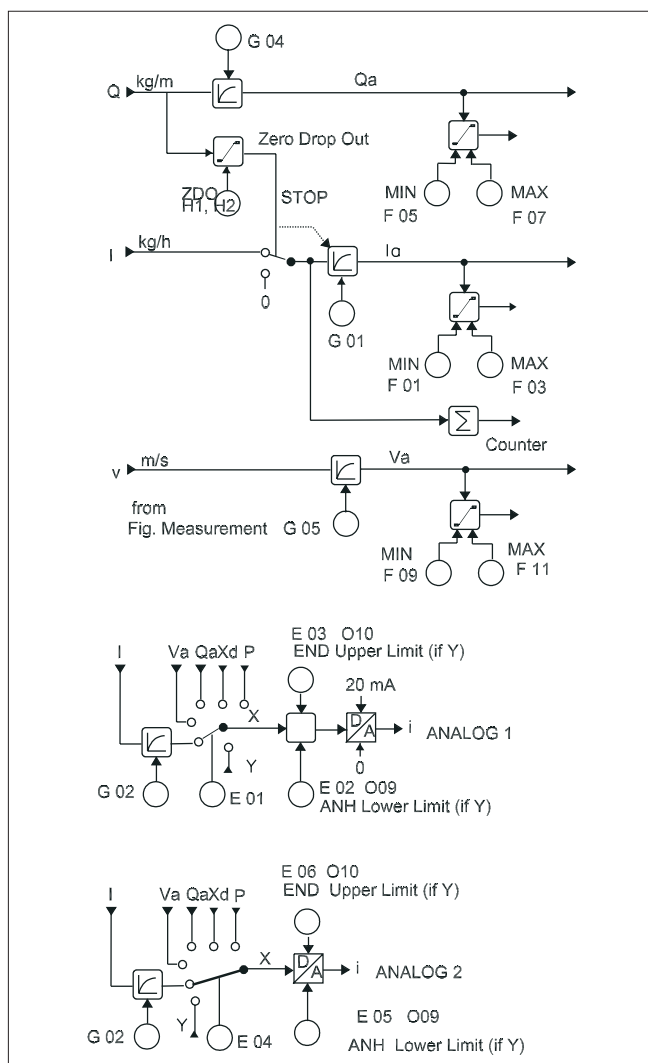
Parameters:

lo Nominal Feed Rate B 02 in t/h

lo serves as reference for limit values and service displays.  
lo has no influence on calibration.

$$IO = Qo \cdot Vo \cdot 3.6$$

Qo = in kg/m  
Vo = in m/s  
lo = in t/h



*Fig.: Displays, Limit Values, Analog Outputs*

Q0    **Nominal Belt Load**    D 01    in kg/m

Computed by INTECONT, Qo serves as reference for limit values and displays.

Nominal platform load  $Q_B = Q_o * L$

L = effective platform length

**Vo**      **Nominal Speed**                      **B 05**      **in m/s**

Reference for limit values; calibration relevant only with scales

1. operating without speed measurement or
2. using adaptive belt influence compensation BIC

### Meanings:

| L | Eff. Platform Length | C 05 | in m |
|---|----------------------|------|------|
|---|----------------------|------|------|

Acquire exact length value and enter into parameter.

| VS | Speed Transducer<br>Characteristic Value | B 04 | in l/m |
|----|------------------------------------------|------|--------|
|----|------------------------------------------|------|--------|

Determining speed measurement normalization, VS is important for scale accuracy.

**Fo L/C Rated Cap. C 04 in kg**

Total for load cell rated capacities. Pivots count as load cells.

$$F_o = (N + M) * NOM$$

NOM = Load cell rated capacity

N = Number of load cells

M = Number of pivots

Fo determines scale accuracy.

| E | L/C Characteristic Value | C 03 | in |
|---|--------------------------|------|----|
|   | mV                       |      |    |

The transmission coefficient is an important value for calibration.

$$U_a = U_e \cdot E \frac{F}{F_o}$$

Ua = Load cell output voltage  
in mV

Ue = Load cell input voltage in V  
= +/- 5 V (10 V)

$$F = \text{Load on load cell}$$

CSD load cells:  $E = 2.85 \text{ mV/V}$

HBM load cells  $E = 2.00 \text{ mV/V}$ 

With certain scale types, E is not calibration.  
For exact value, see type plate.

## H Lever Ratio C 06

The lever ratio determines calibration accuracy. However, it must not be factored into check weight.

$$F = H \cdot QB$$

$$H = \frac{LWB}{LWZ}$$

F = Force applied to load cell in kg  
QB = Load on platform in kg  
LWB = See figure

With weighing platforms with leaf spring parallel guidance, H = 1.

## a Angle a C07 in degrees

Enter belt inclination if load cell is mounted right-angled to belt.  
If load cells are arranged vertical to belt, the angle is always 0.

## QPRF Check Weight C 08 in kg

Used to check theoretical calibration.

As effective check weight, enter the platform load simulated by the check weight.  
Spec Sheet indicates check weight factor f.

$$f = \frac{\text{Simulated Platform Load}}{\text{Actual Check Weight}} \text{ in } \frac{\text{kg}}{\text{kg}}$$

$$QPRF = f \cdot \text{Check Weight}$$

## KOR Span Correction D 02

D 02 is primarily used to calibrate the result of a check using material. This parameter acts only on the belt load measuring branch.

### Analog output:

$$i = ANH + (END - ANH) \cdot \frac{x}{X_{nenn}}$$

$$ANH \leq i \leq 20\text{mA}$$

X = I, V, Q and Xd (see Fig. "Analog Output").

With control magnitude Y, the upper and lower controller limits (O09 and O10) are active.

### Operating Principle - Summary :

Load cell output voltage Ua is amplified and converted into digital value d.

As INTECONT knows transmission factor A0 of analog-to-digital converter and load cell data, the load on load cell can be computed from the output voltage.

Belt load results from platform data and belt inclination.

$$Q(\text{display}) = Ua \cdot A0 \cdot QN \text{ in kg/m}$$

$$QN = \frac{1}{GK \cdot A0} \text{ in } \frac{\text{kg/m}}{d}$$

$$GK = Ue \cdot E \cdot H \cdot \cos(a) \cdot \frac{L}{Fo}$$

Fo = Total of load cell rated capacities in kg

E = L/C characteristic value in m/mV  
GK = Device constant

$$\text{in } \frac{\text{mV}}{\text{kg/m}}$$

Ue = Load cell input voltage in V  
a = Angle in degr.

Belt speed is computed from speed transducer output frequency f.

$$V = \frac{f}{VS} \text{ in } \frac{\text{m}}{\text{s}}$$

The product from Q and V is feed rate I.

$$I = Q \cdot V \cdot 3600 \text{ s/h in kg/h}$$

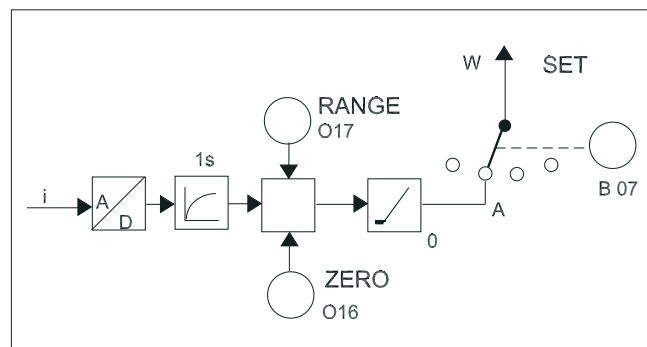
For the tare parameters to be exactly calibrated, range correction takes place before taring. If range normalization QN (Parameters KOR, Fo, etc.) are changed, tare parameters are automatically converted so that zero point is maintained.

0329

### Scales without speed measurement:

For setting program LB, a frequency of  $V0 \cdot VS$  is internally generated.

If one of the two parameters has been changed, recall setting program LB. Only then all other setting programs will run over integer belt circuits.



*Fig.: Analog Setpoint*

### Simulation:

In Simulation Mode, the control circuit internally closes.

Measurement values Q and V continue being acquired, displayed and monitored, and enter into controller via the adaption points. If need be, set adaptations to NO.

Simulated value is displayed as feed rate.

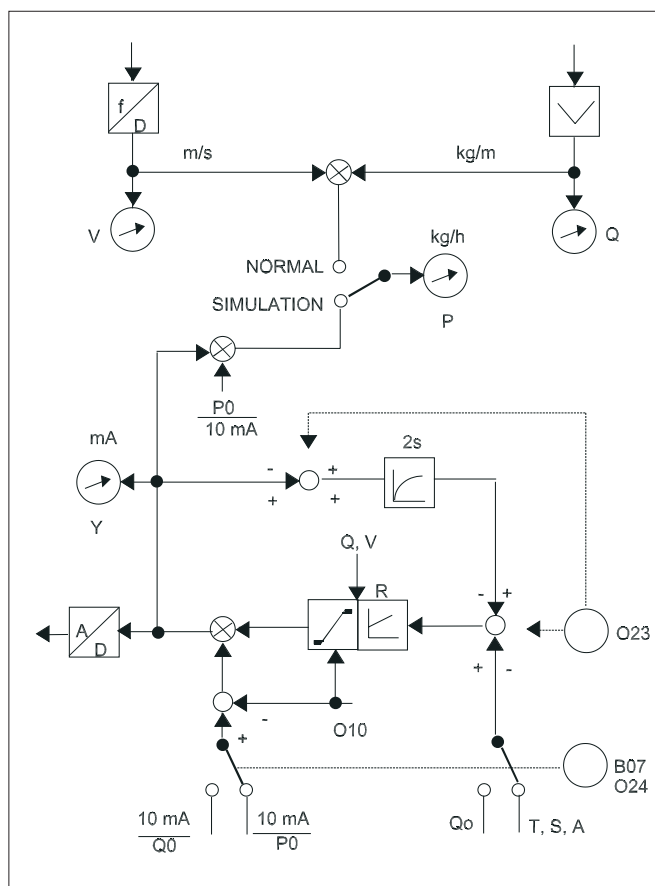
If nominal setpoint is preset, control magnitude Y is 10 mA.

Operating modes, Start/Stop and setpoint sources act  
behave as usual.

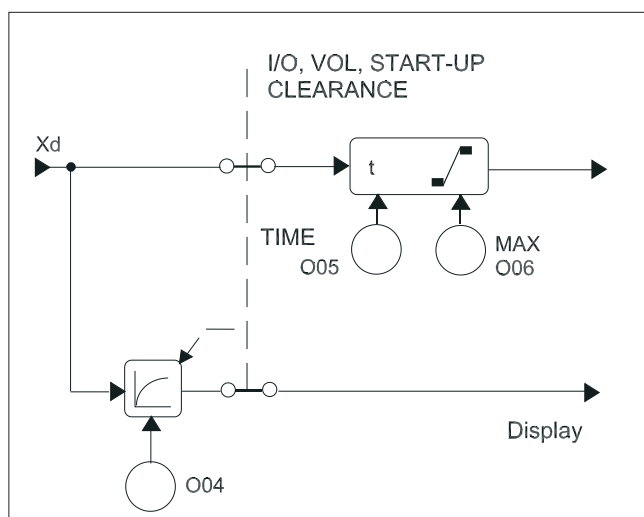
Functions : Test system, particularly controller sequences; learn how to operate INTECONT.

Call : Calibration Functions, START simulation

Display : Event S7



*Fig.: Simulation Mode*



*Fig.: Deviation*

## Data Flow Diagrams: Control

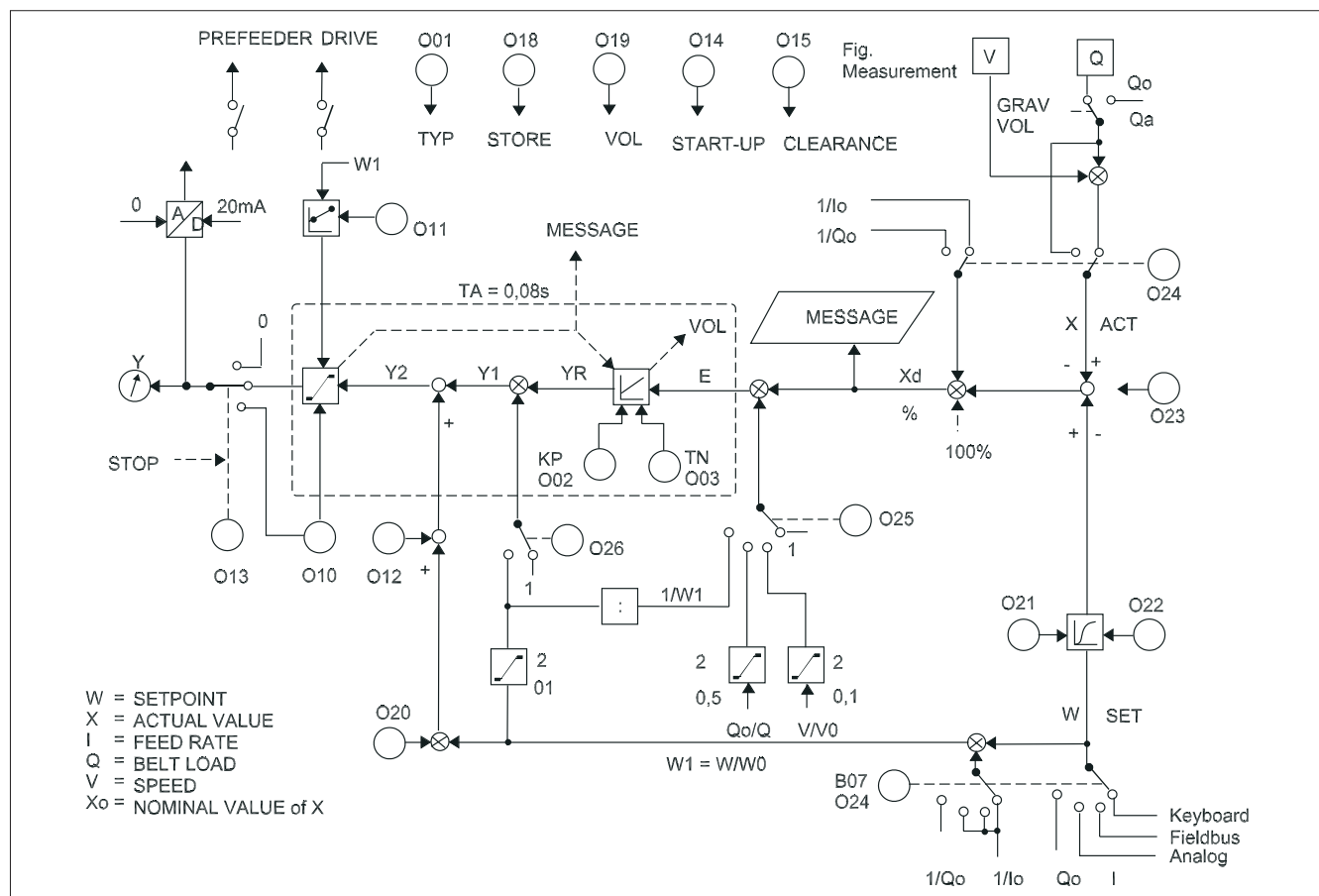


Fig.: Universal Controller

### Parameters:

The universal PI controller with setpoint bypass and adaptive interventions can be matched to the application by parameter setting.

### B 07 Setpoint Source

OP = keyboard  
FB = fieldbus  
AI = analog input

Special setting Q 0 (nominal belt load) is possible only with O24.

OP, FB, AI: for weighfeeder and belt weigher with feed control

Q0: for belt weighers with control for constant belt load (see O24)

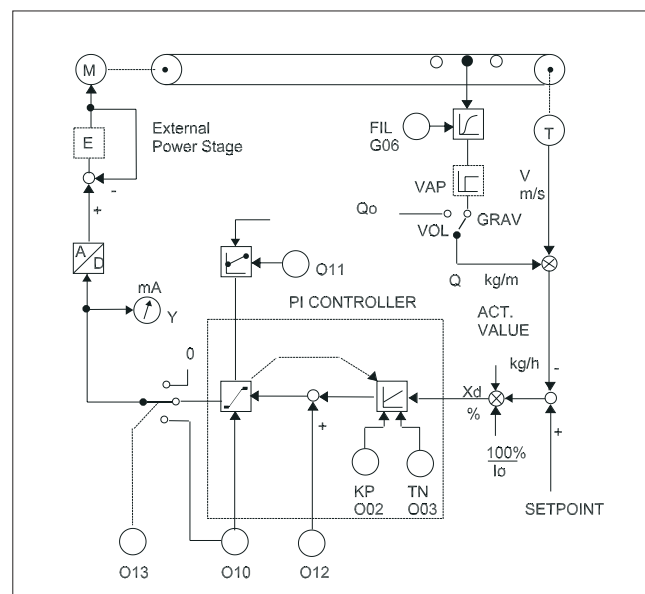


Fig.: FEED

## O 01 Controller Type

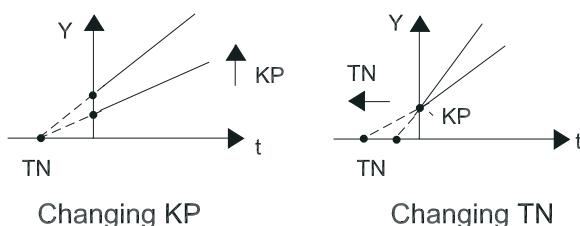
Change over to reduced extent of functions (see Fig. FEED).  
The reduced extent is specially provided for speed-controlled weighfeeders.

## O 02 P-Component KP in mA/%

Reference is nominal feed rate or nominal belt load (Parameter B 07, O 23).

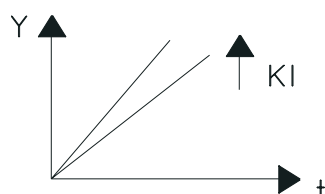
## O 03 Setback Time TN

TN is the time during which the I component causes the same control magnitude change as the P component.



### Special cases:

1.  $KP = 0$  : I-controller  
TN is the reciprocal of integration constant KI, i.e. big value = slow controller.  
 $KI = 1/TN$  in %/(mA/s)
2.  $TN = 0$  : P-controller  
The I-component is removed.
3.  $KP = TN = 0$  : Controller output = elevation



## O 04 ... O09

For monitoring, see Parameter List.

## O10... O13

For limitation and elevation, see Parameter List.

## O14 Start-up in belt circuits

Volumetric Start-up mode after start of scale.  
See Parameter O 18 and further explanations given in this chapter.

## O 15 Clearance in belt circuits

Volumetric clearance mode after scale stop.  
See Parameter O 18 and further explanations given in this chapter.

## O16 ..O17

For analog setpoint input normalizations, see Parameter List.

## O18 Store

Store integral component (O19 = Yconst) or belt load (O19 = Qconst) after scale stop. Value is active upon start-up in gravimetric or volumetric mode.

- NO : Reject  
YES : Store  
YES-A : Store only if scale has not failed due to Alarm.

## O19 Volumetric Mode

Volumetric mode can be determine using Qconst or Yconst.

Qconst:  
Controller continues operating. Current belt speed multiplied by Qconst acts as actual value.  
Qconst can be either the nominal belt load or the value stored after stop (Parameter O18).  
This mode is suitable for speed-controlled weighfeeders.

Yconst:  
Controller is cut off. Setpoint bypass and constant controller output YR are active. YR is either 0 or the integral controller component stored after stop (Parameter O18). This mode is suitable, e.g. for belt weighers with controlled prefeeder.

## O 20 Bypass in mA

Proportional to preset setpoint, the control magnitude is additively superposed by a value. This parameter indicates the bypass value for nominal setpoint.

## O 21 Setpoint Filter T1 in s

1st-order filter for setpoint branch (see O22)

## O 22 Setpoint Filter T2

2nd-order filter for setpoint branch. If bypass O20 is used, set approximate time behaviour of control circuit using O21 and O22. If no bypass is used, use O21 as "setpoint ramp".

## O 23 Set/Act Comparison

Form deviation Xd:

1.  $X_d = \text{Set/Act (W-X)}$
2.  $X_d = \text{Act/Set (X-W)}$

The second case is useful only with belt weighers controlled for constant belt load.

## O24 Set/Act Sources

Select setpoint and actual value.

I : Setpoint = feed rate in accordance with Parameter B 07  
Act. val. = feed rate measured  
Q : Setpoint = nominal belt load Qo.  
Act. Val. = belt load measured

In the second case, setpoint cannot be entered from sources B 07. Upper display shows reading of counter Z1.

## O25 Adaption 1

Controlled adaption of controller input

NO: No adaption  
V : Multiplication by V/Vo  
1/Q: Multiplication by Qo/Q  
1/W: Multiplication by Wo/W

Use adaption to adjust controller to variable loop gains, or improve reference behaviour. For more information, see application examples given in this chapter.

## O26 Adaption 2

Controlled adaption of controller output

NO: No adaption  
W: Multiplication by setpoint  
See O25.

## Power ON:

Reset controller, set setpoint filters O21, O22 to 0., Qold = Qo.

## Stop:

Setpoint filters O21, O22 are still active. Store integral controller component and belt load.

## Stand-By:

If setpoint falls below limit F 19 for more than 3s, scale goes to Stand-by mode.

O19 = Qconst Stop controller and initialise; for Y, see Parameter O13.  
O19 = Yconst Stop controller; for Y, see Parameter O13.  
Remove integral component as a function of Parameter O18, or store.

The "Drive" and "Prefeeder" control outputs go to STOP state, the "START Feeder" output remains in

START state.

Return to normal mode is effected without delay.

## Operating Intervals with O19 = Qconst:

O19 determines the type of volumetric mode, start-up and clearance.

Controller setting O 01 = STANDARD produces the same effect as O19 = Qconst.

**Start-up:** Start command immediately sets "Prefeeder" and "Drive" outputs to ON.  
Controller belt load input is set to nominal belt load  $Q_0$  or stored value  $Q_a$  as a function of Parameter STORE (O18).  
Belt speed is controlled proportionally to setpoint.  
After start-up period, changeover to belt load measurement value takes place automatically, i.e. feed rate is controlled (Gravimetric mode).

**Clearance :** Stop command immediately shuts prefeeder.  
Controller continues operating as in Start-up mode.  
Belt load is not the measurement value but value  $Q_a$  measured and stored last.  
This ensures smooth changeover.  
 $Q_a$  is the value smoothed by display filter. It is stored for next start-up period.  
After clearance, drive automatically cuts off.  
Start command during clearance starts new start-up period.

**Alarm :** Alarms cut off scales without clearance. Only  $Q_a$  is stored.

**Volumetric :** Belt speed is controlled proportionally to setpoint.  
Across controller belt load input, constant value  $Q_0$  (nominal belt load) is available. Volumetric mode does not use any stored belt load values during start-up and clearance.  
In contrast to Gravimetric mode, the triple nominal setpoint can be entered.

**Volumetric Synchronous :** See above, however, not  $Q_0$  but the value last measured before changeover is active.

**Gravimetric :** Control of feed rate  
 $I = Q \cdot v$ .  
 $Q$  and  $v$  are the current measurement values.  
Gravimetric mode is available if

1. no Volumetric mode is selected
2. no Start-up or Clearance mode is available.

Setpoint is limited to nominal value.

## Operating Intervals with O19 = Qconst:

This case is possible only if controller is set to UNIVERS.

**Start-up:** Start command immediately sets "Prefeeder" and "Drive" outputs to ON.  
Controller remains cut off. As a function of Parameter STORE (O18), controller output YR is set to 0 or to the integral component stored.  
External controller drive operates in proportion to preset setpoint.  
After start-up period, controller automatically connects and Gravimetric mode (feed rate control) commences.

**Clearance :** Stop command immediately shuts prefeeder and disconnects controller.  
The integral component is stored for next start-up period.  
Setpoint bypass is superposed by the controller integral component YR acquired last.  
After start-up, drive automatically cuts off.  
A start command output during clearance starts new start-up period.

- Alarms : Alarms cut off scale without clearance period. Only the controller integral component is stored.
- Volumetric : Only setpoint bypass is active. The controller integral component is set to 0, and controller cuts off. Any external control drive is controlled in proportion to setpoint. For max. setpoint, the triple nominal feed rate of scale can be entered.
- Volumetric (Synchronous) : See above, however, the controller integral component remains stored.
- Gravimetric : See O19 = Qconst.

## Display and store modes:

- Volumetric/  
Volumetric
- Synchronous : "v" in upper display field
- Start-up : "v" flashes
- Clearance : "v" flashes
- Keyboard Mode: Keyboard symbol (square) in upper display field frames "v"
- Gravimetric : No symbol, running message = rotating point
- Power OFF : Volumetric, Gravimetric and Keyboard modes are stored.

## Applications:

### 1. Weighfeeder

A weighfeeder is designed to extract material from a surge bin.

Actual feed rate is compared with setpoint; the difference, sent to controller R. The latter keeps varying belt speed until actual value equals setpoint.

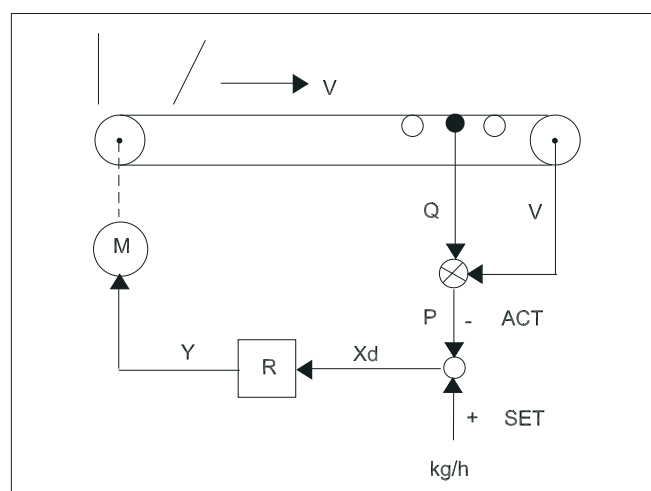


Fig.: Weighfeeder

**Input :** Set Parameter O 01 to STANDARD.  
Only Parameters O 01...O 17 are available; O 18...O 26 cannot be accessed.

**Control diagram:** See Fig. FEED.  
**Gravimetric :** Control of actual feed rate  $Q * V$   
**Volumetric :** Control of belt speed (see "Operating intervals with O19 = Qconst").

The weighfeeder's control structure can also be reached by setting Parameter O 01 to UNIVERS. In addition, you can adapt variations in loop gain caused by varying belt load. (O25 = 1/Q, dashed data flow arrow).

### 2. Belt Weigher With Controlled Material Prefeeder

Feed rate is controlled for setpoint by variation of belt load. Belt speed is normally constant.

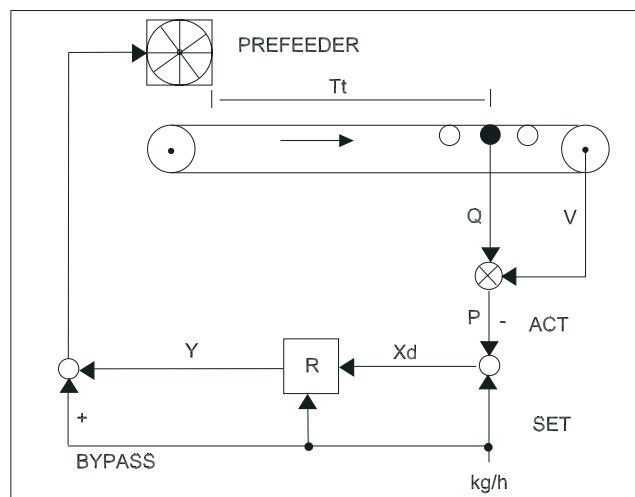


Fig.: Belt Weigher With Prefeeder

Position-controlled prefeeders require an external 3-step controller to be present.

**Input :** Set Parameter O 01 to UNIVERS. All controller parameters O 01...O 26 can be accessed.

**Control diagram:** See Fig. Universal Controller  
**Gravimetric :** Control of actual feed rate  $Q * V$

**Volumetric :** Control of prefeeder via setpoint bypass, no control.

**Parameters:**

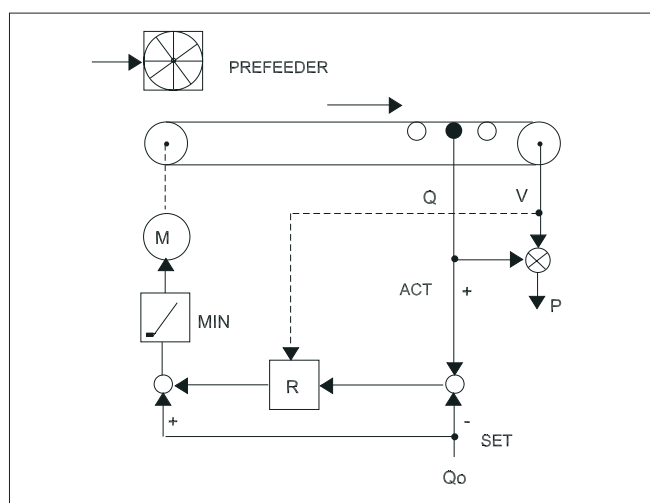
|            |   |                                |
|------------|---|--------------------------------|
| O 01       | = | UNIVERS                        |
| O 02...O18 | = | See Parameter List.            |
| O19        | = | Yconst, Volumetric type        |
| O20        | = | apr. 10...15mA Setpoint Bypass |
| O21        | = | apr. Tt/5 Setpoint Filter T1   |
| O22        | = | apr. Tt/5 Setpoint Filter T2   |
| O23        | = | W-X Set/Act Comparison         |
| O24        | = | I Actual value I = $Q * V$     |
| O25        | = | 1/W Adaption 1                 |
| O26        | = | W Adaption 2                   |

The adaption points enables the controller working point to be quickly adapted to setpoint changes. Parameters KP and TN remain stored. Filters O21 and O22 simulate the time behaviour of the loop (deadtime Tt) to permit bypass to act alone in

case of setpoint steps. If, with setpoint steps, the actual value strongly tends to over or undershooting, adjust filters empirically.

### 3. Belt Weigher With Constant Load

A controlled prefeeder feeds material to a conveyor belt.  
Controller adjusts belt speed so that belt load always corresponds to nominal value  $Q_0$ .



*Fig.: Belt Weigher*

The use of a constant working point of the weighing system ensures high accuracy of measurement. Feed rate at point of discharge corresponds to the prefeeder feed rate.

Input : Set Parameter O 01 to UNIVERS.  
All controller parameters  
O 01...O26 can be accessed.

Control diagram: See Fig. Universal Controller

Gravimetric : Control for constant belt load  $Q_0$ .

Volumetric : Control of belt drive using a signal proportional to  $Q_0$   
(see "Operating intervals with  $O_{19} = Y_{const}$ ")

Setpoint : No input possible

Control range : To avoid standstill of conveyor belt with small belt loads, lower control magnitude is limited to MIN.

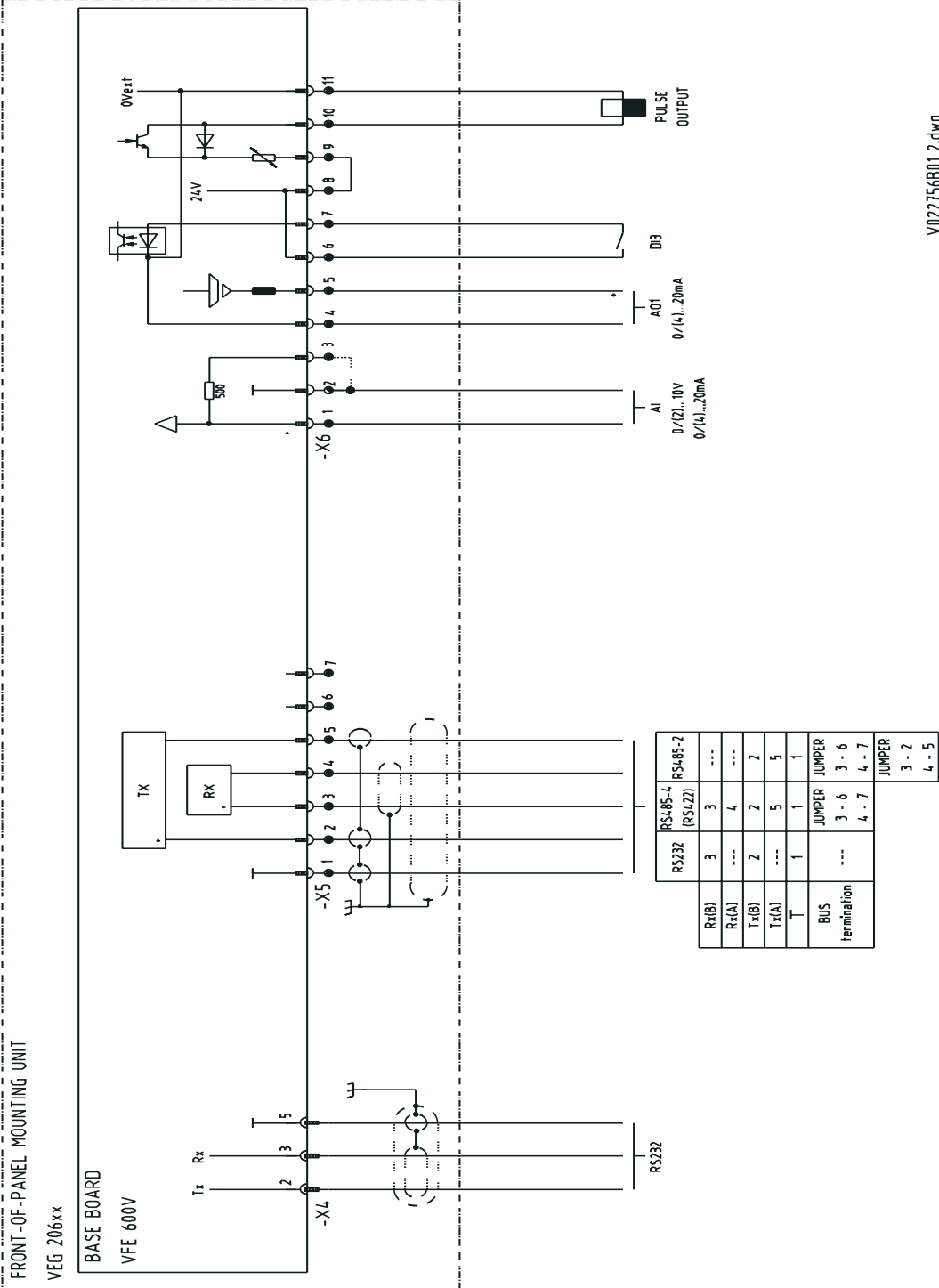
Parameters:

|           |   |                                               |
|-----------|---|-----------------------------------------------|
| O 01      | = | UNIVERS                                       |
| O 01..O09 | = | See Parameter List                            |
| O10       | = | 1mA Lower Limit Y<br>(apr. 0.1 nominal speed) |
| O11...O20 | = | See Parameter List                            |
| O19       | = | Yconst, Volumetric type                       |
| O20       | = | apr. 10...15mA Qo Bypass                      |
| O21       | = | 0 Setpoint Filer                              |
| O22       | = | 0 Setpoint Filter                             |
| O23       | = | X-W Set/Act Comparison                        |
| O24       | = | Q Actual value = Q<br>Setpoint = Qo           |
| O25       | = | V Adaption 1                                  |
| O26       | = | NO No Adaption 2                              |

Control of belt speed changes the dead time from prefeeder to weighing platform. To ensure consistent control quality over a wide range, controller is adapted as a function of belt speed.

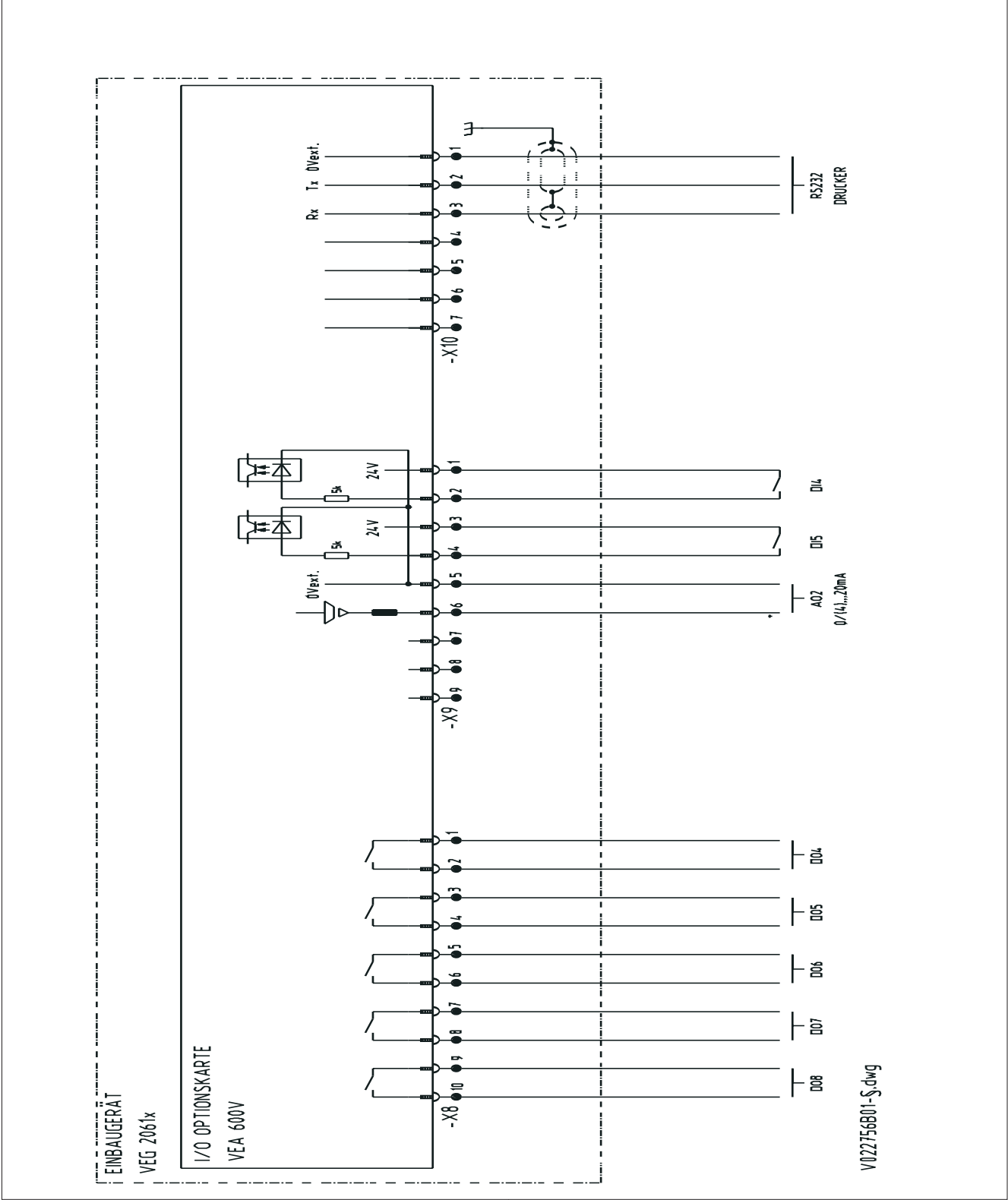
### Base Card





V022756B01\_2.dwg

Option Card



## Namur Inputs:

> 8 mA = Short circuit  
 1.65...8 mA = Level 0  
 0.1...1.65 mA = Level 1 (sensor covered)  
 < 0.1 mA = Cable breakage

## Option Card:

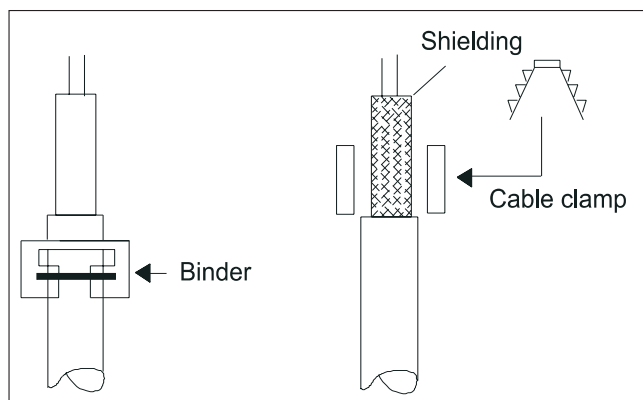
The option card is affixed to the INTECONT cover. Signal exchange with base card is via a pluggable flat cable connector.

If pulled off, system behaves as if no option card was present and only the operating functions and parameters of the base card can be used.

## Cabling Hints:

- All cables are led to the device from the rear (front-of-panel mounting housing) or from below (flush mounting housing) and connected with special connectors. Connectors cannot be confused.
- The INTECONT is isolated from the mounting surface by its plastic housing.  
 The rear earthing connector is internally connected with the electronics zero potential.  
 The protective ground used requires to be hum- and transient-free.  
 The power supply zero potential (0V ext.) is not connected to the earthing connector. In general, no earthing is required.
- Make sure no leakage currents are conducted over cable shields. Therefore shields are applied to one side of INTECONT. The load cell housing is not connected with shields.  
 Shields of data cables to host computer are connected to ground on either side.
- The consumers connected to the relay and pulse outputs require to be radioshielded, e.g. by suppressor diodes or RC elements.
- Analog outputs  
 Long cables, particularly to frequency transducers, can interact on analog outputs. We suggest to use an isolating amplifier.
- Make sure the space between power and measuring cables, if run in parallel, is min. 0.30 m.  
 If this is not possible lay measuring cable in steel conduit. The same holds if the measuring cable is laid over free stretches or close to powerful transmitters (e.g. broadcasting stations).

## ■ Fasten cables.

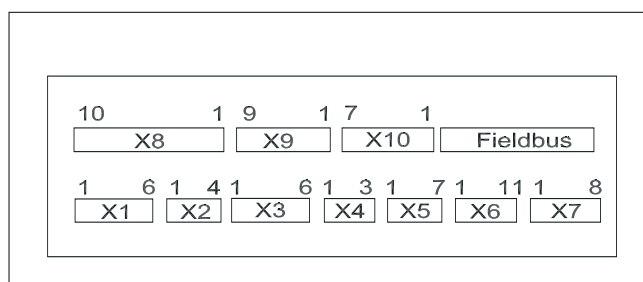


In the rear wall of the INTECONT, under each connector, several slots are provided. The upper slot pair is reserved for the cable clamp which presses the shielded cable against the rear wall. The shield needs not be separately connected. The lower cut-out serves for accommodation of a pull relief, e.g. by cable binders.

## Attention!

Connectors to base card require to be plugged-in with screwed connections up; connectors to option card, with screwed connections down.

## ■ INTECONT rear wall: connectors



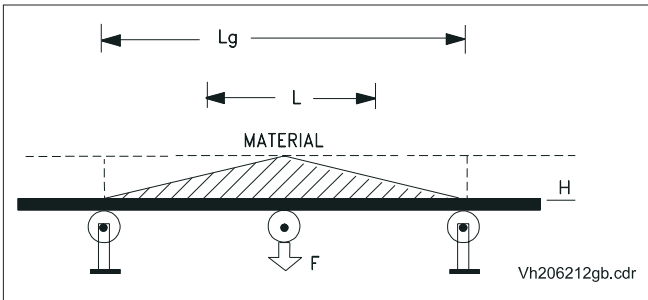
Weighing Plattformen

1. Effective Platform Length

The effective platform length is an important value for calibration. Determined by platform design, it can normally be taken from Spec Sheet. However, we recommend to remeasure length and to enter the exact value into parameter. The hatched areas shows the distribution of load on platform.

- Single-idler platform,  
e.g. DEL, DEM, DAN, BEZ, BED types

Platform load is applied to load cell via a weighed idler. This can be done by using a lever system or, with no levers, by leaf spring parallel guidance.

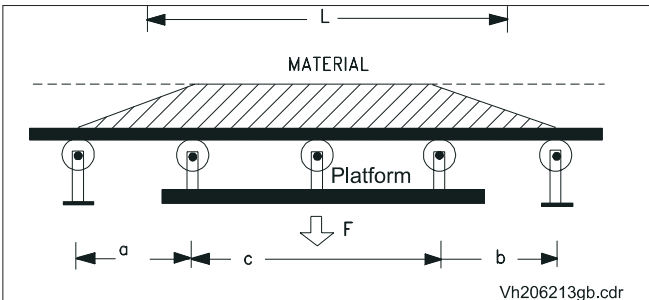


$$L = \frac{L_g}{2}$$

Single-idler platform

- Multi-idler platform,  
e.g. BMP, BMC types

Multiple weighed idlers are combined to form a mechanical unit.

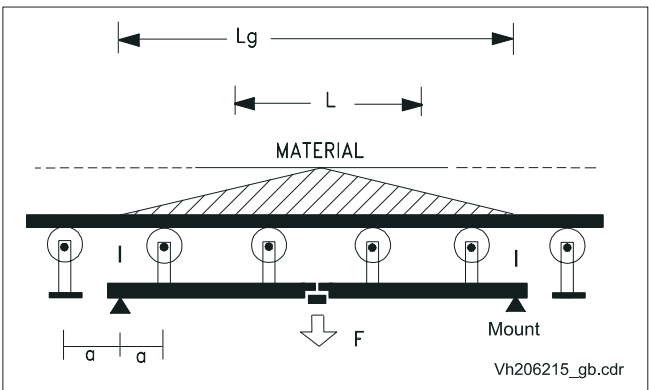


$$L = \frac{a + b}{2} + c$$

Multi-idler platform

- Buckled platform  
e.g. DEA, BMK types

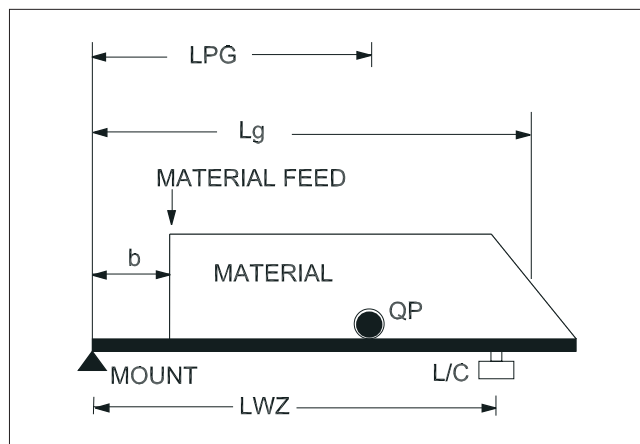
The platform consists of two sections. In mid platform, force is applied to load cell. Instead of weighed idlers, a two-sectional weighing table can be used.



$$L = \frac{L_g}{2}$$

Buckled platform

## ■ One-sided mounting of total belt



$$L = \frac{Lg^2}{2 LWZ} - \frac{b^2}{2 LWZ}$$

On-sided mounting

Lg = length up to mid hillside

## 2. Check Weight QPRF

Simulating a certain belt load, the check weight is used for control of weighing system. It can act on weighing platform direct or via lever system. The weight to be entered into INTECONT is not the actual weight, but the weight acting on platform.

Effective check weight QPRF is the platform load in kg simulated by the check weight.

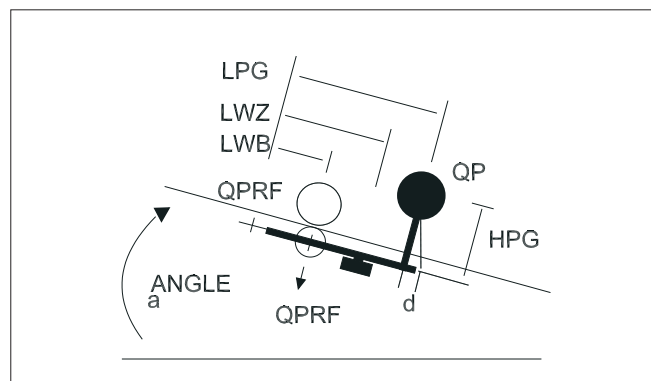
### ■ Single-idler platform, DEL, DEM, DAN types

#### a) Leaf spring parallel guidance:

Theoretically, the check weight can be applied to any point of the mechanical platform equipment. Applied weight QP is active always.

$$QPRF = OP$$

#### b) Joint mounting (e.g. universal spring joint):



$$QPRF = QP \frac{LPG}{LWZ}$$

$$K = QP \frac{HPG}{LWB} \cdot \tan(a)$$

$$\tan(a) \approx 0,018 \cdot a$$

Angle a = 0 : K = 0

Angle a = neg : K = neg

## ■ Multi-idler platform

a) Total platform mounted on load cells:

$$QPRF = OP$$

b) Joint mounting:

see Single-idler platform

## ■ Buckled platform

Normally, check weight acts on center.

$$QPRF = OP$$

## ■ One-sided mounting of total belt

With inclined belts, consider extension, or shortening, of calibration lever, if any, in accordance with single-idler platform.

$$QPRF = QP \frac{LPG}{LWZ}$$

## Chain calibration weight:

If a chain curtain is placed over the platform area, the total weight of chains in weighing area is the effective check weight.

$$QPRF = * L * s$$

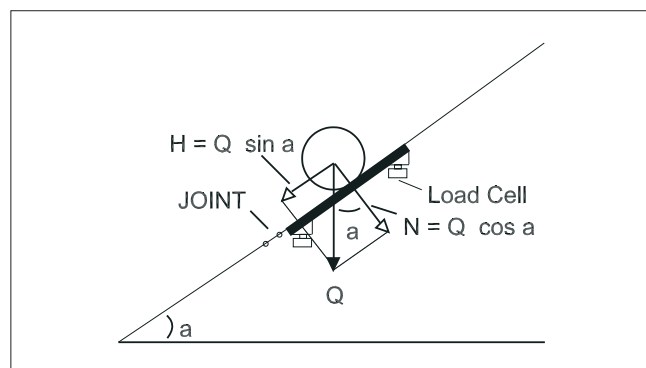
|   |   |                           |         |
|---|---|---------------------------|---------|
| n | = | Number of chains          |         |
| L | = | Effective platform length | in m    |
| s | = | Weight of chain per m     | in kg/m |

## 3. Conveyor Belt Inclination

In certain cases, the inclination of conveyor belt has to be entered into parameter. Decisive is the load cell mounting position.

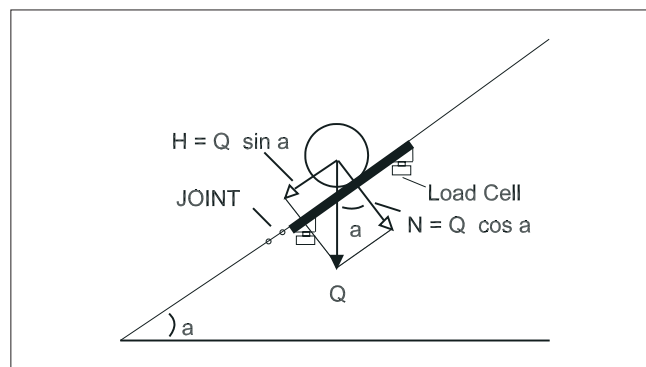
a) Right-angled to conveyor belt

Enter angle a into parameter. Load cell is loaded with  $N = Q * \cos a$ .



b) Vertical to conveyor belt

Enter 0 as angle a. Load cells are loaded with full force Q.



## Printer Connection

Interface : Serial, see "Technical Data".  
Cables : The connection diagram applies to EPSON and CITIZEN printers.

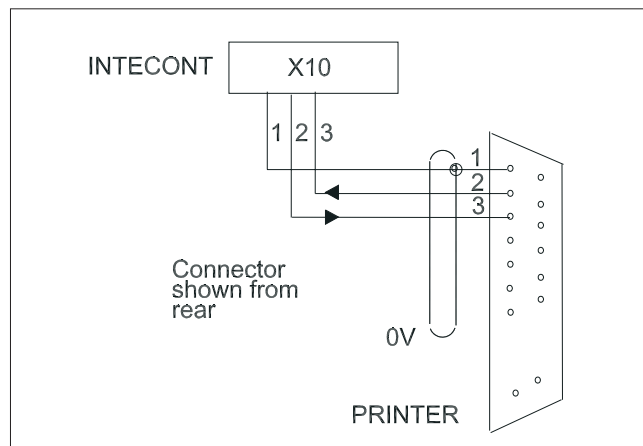


Fig. Printer Cable

EPSON printer: LX 850

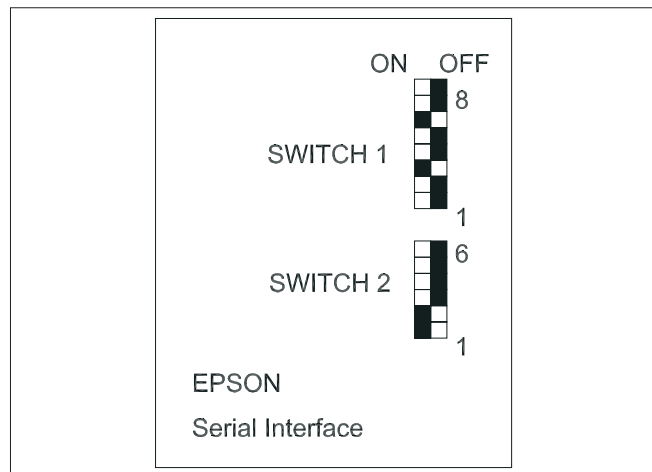


Fig. EPSON Interface

The interface board is equipped with 2 DIP switches used for selection of serial message base data.

To initialize:

Printer is initialised via serial interface, 5s after power-up of INTECONT PLUS.

Switch

|         |   |                            |               |
|---------|---|----------------------------|---------------|
| 1-1     | : | OFF                        | 8-bit message |
| 1-2     | : | OFF                        | No parity bit |
| 1-3     | : | egal                       | Even/odd      |
| 1-4     | : | OFF                        | Postive edge  |
| 1-5...8 | : | OFF, ON, OFF, OFF          | 9600 baud     |
| 2-1...6 | : | ON, ON, OFF, OFF, OFF, OFF | (default)     |

Use Parameter J 01 to set baud rate to 9600. On the right side of printer, there are two additional DIP switches. Diagram view corresponds to top of printer.

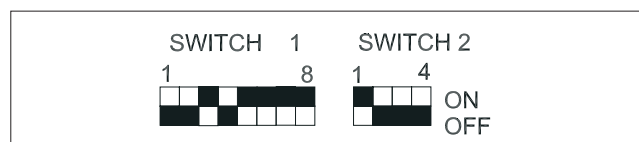


Fig. EPSON Printer

Switch

|         |   |            |                            |
|---------|---|------------|----------------------------|
| 1-1     | : | OFF        | 10 CPI character spacing   |
| 1-2     | : | OFF        | Zero without backslash     |
| 1-3     | : | ON         | Graphics character set     |
| 1-4     | : | OFF        | AUTO hyphenation activated |
| 1-5     | : | ON         | Normal sprint speed .      |
| 1-6...8 | : | ON, ON, ON | USA                        |
| 2-1     | : | ON         | 12" page length            |
| 2-2     | : | OFF        | No single sheet            |
| 2-3     | : | OFF        | Jump over performance      |
| 2-4     | : | OFF        | No AUTO line feed          |

Information on switch position can be printed using the printer self-test function. (Press line feed key while starting printer).

CITIZEN Printer: 120D

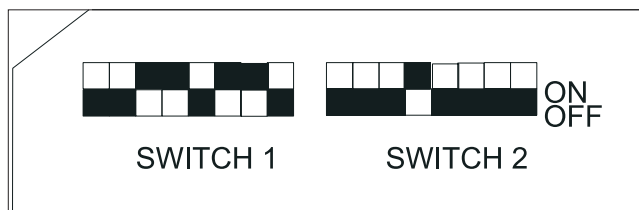


Fig. Serial Interface

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The FIP base and option cards connection diagrams indicate for every digital input and output the relevant INTECONT VEG parameter numbers used for distribution on physical signals.

The tables below compares the connector assignments of FIP and VEG for all physical signals.

## Digital Outputs

| FIP     | Physical Signal | VEG     |
|---------|-----------------|---------|
| X1 1,2  | DA1             | X1 1,2  |
| X1 3,4  | DA2             | X1 3,4  |
| X1 5,6  | DA3             | X1 5,6  |
| X6 1,2  | DA4             | X8 1,2  |
| X6 3,4  | DA5             | X8 3,4  |
| X6 5,6  | DA6             | X8 5,6  |
| X6 7,8  | DA7             | X8 7,8  |
| X6 9,10 | DA8             | X8 9,10 |

## Digital Inputs

| FIP    | Physical Signal    | VEG    |
|--------|--------------------|--------|
| X3 1,2 | DE1                | X3 1,2 |
| X9 1,2 | DE2                | X3 3,4 |
| X9 3,4 | DE3                | X6 6,7 |
| —      | DE4                | X9 1,2 |
| —      | DE5                | X9 3,4 |
| X3 5,6 | Sensor = (DE6)     | X3 5,6 |
| X5 1,2 | V1 (Tacho 1) = DE7 | X7 1,2 |
| X3 3,4 | Vol. Synchron.     | —      |

## Analog Inputs/Outputs

| FIP       | Physical Signal | VEG      |
|-----------|-----------------|----------|
| X9 9,10   | A01             | X6 4,5   |
| —         | A02             | X9 5,6   |
| X11 1,2,3 | AI              | X6 1,2,3 |

## Load Cell

| FIP  | Physical Signal | VEG  |
|------|-----------------|------|
| X5 3 | OUT1            | X7 3 |
| X5 4 | REF1            | X7 4 |
| X5 5 | IN2             | X7 6 |
| X5 6 | IN1             | X7 5 |
| X5 7 | REF2            | X7 7 |
| X5 8 | OUT2            | X7 8 |

## Attention:

Signals IN1 and IN2 are swapped between FIP and VEG. (VEG is compatible to DISOCONT).

## Power Supply

| FIP  | Physical Signal | VEG  |
|------|-----------------|------|
| X2 1 | 24V ext         | X2 1 |
| X2 2 | 0V ext          | X2 2 |
| X2 3 | ext. 0V supply  | X2 3 |
| X2 4 | ext. 24V supply | X2 4 |

## Printer

| FIP  | Physical Signal | VEG   |
|------|-----------------|-------|
| X8 1 | Screen          | X10 1 |
| X8 2 | TX              | X10 2 |
| X8 3 | RX              | X10 3 |

## Pulse Output

| FIP  | Physical Signal | VEG   |
|------|-----------------|-------|
| X9 5 | 24V             | X6 8  |
| X9 6 | Open collector  | X6 9  |
| X9 7 | Open emitter    | X6 10 |
| X9 8 | 0V              | X6 10 |

With VEG, fieldbus is connected via separate fieldbus cards (FIP: Connector X7). See Fieldbus Manual BVH2220.

Connector Size

| FIP |                             |                |
|-----|-----------------------------|----------------|
| X1  | Phoenix contact spacing 5mm | : 6 pins       |
| X2  | Phoenix contact spacing 5mm | : 4 pins       |
| X3  | Phoenix contact spacing 5mm | : 7 pins       |
| X5  | Phoenix contact spacing 5mm | : 8 pins       |
| X6  | Phoenix contact spacing 5mm | : 8 or 10 pins |
| X7  | SUB-D                       | : 9 pins       |
| X8  | SUB-D                       | : 9 pins       |
| X9  | Phoenix contact spacing 5mm | : 12 pins      |
| X11 | Phoenix contact spacing 5mm | : 3 pins       |

| FIP |                             |           |
|-----|-----------------------------|-----------|
| X1  | Phoenix contact spacing 5mm | : 6 pins  |
| X2  | Phoenix contact spacing 5mm | : 4 pins  |
| X3  | Phoenix contact spacing mm  | : 6 pins  |
| X4  | SUB-D                       | : 9 pins  |
| X5  | Phoenix contact spacing mm  | : 7 pins  |
| X6  | Phoenix contact spacing mm  | : 11 pins |
| X7  | Phoenix contact spacing mm  | : 8 pins  |
| X8  | Phoenix contact spacing 5mm | : 10 pins |
| X9  | Phoenix contact spacing mm  | : 9 pins  |
| X11 | Phoenix contact spacing mm  | : 7 pins  |